Owning the Chokepoint: ASML, Alliance Dependence, and the Limits of Europe's Leverage over Washington

Matt DiGiuseppe*

October 29, 2026

Abstract

Since 2019 the Netherlands, home of ASML, the only maker of extreme ultraviolet lithography machines, has aligned its export controls with US demands while extracting little in return, as has Japan. Yet both resist some demands, such as limits on servicing tools installed in China. We explain the pattern with a game-theoretic model in which a security-dependent holder of a chokepoint bargains with its patron while the patron's rival bids for the chokepoint. Every threat is sequentially rational. We verify each result by brute-force computation. A holder's leverage is bounded by its exit premium: the rival's credible offer, net of the costs the patron can credibly impose. The patron's demands are bounded by the holder's slack, and its conditional threats share that one budget. Conditionality is the price of concessions: a patron makes its guarantee more conditional than deterrence requires only to create slack it can spend. When slack is wasting, the patron has a preventive motive to demand control of the chokepoint. Europe's leverage over Washington therefore depends on the conditional part of its dependence and on China's credible promises. Washington's need for ASML matters only through named channels such as rupture risk and hold-up over servicing.

*Working draft, October 29, 2026. Drafting, proofs and numerical checks were prepared with the assistance of an AI system (Claude Code); all results were checked by brute-force computation, and code is available on request.

1 Introduction

On 11 June 2026 eight authors, writing in a personal capacity, published *Europe 2031*, a scenario of what getting artificial intelligence wrong would mean for Europe (Juijn et al., 2026). They call ASML, the Dutch firm that alone builds extreme ultraviolet (EUV) lithography machines, "the only card Europe still holds". In the story's climax, in March 2031, the White House proposes to the leaders of the Netherlands, Germany and France that ASML be folded into a joint Dutch-American holding company in which Washington holds a controlling vote over output, customers and technology transfer. When the three leaders decline, Washington makes the offer public and threatens to drop the whole EU to Tier 3 of its rules on access to frontier AI. Three years earlier in the story, the Dutch had ended ASML's remaining sales and servicing of immersion tools in China under the threat of the US foreign direct product rule, and one character reflects that "we didn't even manage to negotiate a single thing in return".

The scenario is fiction (appendix C sets its episodes beside the model), but its premise describes the record since 2019 (section 2). In 2019, after US lobbying reported by Reuters, the Dutch government let ASML's licence to ship an EUV machine to China lapse without renewal (NL Times, 2020). National licensing of advanced immersion lithography took effect on 1 September 2023 (Government of the Netherlands, 2023). In September 2024 licensing for the older NXT:1970i and NXT:1980i moved from Washington to The Hague (ASML Holding N.V., 2024), a measure on metrology and inspection systems published in January 2025 took effect on 1 April 2025, and in November 2025 the EU put the most advanced immersion tools on the Union control list through Delegated Regulation 2025/2003 (ASML Holding N.V., 2026a). What The Hague received was a status. In December 2024 the Bureau of Industry and Security (BIS) placed the Netherlands on Supplement No. 4, whose listed countries' exporters are excluded from new foreign direct product and de minimis requirements on chipmaking equipment, because "Governments in those destinations ... have the authority to control key SME items of concern" (Bureau of Industry and Security, 2024). BIS can withdraw the status by amending a rule. We code it as a threat left unexercised for as long as Europe controls the items itself, a reward for self-binding and not a transfer (section 8.2).

Yet the holders of these chokepoints resist some demands. In April 2026 bills in both chambers of Congress, the MATCH Act, proposed controls on all immersion lithography, a policy of denial for licences to service covered Chinese fabs, and extended US controls on allies whose own controls were not certified as equivalent within 150 days (Baumgartner, 2026b; Ricketts, 2026). Prime Minister Jetten raised the proposed restrictions with President Trump at a White House dinner on 13 April (NL Times, 2026c), and on 12 May the trade minister told parliament that "the cabinet is against the extraterritorial effect of the American proposal" (Sjoerdsma, 2026, our translation). As of June 2026 both the Netherlands and Japan still let their firms service equipment in Chinese fabs (Zakaria, 2026). Tokyo had put 23 types of chipmaking equipment under licence in 2023 while stating that it was "not aligning itself with the United States' measures" (Allen, 2023; Scanlan-Dyas et al., 2023). Germany reportedly resisted US, Dutch and Japanese pressure to join controls aimed above all at Zeiss, ASML's sole supplier of optics; in one European executive's words, "Germany does not want

to join this club" (Barkin, 2024). South Korea, by contrast, accepted the loss of validated end-user status for the Chinese fabs of Samsung and SK hynix in 2025 (U.S. Department of Commerce, Bureau of Industry and Security, 2025).

Holders that Washington needs comply with little compensation, and they comply with some demands while resisting others. Each of the accounts that should explain this pattern misses part of it. Theories of asymmetric and weaponized interdependence give power to the less dependent party and to the state with jurisdiction over a hub (Hirschman, 1945; Keohane and Nye, 1977; Farrell and Newman, 2019). On that account Washington, which has no substitute for EUV lithography, should be paying The Hague. Accounts of alliance dependence and hierarchy treat compliance as the price of protection already received (Morrow, 1991; Lake, 2009), and Drezner (1999) finds that coercion among allies, though rare, often wins concessions. They fit the compliance but leave the resistance on servicing and extraterritorial reach unexplained. Models of competition among bidders hold that a client courted by two patrons is paid what the losing bidder would offer (Groseclose and Snyder, 1996; Bueno de Mesquita and Smith, 2016), as Kyrgyzstan was when it used a Russian aid package to raise the rent Washington paid for the Manas base in 2009 (Cooley, 2012). Customers in China took 36 percent of ASML's sales in 2024 (ASML Holding N.V., 2026a). A bidding account predicts a price, and The Hague has not been paid one.

We argue that a chokepoint holder's leverage is bounded by its exit premium, and that a patron's demands are bounded by the holder's slack. The exit premium is the rival's credible offer, net of the costs the patron can credibly impose on exit and the value the holder places on alignment; slack is the amount by which the premium falls short of zero. The first bound is a known result: the outside-option principle (Shaked and Sutton, 1984; Sutton, 1986; Binmore et al., 1989) and the one-voter case of sequential vote buying (Groseclose and Snyder, 1996; Dekel et al., 2008). Its substance goes back to Hirschman (1945) and Wagner (1988), and Clayton et al. (2026a) derive a two-hegemon version. The second bound organizes the paper. Clayton et al. (2026b) write that "threats generate power to the extent that they generate slack in the participation constraint". We take their term and ask what slack buys a patron when each of its threats must be sequentially rational, when the holder's outside option is a rival's bid, and when a threat that is carried out is spent. Our answer is a slack principle: whatever instruments back them, the conditional parts of a patron's threats draw on one budget, and every demand the patron makes (on sales, servicing, hedging or control) competes for it. A holder with a negative premium therefore gets nothing for its chokepoint unless the dispute risks rupture, however much its patron needs it. Its patron must ration demands. A demand meets resistance when its cost to the holder exceeds the bite that remains, and China can raise that cost. Its 2026 decree on supply-chain security lets Beijing restrict the trade and investment in China of foreign firms that cut off normal trade with Chinese entities (State Council of the People's Republic of China, 2026), which on our reading attaches a counter-threat to any order to stop servicing Chinese fabs.

In the game of section 4, the United States and China bid for Europe's alignment and one commitment technology governs every threat and promise. After the bidding benchmark (Lemma 1),

Proposition 1 makes China's offer Europe's outside option in random-proposer bargaining with a hazard of rupture. The transfer from Washington is $P^* = \max\{Q, \psi(Q + \beta(V - Q))\}$, where Q is the exit premium, V Washington's dependence, β Europe's probability of proposing and ψ the rupture weight. Corollary 1 states the threshold and names the five channels through which V enters. Proposition 2 proves the slack principle: in every subgame perfect equilibrium of any sequence of ultimatums, Europe's total loss is bounded by the bite of the credible threats, the unconditional parts plus the smaller of the conditional parts and the slack. Corollary 2 turns the bound into a rationing problem whose capacity shrinks as the premium rises toward zero. By Proposition 3, China executes a counter-threat only to crowd out another US demand, and when Washington has a divisible demand China cannot threaten, the counter-threat taxes the patron and leaves Europe's loss unchanged. Proposition 4 makes conditionality a choice with an assurance cost (Pauly, 2024). A patron makes its guarantee more conditional than deterrence requires only to create slack it can spend, so conditionality is the price of concessions, and a stronger Chinese bid can be met with a more conditional guarantee instead of a payment.

Section 6 shows that a wasting slack gives Washington a preventive motive to demand control of the chokepoint while its threats still bite. Proposition 5 weighs the rents a transfer removes and Washington's intrinsic value of control against the issuing cost, Europe's sovereignty cost and China's punishment of the transfer. By Corollary 3, an autonomy investment that Europe cannot commit to forgo strengthens the motive, and a credible Chinese threat against a transfer can protect European ownership without being carried out. Corollary 4, remark 1, and Proposition 6 cover the timing of demands, self-binding and refusal. Section 7 drops the unitary holder (remarks 2 to 4) and gives Washington's dependence its most direct role: interrupting servicing during a dispute helps Europe only if the US buffer of tools and know-how is short and only for the servicing Washington cannot compel (Proposition 7 and Corollary 5).

Section 8 is a plausibility probe. It derives predictions that separate the slack logic from the rival accounts (table 3), reads the Dutch record and the Nexperia dispute of 2025 through the model (section 8.3), and compares holders in Japan, South Korea, Germany and Taiwan (section 8.4). The record is consistent with the jurisdictional channel: Washington has coerced mainly by granting and withdrawing statuses it can change by rule. It leaves the security channel largely untested. We found no public record that links European chip controls to US force posture, so the evidence cannot show how conditional the guarantee is on export controls, and section 8.5 describes the designs that could.

We contribute to four literatures.

1. *Geoeconomics and coercion.* In Clayton et al. (2026a), pooled threats are enforcement threats that raise the target's inside option and are never spent, and threats to cut a target off appear only in their Supplement B.3.1. The bidding models cited above have carrots only, and Lacy and Niou (2004) model linkage through preferences and information. We put a rival's bid inside the participation constraint, split each instrument into conditional and unconditional parts, and let executed threats be spent. The result is a theory of how a patron rations coercion

across demands as an ally's outside option improves (Propositions 2 and 3 and Corollary 2).

2. *Alliance politics.* Alliance theory already separates an ally's dependence from the patron's commitment (Snyder, 1984, 1997), and abandonment threats are known to work when they are conditional and paired with assurance (Gerzhoy, 2015; Blankenship, 2023). We make conditionality a costly choice and show that its value to the patron is the slack it creates against the rival's credible offer (Proposition 4). Base politics has the same structure with territory as the asset (Cooley, 2008); the ASML case adds nested jurisdiction, hold-up through servicing and preventive control.

3. *Shifting power and control rights.* Preventive war answers a rising state's inability to commit (Fearon, 1995; Powell, 2006; Krainin, 2017), states try to stop investments in future bargaining power (Debs and Monteiro, 2014; Bas and Coe, 2016; Benson and Smith, 2023), and control rights are the remedy for hold-up (Grossman and Hart, 1986; Lake, 1996; Cooley and Spruyt, 2009). We show that a patron's power to demand control is capped by the holder's current slack, so the demand comes in windows while that slack lasts (Proposition 5 and Corollary 4). When Washington places little intrinsic value on control, an accepted transfer is inefficient in the way preventive war is: Europe bears its sovereignty cost so that Washington avoids rents that are only transfers.

4. *Weaponized interdependence.* Farrell and Newman (2019) make jurisdiction over a hub and the domestic institutions to use it the conditions for weaponizing it. When the holder of a hub bargains with a state that depends on it, we add a third: the holder's exit premium must be positive.

Section 2 describes ASML's position and the control sequence, and section 3 the work we build on. Sections 4 to 7 present the model and results. Section 8 reads the record and the comparison cases through the model, and section 9 draws implications and states the scope conditions (section 9.1). Proofs are in appendix A and numerical checks in appendix B.

2 The Chokepoint and Its Owners

2.1 A chokepoint

ASML is the only firm that makes extreme ultraviolet (EUV) lithography machines, on which the leading-edge logic processes used for the most advanced US AI accelerators depend (Miller, 2022; ASML Holding N.V., 2026a). In October 2026 its chief executive said that planned EUV supply for 2027 and 2028 would not fully meet demand (ASML Holding N.V., 2026b).

China cannot buy EUV machines. It buys immersion deep-ultraviolet (DUV) tools instead, and Fedasiuk and Torres (2026) estimate that Chinese entities took 70 percent of ASML's immersion DUV sales in 2024. Exposing the same layer several times (multiple patterning) lets these tools print features that would otherwise need EUV, and Chinese fabs have used that route toward the advanced nodes on which domestic AI accelerators are made, "printing its way" toward self-sufficiency in the

AEI authors' phrase (Fedasiuk and Torres, 2026). Customers in China accounted for 26.3 percent of ASML's total net sales in 2023, 36.1 percent in 2024 and 29.1 percent in 2025 (ASML Holding N.V., 2026a). Each sale also starts a servicing relationship. Installed-base management (service, upgrades and field options) made up 25.1 percent of ASML's 2025 sales (ASML Holding N.V., 2026a), and Fedasiuk and Torres (2026) estimate that several hundred ASML immersion tools run in China, each needing specialized maintenance about every six months, often from ASML staff on site. Section 7.4 takes up what interrupting such service would cost.

2.2 A nested chokepoint

ASML is a Dutch company, but its EUV machine depends on suppliers and staff in Germany and the United States. Carl Zeiss SMT is its sole supplier of critical optics, without which ASML says it "would effectively cease to be able to conduct our business" (ASML Holding N.V., 2026a). At the end of 2025 ASML had 8,242 employees in the United States, 18.7 percent of its workforce (ASML Holding N.V., 2026a), and US-incorporated subsidiaries that include Cymer, LLC, the San Diego maker of its EUV light sources (ASML Holding N.V., 2026c). Because the Export Administration Regulations count any juridical person organized under US law and any person in the United States as a "U.S. person" (Bureau of Industry and Security, 2026), these entities and their staff fall under US rules that restrict support for advanced-node fabs in China (ASML Holding N.V., 2026a).

Washington also reaches the machines. An October 2023 rule set the de minimis threshold of US-origin content at zero for certain immersion tools bound for China (Office of the Federal Register, 2026; U.S. Department of Commerce, Bureau of Industry and Security, 2023). A rule effective on 2 December 2024 added a foreign direct product (FDP) rule and a de minimis rule specific to semiconductor manufacturing equipment (SME). Its new Supplement No. 4 to Part 742 lists 33 countries: 25 EU members (all but Cyprus and Malta), Australia, Canada, Iceland, Japan, New Zealand, Norway, Switzerland and the United Kingdom, but not South Korea or Taiwan. The rule's new country-wide licence requirements do not apply to items caught only by the SME FDP or SME de minimis rule when an entity located in a listed country exports them from abroad, unless that entity is headquartered in Country Group D:5 or Macau (Bureau of Industry and Security, 2024). BIS gave its reason: the governments listed "are members of multilateral export control regimes ... and have the authority to control key SME items of concern" (Bureau of Industry and Security, 2024). BIS can amend the list by rule. Since the exclusion is conditioned on the listed government's own capacity to control the items, we code it as a threat left unexercised, equivalently a reward for self-binding, and not as a transfer (section 8.2).

2.3 Europe's dependencies

Security. Europe relies on the United States for nuclear deterrence, and Bruegel and the Kiel Institute estimate that defending Europe without US support would take about EUR 250 billion a year in additional conventional spending (Burilkov and Wolff, 2025). The guarantee is graduated. Its core is protected by statute: Section 1250A of the FY2024 National Defense Authorization Act

bars withdrawal from the North Atlantic Treaty without the consent of two-thirds of the Senate or an Act of Congress (22 U.S.C. § 1928f). Posture, by contrast, is cheap to move at the margin. On 1 May 2026, after Chancellor Merz said the "Americans clearly have no strategy" on Iran, Defense Secretary Hegseth ordered about 5,000 troops out of Germany, and a Pentagon official cited "recent German rhetoric" (Martinez, 2026). Later that month the US brigade combat teams based in Europe were cut from four to three, and in September the Pentagon was weighing a drawdown of 25,000 or more troops (De Luce et al., 2026). Dependence also runs the other way. Spain was already refusing US use of jointly operated bases for the Iran war when, on 30 March 2026, it closed its airspace to the US aircraft involved, and Secretary of State Rubio said "all of it's going to have to be reexamined" (Roque, 2026). Washington needs European access, and a European government can deny it. Section 5.3 interprets these episodes.

Frontier AI. Washington has treated access to advanced AI chips as an instrument of policy. The AI Diffusion Rule of January 2025 put ten EU members in its favoured group and left the other seventeen to country allocations (Bureau of Industry and Security, 2025). On 13 May 2025, before its compliance date, BIS announced its rescission, saying it would have "undermined U.S. diplomatic relations with dozens of countries by downgrading them to second-tier status." In the August 2025 joint statement on trade the EU stated its intention to buy at least $40 billion of US AI chips and to adopt technology-security requirements aligned with those of the United States, and Washington said it would facilitate those exports once the requirements were in place (The White House, 2025d). The National Security Strategy released in December 2025 says Washington stands ready to help, "potentially through more favorable treatment on commercial matters, technology sharing, and defense procurement", countries that take more responsibility for their regional security and "align their export controls with ours" (The White House, 2025e).

China. By customer location, China was ASML's largest market in 2025 (ASML Holding N.V., 2026a). Beijing has also built instruments aimed at those who restrict it. Its licensing regime of 4 April 2025 for seven medium and heavy rare earths and magnets containing them remains in force (Ministry of Commerce of the People's Republic of China and General Administration of Customs of the People's Republic of China, 2025a; Benchmark Mineral Intelligence, 2026). The broader, extraterritorial controls of October 2025 (Mayer Brown, 2025) were suspended after the Busan summit of 30 October 2025 until 10 November 2026 (The White House, 2025b; Ministry of Commerce of the People's Republic of China and General Administration of Customs of the People's Republic of China, 2025b). In September 2026 the two governments agreed to extend the arrangement that paused them to 10 January 2027, although no legal act extending the suspension itself had appeared by late October (Ministry of Commerce of the People's Republic of China, Department of American and Oceanian Affairs, 2026). State Council Decree 834 of April 2026 authorizes countermeasures against foreign firms that cut off normal trade with Chinese entities (State Council of the People's Republic of China, 2026), and Decree 835 of 13 April 2026 creates a Malicious Entity List, prohibition orders, private damages suits and fines for those who implement

foreign extraterritorial measures (Li et al., 2026). Its first application was aimed at Brussels: on 15 May 2026 China's Ministry of Justice determined that the Commission's investigative measures in its Foreign Subsidies Regulation probe of Nuctech "constitute improper extraterritorial jurisdiction measures" and barred anyone from implementing them (Cleary Gottlieb Steen & Hamilton LLP, 2026; Ministry of Commerce of the People's Republic of China, 2026). Beijing has also executed a targeted counter-threat against a Dutch step taken under US pressure, in the Nexperia case (section 8.3).

2.4 Who decides in Europe

Under Article 12(2) of the EU dual-use regulation, export licences are granted by the competent authority of the member state where the exporter is established (European Parliament and Council of the European Union, 2021). Other member states have only non-binding consultation rights, so licences for ASML's shipments to China are a Dutch decision. The US-facing direction is collective. Union general export authorisation EU001 covers exports of listed dual-use items to the United States and eight other destinations, and a member state may bar an exporter from it only on "reasonable suspicion about the exporter's ability to comply" (Article 12(7)) (European Parliament and Council of the European Union, 2021). Tools left on the Dutch national list still need a Dutch licence to leave the EU (ASML Holding N.V., 2026a), but EUV and the most advanced immersion tools ship to US fabs under EU001. Turning them against Washington would take an EU decision, under the Anti-Coercion Instrument (European Parliament and Council of the European Union, 2023) or by amending EU001 (section 7.3). Ownership is guarded at two levels. ASML's independent preference-share foundation can acquire preference shares that would halve the voting power of ordinary shares when, in its board's view, the interests of ASML or its stakeholders are at stake (ASML Holding N.V., 2026a). The Dutch Vifo Act, in force since 1 June 2023, requires notification of acquisitions that reach 10 percent of the votes in firms holding "very sensitive" technology, a category whose definition covers machines for chip manufacturing (Staten-Generaal, 2022; Kingdom of the Netherlands, 2023).

The controls themselves arrived in steps, and from 2023 each step followed a US move. In 2019 the Dutch government let a licence for an EUV shipment to China lapse, reportedly after US lobbying (NL Times, 2020). After the US rules of October 2022, it published national licensing for the most advanced immersion tools (NXT:2000i and later) on 30 June 2023, in force from 1 September 2023 (Government of the Netherlands, 2023), and at the end of 2023 it partly revoked a licence covering a small number of Chinese customers (DutchNews.nl, 2024). After the US rule of October 2023, it extended licensing from 7 September 2024 to the older NXT:1970i and NXT:1980i, with their specially designed parts, software and technology, so that ASML now applied to The Hague rather than Washington for these shipments (ASML Holding N.V., 2024). After the December 2024 rule, it published controls on metrology and inspection systems on 15 January 2025, in force from 1 April 2025 (ASML Holding N.V., 2026a). On 15 November 2025 Commission Delegated Regulation (EU) 2025/2003 put the most advanced immersion tools on the EU list. ASML reported

that this "did not mean a change" because the tools already needed a Dutch licence, and the licence decision stayed in The Hague (ASML Holding N.V., 2026a; European Parliament and Council of the European Union, 2021). Servicing of the installed base in China remained permitted at least into mid-2026 (Zakaria, 2026). The MATCH Act, introduced in Congress in April 2026 and not enacted by late October, would ban immersion-tool sales anywhere in China and servicing at listed Chinese fabs, giving allies 150 days to align before Washington acted alone (Baumgartner, 2026a,b). The Dutch government told parliament that it opposed the bill's extraterritorial reach (Sjoerdsma, 2026), and the trade minister said it "could potentially put our national security in American hands" (NL Times, 2026b). Section 8.3 reads this sequence through the model.

3 Related Literature

3.1 Interdependence and chokepoints

Hirschman (1945) located influence in the losses a partner would suffer if trade stopped, and Keohane and Nye (1977) separated sensitivity, the cost of a cutoff before anyone adjusts, from vulnerability, the cost that remains afterward. Farrell and Newman (2019) moved the argument to networks and attached two conditions. A state must have "physical or legal jurisdiction over hub nodes" (p. 56), and it must have the "legal and regulatory institutions" to exploit them (p. 57); where hubs are spread across jurisdictions, the states involved must coordinate (pp. 56, 58). Farrell and Newman (2023) trace how Washington built that capacity, and Kalyanpur and Newman (2019) show that jurisdiction is itself a policy choice: harmonized EU rules let European firms escape the reach of US securities law. Semiconductors offer the clearest case. The foreign direct product rule let Washington force foreign chipmakers to choose between US tools and Huawei as a customer (Bown, 2020), and US firms routed sales through third countries until that rule closed the gap (Chen and Evers, 2023, 194–195). Treating a toolmaker as a hub is our extension, since Farrell and Newman's evidence comes from finance and the internet. On that extension ASML meets the network condition, but jurisdiction over it is shared with Washington (section 2), and the European institutions that could turn it against the United States are collective (section 7.3).

These works disagree about whether control of a hub or a supply yields lasting power. For Farrell and Newman the power comes from network position and the institutions behind it, and targets' responses, such as isolating themselves or reshaping the network, come later (Farrell and Newman, 2019, 76). Gholz and Hughes (2021) find that Japan absorbed China's 2010 rare-earth embargo through market adjustment, and argue that cost measures that ignore market structure overstate a sender's power. Work on chokepoints as depleting assets sides with Gholz and Hughes: activation induces indigenization and rerouting (Tan et al., 2025; Rajan, 2026). Our model lets the exit premium change over time (section 6) and asks what follows. Following Wagner (1988), we treat asymmetric dependence as a source of political influence only if it changes a party's credible alternative, and ask when Europe's alternative is worth anything. Drezner (1999) adds the relevant asymmetry among senders: allies rarely coerce each other, but when they do they win significant

concessions, because they expect little future conflict.

3.2 Alliance coercion and dependence

Alliance theory already separates an ally's dependence from the credibility of its patron's threats. Snyder (1984, 1997) distinguishes the fear of abandonment from the fear of entrapment, and a protégé's need for its patron from the patron's own strategic interest in defending it. Morrow (1991) treats asymmetric alliances as trades of autonomy for security, and Lake (2009) treats the patron's authority as the price of protection already received, sustained partly by legitimacy. On those accounts compliance needs no fresh threat, a rival reading we take up in section 8.1. Leeds (2003) finds that commitments invoked in war between 1816 and 1944 were honoured about three-quarters of the time, with major powers likelier to renege, and Fang et al. (2014) show that a target that values its alliance can be led to concede by its ally's advice.

The coercion literature disagrees about which instrument worked in the canonical case. Gerzhoy (2015) credits conditional threats of military abandonment, paired with assurance, for stopping West Germany's nuclear ambitions. Lanoszka (2018) finds alliance coercion hard and argues that an ally's economic and technological reliance on Washington is the stronger instrument. Our model contains both kinds. Pauly (2024) shows why assurance is hard: what makes a threat credible undermines the promise not to punish compliance. Blankenship (2021) finds that allies in locations Washington values contribute less to the common defence, consistent with less credible abandonment threats, and Blankenship (2023, 111–119) shows Iceland extracting compensation and fishing concessions for Keflavik. Carson et al. (2025) argue that access to allied territory is a benefit of alliance that a less committed patron may have to buy with cash, and Wolford and Kim (2017) show that a patron's threats against a politically valuable ally can lack credibility, which is our self-harming-sanctions channel.

Base politics studies a host that holds an asset its patron needs. Political change in hosts reopens basing deals (Cooley, 2008), compensation shapes base stability (Calder, 2007), and sovereignty transfers are incomplete contracts that stay open to renegotiation (Cooley and Spruyt, 2009). In 2009 Kyrgyzstan announced the closure of the US base at Manas alongside a Russian aid package, collected the first Russian payment, and then signed a new deal with Washington at a higher rent (Cooley, 2012, ch. 7). The nonproliferation literature studies the reverse problem, a patron restraining an ally's sensitive exports. Kroenig (2009) finds that sensitive nuclear transfers follow suppliers' strategic position and finds no support for commercial motives, while Gray (2012) shows Bonn putting commercial rivalry first in its 1975 sale to Brazil, keeping the deal and promising only future caution. Miller (2014) shows that credible US sanctions deterred nuclear programmes among states dependent on Washington, so that successes stay off the record, and Coe and Vaynman (2015) model rival superpowers colluding to coerce. We add the interaction between the patron's conditionality and a rival's bid, the requirement that conditionality attach to the specific issue, and conditionality as a choice with an assurance cost (section 5.3).

3.3 Rival bidders and geoeconomics

That a contested client earns what the losing bidder would pay is old. It is the one-voter case of sequential vote buying (Groseclose and Snyder, 1996), extended to finite legislatures by Banks (2000); Dekel et al. (2008, Prop. 2) show that the winner pays the loser's value net of the client's preference, a price pinned down only in undominated strategies. Menu auctions (Bernheim and Whinston, 1986) and their application to lobbying (Grossman and Helpman, 1994) give the simultaneous-offer analogue, and Bueno de Mesquita and Smith (2016) show that a dominant aid donor captures the surplus of an aid-for-policy deal while a rival donor's entry raises what it must pay. The outside-option principle supplies the threshold: a threat to walk away matters only when the alternative beats continued bargaining (Shaked and Sutton, 1984; Sutton, 1986; Binmore et al., 1989). Lacy and Niou (2004) model sanctions as coercive linkage under two-sided incomplete information and show that threats likely to succeed work before sanctions are imposed. Jehiel et al. (1996) show that with identity-dependent externalities a seller extracts surplus even from buyers who do not acquire the good.

Recent geoeconomics locates power in threats. Clayton et al. (2026a) show that a hegemon gains by pooling threats across relationships, bounded by the target's participation constraint, and Clayton et al. (2026b) state the principle we build on: "Threats generate power to the extent that they generate slack in the participation constraint." The term slack is theirs. Their pooled threats are enforcement threats that raise the target's inside option; threats to cut off a target that refuses appear only in a supplement (B.3.1), where they are partly self-defeating, and a two-hegemon extension (B.2) drives transfers to zero when the hegemons' threats overlap. In Clayton et al. (2026c) a hegemon gains by committing to limit its own coercion. Broner et al. (2025) and Becko et al. (2025) model alignment among many small countries facing committed instruments. The rival's effect is contested. Rival offers come when a protégé's relations with its patron sour (Yang, 2026), and a rival's entry raised the US price in the aid market (Bueno de Mesquita and Smith, 2016), yet Camboni and Porcellacchia (2025) find that China's growth can lead countries in the US sphere to offer Washington more.

Each of these works lacks a result we derive. In Groseclose and Snyder (1996) and Dekel et al. (2008), whose bidding outcome is our benchmark (Lemma 1), the client is passive; once the holder can propose and talks can rupture, the patron's dependence on the chokepoint returns through the holder's agenda power (Proposition 1). No principal in a menu auction (Bernheim and Whinston, 1986) can move the agent's alternative. Ours can, and every conditional threat it holds then draws on one budget equal to the holder's slack (Proposition 2). Bueno de Mesquita and Smith (2016) give rival donors carrots only. Once the patron has sticks and the rival has counter-threats, China's threats lower the patron's payoff and can leave the holder's loss unchanged, and China executes one on the equilibrium path only to crowd out other demands (Proposition 3). Linkage in Lacy and Niou (2004) works through preferences and information; in our complete-information game it works through the conditional part of each instrument, and a cut once executed is spent (Lemma 2). The threats in Clayton et al. (2026a) are pooled enforcement threats, never spent, and their contracts

are non-exclusive. Ours are cut-off threats that must each be sequentially rational, the rival's bid sits inside the participation constraint, and the holder chooses one alignment. Where the hegemon of Clayton et al. (2026c) gains by binding itself, our patron chooses how conditional to make its guarantee and pushes conditionality past what holds the premium at zero only to buy concessions (Proposition 4), and it is the holder that may bind itself (remark 1).

3.4 Commitment, control and prevention

Anticipated shifts in power create commitment problems that can make bargaining fail under complete information (Fearon, 1995; Powell, 2006), and Krainin (2017) shows that slow but persistent shifts suffice and that the preventive act may be delayed. Fearon (1996), an unpublished manuscript, studies bargaining over objects that shape future bargaining power and finds efficient, gradual transfers when power varies continuously with holdings, and breakdown when it jumps. Control of ASML is such an object. Because Fearon's logic carries weight in our analysis of control, we note that the manuscript has not been peer reviewed. Uncertainty about a rising state's investment can itself produce preventive war (Debs and Monteiro, 2014), an observer that monitors imperfectly waits until the shift looks imminent (Bas and Coe, 2016), and potential power can instead be bought off with concessions (Spaniel, 2019). Benson and Smith (2023) show that allies can offset an alliance-induced shift with concessions, and that the conditions that make prevention attractive also make it unlikely to be carried out.

Control rights come from the theory of the firm: ownership allocates residual rights over an asset when contracts are incomplete (Grossman and Hart, 1986; Hart and Moore, 1990). Lake (1996) orders relations from alliance to empire by these rights, Frieden (1994) ties colonial control to investments whose rents are easy to seize, and in Harstad (2016) conservation abroad is leased while conservation at home is bought, with leasing dominant when the seller's consumption value and the buyer's cost of protecting what it owns are large. Sold tools are a wasting lever, since a durable-goods seller cannot commit (Coase, 1972; Bulow, 1982), while control of maintenance works like leasing and restores commitment (Morita and Waldman, 2004); servicing is therefore a renewable lever (section 7.4). Leaders can tie their hands with audience costs and other public stakes (Schelling, 1960; Fearon, 1994, 1997); in our model every threat and promise is staked in this way, under one commitment technology (section 4.2). The conditional part of the threat behind the patron's preventive demand for control is capped by the holder's current slack, so it comes while the holder is weak but expected to strengthen (Proposition 5 and Corollary 4). The transfer destroys sovereignty value and invites China's punishment in order to avoid paying rents that are pure transfers, which makes it the analogue of preventive war, and it binds only as far as the control it confers can be enforced (Corollary 3).

4 A Model of Chokepoint Bargaining

The model has three players: Europe (E), the United States (U) and China (C). Europe decides whether ASML's most advanced tools, and the engineers who keep them running, reach Chinese fabs. In practice that decision is split among the Dutch licensing authority, the firm and the European Union. We treat Europe as one actor here and separate the three in section 7. The model prices Europe's chokepoint and, when the price is zero, measures the stock of slack that Washington can spend on other demands (section 5).

4.1 Setup

Alignment. Europe chooses an alignment $a \in \{A, X\}$. Under Atlantic alignment (A) ASML supplies only fabs the United States accepts and complies with US denial requirements. Exit (X) is a broad realignment of the chokepoint: Europe lets ASML sell and service EUV and advanced immersion DUV tools for Chinese fabs against US objection. Exit is distinct from hedging, which we define as continued sales and servicing of legacy immersion tools in China. Hedging is the margin of the real disputes since 2023 (section 2), and we treat it as a secondary choice in Proposition 3(b). The security results below concern exit in the broad sense. We do not assume that Washington would make its defence of Europe depend on a dispute over legacy servicing, and the quantities we discuss in section 8 refer to the same broad meaning.

Stakes. The United States loses $V > 0$ if Europe exits. V is the patron's dependence on the chokepoint: what it gives up in the AI race if Chinese fabs gain access to ASML's frontier tools. China gains $W > 0$ from Europe's exit, of which a share $\theta \in [0, 1]$ can be blocked by US jurisdiction over China-bound flows (the foreign direct product rule, de minimis rules and controls on US-made subsystems). Because that jurisdiction targets shipments to China and leaves ASML's supply to US-aligned fabs untouched, we treat it as credible for every V and let θ capture its reach.

Instruments and the baseline. Washington controls instruments j: the core defence guarantee, force posture and enablers, access to frontier AI, and ASML's access to US inputs, persons and customers. After Europe chooses a, the United States sets each instrument at a level $x_j(a)$ that is sequentially rational given any commitments it has made, and Europe values instrument j at $v_j(x_j)$. The conditional part of instrument j is $d_j = v_j(x_j(A)) - v_j(x_j(X)) \geq 0$. The baseline is a loyal Europe receiving $x_j(A)$, Washington's ex post optimal provision to a loyal ally, with no transfers. A transfer is a promised increment above this baseline. A threat is a withdrawal below it. Each instrument is therefore counted once: access to frontier AI enters as the AI instrument, with conditional part d_{ai}, and only access beyond $x_{ai}(A)$ would count as a transfer.

Section 9.1 states the conditions under which we expect the model to apply.

4.2 Commitment

One commitment technology governs every promise and every threat. Before Europe moves, each great power i may attach a stake $m \leq \bar{m}_i$ to any public conditional commitment. It forfeits the stake if the condition is met and it does not act. The stake is a reduced form for the audience costs of backing down (Fearon, 1994); in Fearon's terms, it ties hands rather than sinking costs (Fearon, 1997). Each commitment carries its own stake, capped at $\bar{m}_i$. The cap is a limit per commitment, not a budget shared across commitments.

A promise to pay τ after Europe has moved has no ex post value to the promiser, so it is honoured if and only if $\tau \leq \bar{m}_i$. A threat whose execution costs the threatener e ex post is credible if and only if $e \leq \bar{m}_i$. Creating a stake costs nothing; only forfeiting one does. A threat that deters is never carried out, so it costs its author neither the stake nor e. A threat that is executed costs e, and because the baseline is Washington's ex post optimum, $e \geq 0$ for every withdrawal from a loyal Europe. That asymmetry matters in section 5, where threats are executed when Europe refuses a demand and an executed threat is spent.

Security uses the same technology. Washington can stake $m_S \leq \bar{m}_U$ on a public threat to abandon a realigned Europe. The difference is that such a threat has a price before any crisis arrives: a broad withdrawal rule makes the guarantee less valuable to allies who stay loyal. We introduce that assurance cost in section 5.3. It is why the conditionality of the guarantee is a choice for Washington and why the choice is interior.

4.3 Security, Other Instruments and the Exit Premium

The core guarantee. In a crisis, whose probability we fold into S_k, Washington defends Europe if its net stake in doing so exceeds a crisis-specific cost ξ, drawn from a continuous distribution G with density g. Its stake in defending a realigned Europe is $s_X = s_0 + s_B$, where s_0 is its own interest in European security and s_B is the value to it of European territory, airspace and bases. Its stake in defending a loyal Europe is $s_A = s_X + s_L$, where $s_L \geq 0$ is the added value of a loyal ally, for example Washington's reputation with other chokepoint holders whose own alignment it needs. After exit, defending Europe also forfeits the stake m_S. The conditionality of the core guarantee is

$$\kappa = G(s_A) - G(s_X - m_S), \qquad s_A = s_X + s_L, \quad s_X = s_0 + s_B, \tag{1}$$

and its largest value, at $m_S = \bar{m}_U$, is $\bar{\kappa} = G(s_X + s_L) - G(s_X - \bar{m}_U)$. With S_k denoting Europe's crisis-weighted stake in core defence, the conditional part of the guarantee is κS_k. Conditionality now has two sources that the model makes explicit: the value of loyalty to Washington (s_L) and its public commitment to abandon an ally that realigns (m_S). Dependence enters only through S_k, which is multiplied by κ.

Access and the silicon shield. The term s_B carries the access channel. Allies give Washington permission to operate from their territory, airspace and waters, and Carson et al. (2025) argue that

a less committed United States "may find itself relying on hefty cash payments to try to buy access". A Washington that needs European access has a reason to defend Europe whatever ASML does, which raises s_X. The same may be true of the chokepoint itself. If Washington would defend the territory on which ASML sits because it needs the firm, s_B rises with V, a silicon shield. Neither channel has a fixed sign. By Proposition 4(d), $d\bar{\kappa}/ds_X = g(s_X + s_L) - g(s_X - \bar{m}_U)$, so $\bar{\kappa}$ falls with s_X when the density g of crisis costs is nonincreasing on $[s_X - \bar{m}_U,\ s_X + s_L]$ (for example when Washington would almost surely defend a loyal Europe), and it can rise otherwise. Europe's own power to deny access is a hold-up instrument, and we take it up in section 7.4.

Posture and enablers. Beyond the core guarantee, Washington provides a continuous flow of posture and enablers: forces stationed in Europe, rotations, intelligence sharing, and munitions and spare parts. These are cheap to adjust at the margin and easy to withdraw from one ally at a time, so they can be conditional without any change in the treaty commitment. Their conditional part is d_p. The posture moves of 2026 (section 2) are changes in this flow, and we read them in section 5.3.

The AI instrument. Washington allocates frontier models, inference and advanced chips continuously. The conditional part of the AI instrument is $d_{ai} = v_{ai}(x_{ai}(A)) - v_{ai}(x_{ai}(X))$. Scarce compute makes it cheap for Washington to cut an exiter's allocation, which raises d_{ai}. Extreme scarcity may also lead Washington to ration loyal allies, which lowers $x_{ai}(A)$. With all-or-nothing denial, rationing would remove the conditional part entirely. With continuous allocations it does so only if loyal allies are cut to the exiter's level. Rationing loyal allies is itself costly for Washington. The January 2025 AI Diffusion Rule placed ten EU member states among its favoured destinations and held the other seventeen to per-country allocations (Bureau of Industry and Security, 2025). In May 2025, before compliance was due, the Bureau of Industry and Security announced that it would rescind the rule, citing among other costs the damage to US relations with dozens of countries that the rule downgraded to second-tier status. We read the rescission as evidence that tiering loyal allies is costly, which keeps d_{ai} positive even when compute is scarce.

Firm-level penalties. Washington can also penalize ASML itself, by entity listing or by denying US-origin inputs and US-person support for all of the firm's production, including tools bound for US-aligned fabs. This costs Europe $f \geq 0$. Because it disrupts supply to Washington's own fabs and those of its allies, the penalty harms the United States. Its ex post gain from penalizing an exiting ASML is $\gamma - \varphi V$, where $\gamma \geq 0$ is the added denial benefit and φV, with $\varphi > 0$, the self-inflicted cost. The threat is credible if and only if its execution cost is within the cap, so

$$d_f = \chi_f f, \qquad \chi_f = \mathbf{1}\{\varphi V \leq \gamma + \bar{m}_U\}.$$

This is the mechanism of Wolford and Kim (2017), in which allies that are politically valuable to their patron face retaliation threats that are not credible. Here it applies only to penalties on the firm. Jurisdiction over China-bound flows remains credible for every V.

China's offer. China can promise Europe a transfer if Europe exits: market access, credit, co-production or relief from its own export controls. A promise of τ_C costs China τ_C, and Europe values it at $\lambda_C \tau_C$ with $\lambda_C \in (0, 1]$, which captures the lower value to Europe of what China can offer in kind. China's promises are subject to the same technology, with cap $\bar{m}_C$. China may also value Washington's loss: with a relative-gains weight $\rho \geq 0$, the most it would pay for exit is $W_C = (1 - \theta)W + \rho V$. Its credible offer, in units of European utility, is

$$\omega = \lambda_C \min\{W_C, \bar{m}_C\}. \tag{2}$$

If the race is close to zero-sum, so that ρV is large, the cap binds. Then λ_C and $\bar{m}_C$, how far Europe trusts and values Chinese promises, set ω, and China's interest in the outcome does not.

Chinese punishment of alignment. We do not let China punish Atlantic alignment as such. A threat to punish an aligned Europe is carried out on the equilibrium path whenever Europe aligns despite it, so China makes such a threat only if carrying it out pays for itself ex post or if the threat changes Europe's choice. If it were made and self-rewarding, it would add its cost to Europe to Q. China's observed coercion looks different. It targets firms and sectors in response to specific acts, as in its informal sanctions after South Korea's THAAD deployment (Lim and Ferguson, 2022), and the 2025 Nexperia export ban followed a specific Dutch order (section 8.3). We model Chinese threats of that kind as counter-threats attached to particular concessions (Proposition 3) and to a transfer of control (section 6), and we set untargeted punishment of alignment to zero.

The exit premium. Exit costs Europe the conditional parts of Washington's instruments plus an intrinsic cost $\eta \geq 0$, the value Europe places on keeping frontier capabilities from a rival that supports Russia's war:

$$c = \kappa S_k + d_p + d_{ai} + d_f + \eta.$$

Definition 1 (Exit premium)**.** Europe's exit premium is China's credible offer net of the credible cost of exit,

$$Q = \omega - c = \lambda_C \min\{(1 - \theta)W + \rho V,\ \bar{m}_C\} - \kappa S_k - d_p - d_{ai} - \chi_f f - \eta. \tag{3}$$

Europe's slack is $\Lambda = [-Q]^+$.

The premium prices Europe's alternative to alignment. It follows Hirschman (1945), for whom a state's influence over a partner depends on what the partner would lose by leaving. Slack is the amount by which that alternative falls short. We take the term from Clayton et al. (2026a,b), for whom "threats generate power to the extent that they generate slack in the participation constraint" (Clayton et al., 2026b). Table 1 collects the notation.

Timing. Figure 1 shows one period. The great powers first announce their conditional promises and threats and attach stakes. Washington and Beijing then bid for Europe's alignment, and

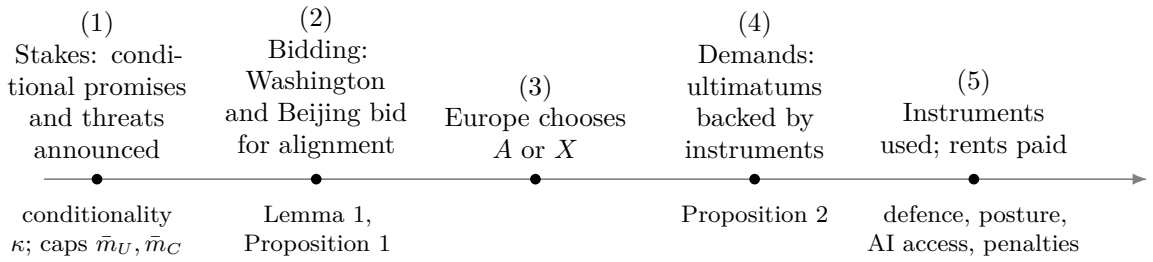


Figure 1: Timing of one period. Each threat in stage (5) is carried out only if doing so is sequentially rational given the stakes announced in stage (1). Section 6 links periods through control rights.

Europe chooses. Washington may next issue demands backed by its instruments (section 5). Finally, instruments are used and rents paid; each threat is carried out only if that is sequentially rational given the stakes. Section 6 links periods through control rights.

4.4 The Bidding Benchmark

We begin with an open contest. Washington and Beijing bid for Europe's alignment in an ascending auction, so either can raise its bid until neither wants to. Write $\bar{z}_U = \min\{V, \bar{m}_U\}$ for Washington's largest credible bid.

Lemma 1 (Bidding benchmark)**.** *Suppose the United States and China bid for Europe's alignment in an ascending (clock) auction in which bids are staked promises, Europe values China's promise τ_C at $\lambda_C \tau_C$, Europe chooses its alignment after the auction, and instruments are then used. Let $\bar{z}_U = \min\{V, \bar{m}_U\}$ and $Q = \lambda_C \min\{W_C, \bar{m}_C\} - c$.*

(i) After alignment a, the United States provides $x_j(a)$ for every instrument j, so exit lowers Europe's payoff, before transfers, by $c = \sum_j d_j + \eta$.

(ii) A promise of τ by power i is honoured if and only if $\tau \leq \bar{m}_i$. Bidding up to one's maximal credible bid (in units of Europe's payoff, $\bar{z}_U$ for the United States and Q for China) is weakly dominant. In weakly undominated strategies, with an indifferent Europe aligning:

(a) if $Q < V$ and $Q \leq \bar{m}_U$, Europe aligns and the United States pays $[Q]^+$;

(b) if $Q > \min\{V, \bar{m}_U\}$, Europe exits, and China's promise gives Europe $\min\{V, \bar{m}_U\}$ net of c.

In both cases Europe's payoff is $[\min\{Q, V, \bar{m}_U\}]^+$.

(iii) If $\bar{m}_U < Q < V$, every binding transfer $\tau \in [Q, V)$ from the United States to Europe would make both Europe and the United States strictly better off than the equilibrium (inefficient exit from limited US commitment).

The losing bidder sets the price, so Washington pays exactly enough to make China's best credible offer unattractive. None of this is new. It is the one-voter case of sequential vote buying

Table 1: Notation for the baseline model

Symbol	Meaning	Symbol	Meaning
E, U, C	Europe, United States, China	$a \in \{A, X\}$	Atlantic alignment or exit
V	US loss from exit (dependence)	W, θ	China's gain from exit; share blocked by US flow jurisdiction
ρ	China's relative-gains weight	W_C	most China would pay, $(1-\theta)W + \rho V$
$\bar{m}_U, \bar{m}_C$	largest stake on one commitment	λ_C	Europe's value of a unit of China's promise
ω	China's credible offer, eq. (2)	$x_j(a)$, d_j	provision of instrument j; its conditional part
s_0, s_B	US own interest in European security; access value	s_L	added US value of a loyal ally
s_A, s_X	US stake in defending a loyal, realigned Europe	m_S	US stake on abandoning a realigned Europe
G, g	distribution and density of crisis costs ξ	κ, $\bar{\kappa}$	conditionality of the core guarantee, eq. (1); its maximum
S_k	Europe's crisis-weighted stake in core defence	d_p, d_{ai}	conditional parts of posture and of the AI instrument
f, γ, φ	firm penalty; US denial benefit; self-harm per unit of V	χ_f	credibility of firm penalties
η	Europe's intrinsic cost of exit	c	credible cost of exit
Q	exit premium, eq. (3)	Λ	slack, $[-Q]^+$
β	Europe's probability of proposing	λ, r	rupture hazard; discount rate
ψ	rupture weight, $\lambda/(\lambda + r)$	P^*	equilibrium transfer flow to Europe

(Groseclose and Snyder, 1996) and the English-auction outcome in undominated strategies of Dekel et al. (2008, Prop. 2), close kin to menu auctions (Bernheim and Whinston, 1986) and to competing aid donors (Bueno de Mesquita and Smith, 2016). In the two-hegemon extension of Clayton et al. (2026a, Supplement B.2), hegemons whose threats overlap fully bid transfers down to zero, as in Bertrand competition. Subgame perfection alone does not pin down the price. Appendix A shows that the alignment outcome nonetheless holds in every subgame-perfect equilibrium of an alternating-raise auction, and that weak dominance selects the price $[Q]^+$. What the lemma adds is the content of Q: China's bid enters Europe's participation constraint, and every cost of exit in c is a conditional part that survives sequential rationality. When $Q \leq 0$, Europe receives nothing however large V is. Part (iii) is the one place where the commitment cap binds on promises: if Washington cannot stake enough, Europe exits although both would prefer a deal.[1]

4.5 Agenda Power and Rupture Risk

The bidding benchmark gives Europe no voice in the price. We now let Europe and Washington bargain. After the bidding stage fixes ω, and hence Q, the two bargain over a perpetual signed transfer flow P: $P > 0$ means Washington pays Europe and $P < 0$ means Europe concedes.

[1]We value transfers at par. If delivering value P to Europe costs Washington $\phi_U P$ with $\phi_U \leq 1$, as for an exemption from an extraterritorial rule, then Washington keeps Europe aligned if and only if $\phi_U Q < V$ (when its commitment cap does not bind). Cheap in-kind transfers widen the range in which Washington wins without changing the price to Europe. Section 8.2 sets out how we distinguish such transfers from threats left unexercised.

Agreements are binding. Offers are made every Δ units of time. In each round Europe proposes with probability $\beta \in [0,1)$ and Washington otherwise, and the responder accepts or rejects. After rejecting a US offer, Europe may exit, which yields flows Q to Europe and $-V$ to Washington forever. If Europe does not exit, the dispute ruptures into exit with probability $\mu = 1 - e^{-\lambda\Delta}$, where $\lambda \geq 0$ is the rupture hazard; otherwise the status quo continues with flows $(0,0)$ and both discount at rate $r \geq 0$. Rupture is realignment that neither government chooses at the bargaining table, for instance after a change of government in The Hague or an escalation that neither capital intends. It is a threat that leaves something to chance (Schelling, 1960). The disagreement point is the aligned status quo with no transfer, so P replaces the bidding stage's payment. Write $\upsilon = e^{-(r+\lambda)\Delta}$ and $\psi_\Delta = \mu/(1-\upsilon)$, which tends to the rupture weight $\psi = \lambda/(\lambda + r)$ as $\Delta \to 0$.

Proposition 1 (Bargaining with an outside option and rupture risk)**.** *Let $Q < V$, $\beta \in [0,1)$, $r, \lambda \geq 0$ and $r + \lambda > 0$.*

(i) For every $\Delta > 0$, every subgame-perfect equilibrium gives Europe the same payoff $x^*(\Delta)$ and the United States $-x^*(\Delta)$ (closed forms in appendix A).

(ii) If $\lambda > 0$, the subgame-perfect equilibrium is unique (up to Europe's exit choice at one point of indifference, which does not affect payoffs), and agreement is immediate. If $\lambda = 0$, there are also stationary equilibria with delay, with the same payoffs.

(iii) As $\Delta \to 0$, $x^*(\Delta)$ and the equilibrium offer and demand converge to

$$P^* = \max\{Q,\ \psi(Q + \beta(V - Q))\}, \qquad \psi = \frac{\lambda}{\lambda + r} \in [0,1]. \tag{4}$$

If $r > 0$, the second term is the larger one if and only if $Q < \hat{Q} \equiv \lambda\beta V/(r + \lambda\beta)$.

Europe's exit right is placed after it rejects a US offer.[2] Figure 2 plots P^* against Q for three parameter sets.

Corollary 1 (Threshold and channels)**.** *Under the conditions of Proposition 1:*

(a) $P^* \geq Q$: Europe never concedes more than its slack, $-P^* \leq \Lambda = [-Q]^+$. (For $\Delta > 0$ the value $x^*(\Delta)$ can be below Q, by at most $\beta(1-\upsilon)\left[Q - \psi_\Delta V\right]^+/(1-\beta\upsilon)$.)

(b) If $\lambda = 0$, $P^* = [Q]^+$; if $\beta = 0$, $P^* = \max\{Q, \psi Q\}$. In either case $Q \leq 0$ implies $P^* \leq 0$.

(c) $P^* > 0$ if and only if $Q > 0$, or $\lambda > 0$ and $Q + \beta(V - Q) > 0$.

(d) Holding Q fixed, $\partial P^*/\partial V = \beta\psi$ where the second term binds ($Q < \hat{Q}$) and 0 where $Q > \hat{Q}$. If Q itself depends on V, then $dP^*/dV = \psi[\beta + (1-\beta)\, dQ/dV]$ in the first case and dQ/dV in the second.

[2] If Europe could also exit right after the United States rejects Europe's own demand, the stationary equilibrium would have the same frequent-offer limit P^*. But when the outside option binds ($Q > \lambda\beta V/(r + \lambda\beta)$), there would also be nonstationary equilibria in which Europe obtains up to V: a US rejection can be followed by a continuation that holds Europe to its outside option, which makes exit after the rejection credible, so the United States accepts any demand up to V. Allowing exit only after Europe rejects a US offer is what pins down the outside-option principle (Binmore et al., 1989). On the timing of outside options see Shaked (1994) and Ponsatí and Sákovics (1998).

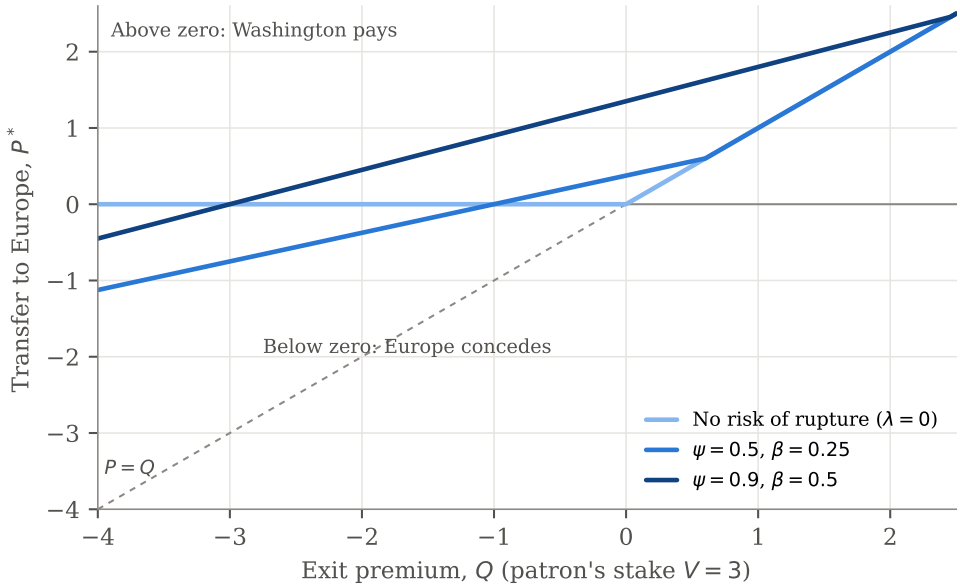


Figure 2: Equilibrium transfer to Europe in the master bargaining game (Proposition 1) as a function of the exit premium, with $V = 3$. Without rupture risk ($\lambda = 0$) the transfer is $[Q]^+$. With rupture weight $\psi = 0.5$ and $\beta = 0.25$, Washington pays Europe whenever $Q > -1$; with $\psi = 0.9$ and $\beta = 0.5$, whenever $Q > -3$. Below those values Europe concedes. Every curve lies on or above the dashed line $P = Q$ (Corollary 1(a)).

(e) V enters P^* directly only through the term $\beta\psi V$ (agenda power under rupture risk), and indirectly only through Q: through the rival's valuation, ρV in W_C when $W_C < \bar{m}_C$ ($\partial\omega/\partial V = \lambda_C\rho$); through self-harming sanctions, $\chi_f = \mathbf{1}\{\varphi V \leq \gamma + \bar{m}_U\}$ (a rise in V past $(\gamma + \bar{m}_U)/\varphi$ raises Q by f); and through the silicon shield, $\kappa = G(s_X + s_L) - G(s_X - m_S)$ with $s_X = s_0 + s_B(V)$ (which equals $\bar{\kappa}$ when $m_S = \bar{m}_U$), with a sign that depends on g. Proposition 7 adds the fifth channel, hold-up.

Nesting: $\lambda = 0$ gives $[Q]^+$ (the outside-option principle); $r \to 0$ with $\lambda > 0$ gives $Q + \beta(V - Q)$; $\beta = 0$ gives $\max\{Q, \psi Q\}$ (Washington proposes); $\beta \to 1$ and $\lambda \to \infty$ give V.

The threshold in part (b) is the outside-option principle (Shaked and Sutton, 1984; Sutton, 1986; Binmore et al., 1989). Without rupture risk, a holder whose exit premium is negative gets nothing for its chokepoint, whoever proposes. The substantive point is old as well. Wagner (1988) showed that asymmetric interdependence yields influence only when trading economic resources for political concessions leaves both sides better off than bargaining over the economic gains alone, a condition that "is independent of the degree of asymmetry in the economic relationship, or its direction". We claim no novelty for the threshold; our contribution lies in what Q contains and in how Washington spends a negative Q (section 5).

Why does V drop out when $\lambda = 0$? Moving first gives Washington no advantage in the bidding benchmark, where both powers can revise their bids. Moving first against a rival bidder is, if

anything, a disadvantage in sequential vote buying (Groseclose and Snyder, 1996). V drops out because Europe lacks agenda power under rupture risk. A Europe that proposes in almost every round still cannot charge for V when $\lambda = 0$, because its only threat is exit, and exit is credible only when $Q > 0$. When $\lambda > 0$, a rejected demand risks a rupture that costs Washington V, so Europe's demands carry weight in proportion to $\beta\psi$. Washington's loss from exit is an externality that Europe's choice imposes on it; the agenda-power channel is a binary instance of the identity-dependent externalities in Jehiel et al. (1996), through which a seller extracts surplus from parties that do not buy.

The master game nests the protocols one might propose. With $\lambda = 0$ it reproduces the bidding benchmark's price for every β. As $r \to 0$ it becomes a random-proposer version of the breakdown model in Binmore et al. (1986), and Europe's share of the gap between Q and V equals its proposal probability. The corner $\beta \to 1$, $\lambda \to \infty$ gives Europe all of V. That corner is the version of the argument implied when Juijn et al. (2026) call ASML "the only card Europe still holds" (section 1). It requires Europe to make take-it-or-leave-it demands whose rejection ends in realignment with near certainty, which is a commitment Europe has not shown it can make.

Part (a) links the bargaining game to the rest of the paper. Rupture risk cuts both ways. When $Q < 0$, a rupture would leave Europe worse off than the aligned status quo, and Washington can extract part of the value to Europe of avoiding it: $P^* < 0$ if and only if $\lambda > 0$ and $Q + \beta(V - Q) < 0$ (with $V = 3$, $\beta = 0.25$ and $\psi = 0.5$, that is $Q < -1$ in fig. 2). But $P^* \geq Q$ always, so Europe never concedes more than its slack Λ. Slack bounds what Washington can obtain from Europe in exchange for keeping it aligned, and section 5 shows that it bounds what Washington can obtain by any combination of conditional threats.

5 The Slack Principle

Lemma 1 and Proposition 1 price the chokepoint when Europe's exit premium is positive. On our reading of the record since 2019 (section 8), it has not been, and Washington has still asked Europe for a great deal. This section asks how much a patron can obtain from an ally whose premium is negative. The answer is the ally's slack, $\Lambda = [-Q]^+$, plus whatever the patron can threaten that does not depend on the ally's alignment. The idea that a coercer's power lies in the slack of the target's participation constraint belongs to Clayton et al. (2026b) and Clayton et al. (2026a). We extend it to threats that cut off an ally for whom a rival is bidding.

5.1 Bite, budgets and assurance

Consider one period in which Europe is aligned, its premium is Q, and it receives $[Q]^+$ from the bidding stage. Washington has a finite set of demands. Demand k costs Europe $t_k > 0$ and is worth $w_k > 0$ to the United States. A concession is *reversible* if exit would undo it, so that it lowers Europe's aligned payoff and leaves its exit payoff unchanged. Policy alignment is of this kind. A limit on servicing Chinese fabs is reversible, because a Europe that exited would resume servicing;

so are tariff cuts and export-control rules. A concession is *sunk* if it lowers both payoffs: payments, purchases, and transfers of assets or of control over them.

Washington backs its demands with instruments (section 4.3). Withdrawing a set J of instruments from an aligned Europe for the period lowers Europe's aligned payoff by $\Delta^A(J)$ and its exit payoff by $\Delta^X(J)$, with $0 \le \Delta^X(J) \le \Delta^A(J)$ and both nondecreasing in J. We call

$$u(J) = \Delta^X(J) \qquad \text{and} \qquad d(J) = \Delta^A(J) - \Delta^X(J)$$

the unconditional and conditional parts of J. The conditional part is what Europe would lose by exiting anyway; for a single instrument it is the d_j that enters the exit cost c in eq. (3). The unconditional part is what Washington would still provide to a Europe that had exited and can threaten to take away. We impose no additivity: the AI instrument and US force posture may be complements or substitutes in Europe's payoff.

Washington's credible set J^c is fixed at the start of the period, and every subset of it is a credible threat, as with one stake per instrument under the commitment technology of section 4.2. Washington issues *adaptive ultimatums*: any sequence of demands, each naming unconceded demands and a threat set of instruments not yet withdrawn, each chosen after seeing Europe's earlier responses. Europe accepts or refuses. A refusal executes the threat at cost $e(J) \ge 0$ to Washington, and executed instruments stay withdrawn for the period, so a threat once carried out is *spent*. A threat that deters is not spent and can back the next demand. At the end of the period the bidding contest is replayed. If J_w has been withdrawn and Europe has made reversible concessions T_{r}, Washington must pay $[Q + d(J_w) + T_{\mathrm{r}}]^+$ to keep Europe aligned. For a set S of concessions, $T_{\mathrm{r}}(S)$ and $T_{\mathrm{s}}(S)$ are its reversible and sunk totals and $W(S) = \sum_{k \in S} w_k$ its value to Washington.

Proposition 2 (Slack principle)**.** *Let* $\Lambda = [-Q]^+$, $u(J) = \Delta^X(J)$, $d(J) = \Delta^A(J) - \Delta^X(J)$ *and* $B(J) = u(J) + \min\{d(J), \Lambda\} = \min\{\Delta^A(J), \Delta^X(J) + \Lambda\}$. *Every subset of* J^c *is a credible threat, no other instrument can be withdrawn during the period, executing a withdrawal of* J *costs the United States* $e(J) \ge 0$*, and the United States retains Europe after every history.*

(a) Bite. *Refusing an ultimatum backed by* J*, issued when nothing has been conceded or withdrawn, costs Europe* $B(J)$*, net of the rent the United States must then pay.* B *is nondecreasing in* J*. If* $Q \ge 0$*, then* $B(J) = u(J)$*: only unconditional parts bite.*

(b) Reversible concessions. *Europe's net loss from reversible concessions totalling* T *is* $\min\{T, \Lambda\}$*. Faced with a single ultimatum that asks for reversible concessions* T_{r} *and sunk concessions* T_{s} *and is backed by* J*, Europe accepts if and only if* $\min\{T_{\mathrm{r}}, \Lambda\} + T_{\mathrm{s}} \le B(J)$.

(c) One budget.

(i) In every subgame perfect equilibrium of every adaptive sequence of ultimatums, and for every Q*, Europe's total net loss is at most* $B(J^c)$*. Conditional instruments share one budget:* $B(J) - u(J) \le \Lambda$ *for every* J*, and reversible concessions cost Europe at most* Λ *in total, however they are backed.*

(ii) The United States' equilibrium payoff is at most $V_1 = \max\{W(S) - [Q + T_{\mathrm{r}}(S)]^+ : \min\{T_{\mathrm{r}}(S), \Lambda\} + T_{\mathrm{s}}(S) \le B(J^c)\}$*, the payoff of the best single ultimatum backed by* J^c*.*

(iii) If the United States can commit to a single ultimatum, one bundled ultimatum backed by J^c *yields* V_1*. Europe's loss then equals* $B(J^c)$ *if the United States has a divisible sunk demand of positive value and sufficient size.*

(d) Assurance. *Without that commitment the United States still obtains* V_1 *if all demands are reversible and* $B(J^c) \ge \Lambda$*, or if some best single ultimatum contains all of its demands. (In every equilibrium if* $e(\{j\}) > 0$ *for every* j*; in some equilibrium otherwise.) In general it can obtain strictly less. Threats that deter are not spent, so the United States reuses them after Europe concedes, and Europe anticipates this. With one instrument with* $d = 0$ *and* n *identical sunk demands of cost* τ*, the United States obtains exactly* $n \bmod (b+1)$ *of them, where* $b = \lfloor B(J^c)/\tau \rfloor$*.*

The proof is in appendix A, and appendix B reports the brute-force checks. Part (a) has a simple logic. Withdrawing a conditional part from an aligned Europe takes away something Europe would lose by exiting anyway, so it raises Europe's premium by $d(J)$. Up to Λ the withdrawal hurts Europe. Beyond Λ the premium turns positive and Washington must pay rent to keep Europe aligned, which hands the excess back. A reversible concession works the same way, which is part (b): it lowers only the aligned payoff, so past Λ Europe would exit unless compensated. Sunk concessions and unconditional parts escape the cap because they hurt Europe whatever it does. So once Europe's premium is positive, a threat to withhold what Washington would withhold from an exiting Europe in any case has no bite at all.

In panel (a) of fig. 3 the unconditional threat is the smaller one, yet it is the only one that still works as Λ falls toward zero. Panel (b) illustrates part (c)(i). A coercer that prices each conditional threat on its own counts the same slack twice: at $\Lambda = 1$ the separate bites sum to 2 and the joint bite is 1.

The bound in (c)(i) holds because Europe can always refuse every ultimatum, and the worst that refusal brings is the withdrawal of a subset of J^c. It needs a fixed credible set: if a refusal could make a new instrument credible, Europe's loss could exceed the bite of the initial set. Under the stake technology of section 4.2 an instrument is credible when its execution cost is at most $\bar{m}_U$, whatever has happened before, so the set does not move.

What is new here. The participation constraint and its slack are Clayton et al.'s: "Threats generate power to the extent that they generate slack in the participation constraint" (Clayton et al., 2026b). In their framework the pooled threats are enforcement threats. They raise the target's inside option, and the hegemon charges for the surplus they create; threats to cut off a target that refuses appear only in the supplement (Section B.3.1) of Clayton et al. (2026a), where they are partly self-defeating. Washington's instruments against Europe are cut-off threats aimed at an ally that a rival is bidding for. Taking them seriously yields three results. Only the conditional part of

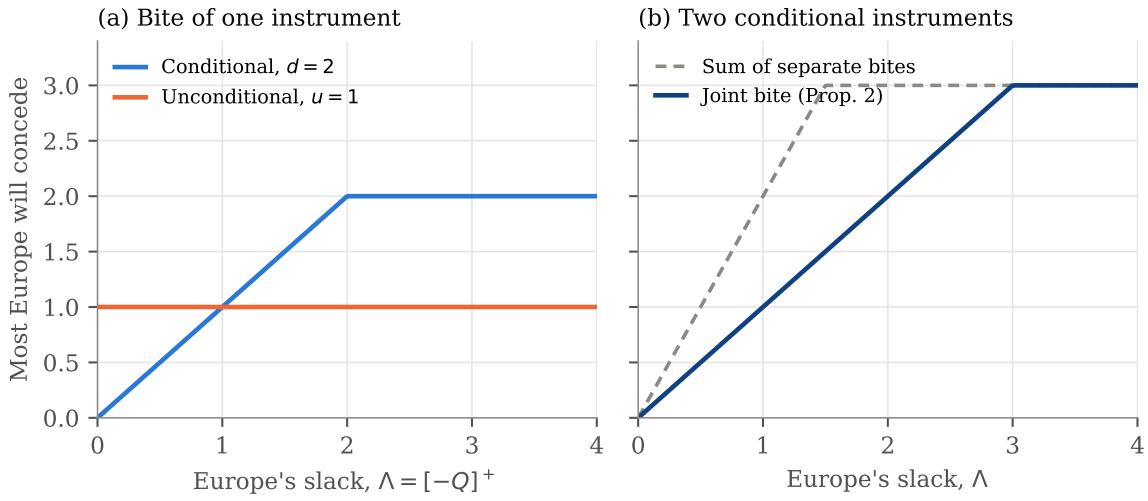


Figure 3: The slack budget. Panel (a): the bite of a conditional instrument with $d = 2$ rises one for one with Europe's slack Λ until $\Lambda = 2$; an unconditional instrument with $u = 1$ bites the same at every Λ. Panel (b): two conditional instruments with $d = 1.5$ each. The sum of their separate bites (dashed) overstates their joint bite $\min\{3, \Lambda\}$ at every $\Lambda < 3$. Bites from Proposition 2(a).

an instrument is capped by slack. Conditional bites are not additive, so instruments that deter the same exit compete for one budget. And a coercer that cannot commit to stop demanding obtains less (part (d)), because a threat that deters is not used up and will back the next demand. Europe concedes only if the total it expects to concede, counting what the unspent threat will extract later, stays within the bite; with $b + 1$ identical sunk demands, a single ultimatum would obtain b of them and the adaptive sequence obtains none. The problem is an old one in the theory of strategic moves (Schelling, 1960): a threat meant to induce compliance works only if the target believes that compliance will end it. Part (d) puts it in a form close to Pauly's assurance dilemma: steps that make a coercer's threats credible weaken its assurance that compliance ends the punishment (Pauly, 2024). Washington can buy the commitment with a staked promise to make no further demand in the period, which is credible if $\bar{m}_U$ exceeds the value of further extraction.[3]

Washington's choice of demands follows from parts (b) to (d).

Corollary 2 (Rationing)**.** *Suppose all demands are reversible and either the United States can commit to a single ultimatum or $B(J^c) \geq \Lambda$. Then its optimal demand set solves*

$$\max_{y \in \{0,1\}^K} \sum_k w_k y_k - [\textstyle\sum_k t_k y_k - \Lambda]^+ \quad \textit{s.t.} \quad \min\Big\{ \sum_k t_k y_k, \Lambda \Big\} \leq B(J^c).$$

This is the knapsack $\max \sum_k w_k y_k$ *s.t.* $\sum_k t_k y_k \leq \min\{\Lambda, B(J^c)\}$ *if* $B(J^c) < \Lambda$*, or if demands are*

[3]Spent threats can also serve as a commitment device. Take two instruments with $u = 1.2$ and $u = 1.0$ ($d = 0$), sunk demands with $t = 1$ ($w = 5$) and $t = 1.5$ ($w = 1$), and execution cost 0.1 for each instrument. A single ultimatum yields 5. In every subgame perfect equilibrium of the adaptive protocol, Washington first issues an ultimatum that Europe refuses, which burns one instrument, and then obtains the first demand, for a payoff of 4.9.

divisible ($y \in [0,1]^K$) and $w_k \leq t_k$ for every k. In general the United States may exceed Λ and pay the excess back as rent. The free capacity $\min\{\Lambda, B(J^c)\}$ falls as Q rises toward 0. Demands from every issue area draw on it.

The corollary is a theory of how a patron rations coercion across its demands on an ally. Export-control alignment and limits on servicing in China are reversible concessions (section 2). So are the trade terms of August 2025, under which the EU said it would eliminate tariffs on US industrial goods and adopt technology-security requirements in line with those of the United States (The White House, 2025d). The NATO commitment to spend 5 percent of GDP on defence and related items by 2035 (North Atlantic Treaty Organization, 2025) and the EU's stated intention to increase procurement of US military equipment (The White House, 2025d) are closer to sunk, since they cost Europe whatever it later does with ASML. A demand on Europe's digital rules would draw on the same budget. For Washington these demands are substitutes. The reversible ones share the capacity $\min\{\Lambda, B(J^c)\}$, so a concession extracted on tariffs leaves less to extract on servicing, and the sunk ones draw down the unconditional part of the bite. Washington links the issues openly. At The Hague in June 2025, Trump answered Spain's refusal of the spending target by saying "We're negotiating with Spain on a trade deal. We're going to make them pay twice as much" (RTÉ News, 2025). In the model this moves one budget between demands. The budget shrinks as China's credible offer rises, so a patron facing a holder whose premium approaches zero must drop demands, starting with those of lowest value per unit of European cost.

5.2 Counter-threats and hedges

We do not let China punish alignment as such. Its coercion of US allies has been aimed at specific acts. When Washington pressed Tokyo in 2024 to restrict Tokyo Electron's sales and servicing in China, Beijing threatened "severe economic retaliation" (Bloomberg, 2024), and the first use of its new regulation on countering foreign extraterritorial jurisdiction was a May 2026 determination against the Commission's investigation of Nuctech under the Foreign Subsidies Regulation (Cleary Gottlieb Steen & Hamilton LLP, 2026). We therefore model Chinese coercion as counter-threats attached to particular concessions, announced before Washington makes its demands. We also consider a hedge: legacy-tool sales and servicing worth y_H to Europe and w_H to Washington if stopped. Washington can stop the hedge in two ways. It can demand it as a concession backed by J^c (linkage), or it can threaten to block the hedged flows with its jurisdiction over them, at net ex post cost e_F.

Proposition 3 (Counter-threats and hedges)**.** *Maintain the assumptions of Proposition 2.*

(a) Counter-threats. *Before the United States moves, China chooses credible counter-threats $r_k \in [0, \bar{r}_k]$. If concession k is made and $r_k > 0$, China executes at cost $e_k^C > 0$ and Europe bears the sunk loss r_k. China loses $z_k \geq 0$ when concession k is made. The United States makes a single ultimatum backed by J^c, and its optimal bundle is unique at the profiles compared below.*

(i) A counter-threat adds r_k to the sunk cost of k: Europe accepts S if and only if

$$\min\{T_{\mathrm{r}}(S), \Lambda\} + T_{\mathrm{s}}(S) + \textstyle\sum_{k\in S} r_k \leq B(J^c).$$

(ii) China does better than with no threats only if the United States' demand set changes.

(iii) If a counter-threat on k is executed in China's optimal profile, then dropping it changes the United States' demand set to one that contains a demand outside the equilibrium set. China executes on path only to crowd out other demands, and this occurs for an open set of parameters.

(iv) The United States' payoff is weakly lower under any profile of counter-threats.

(v) Europe's net loss is at most $B(J^c)$ with or without counter-threats. It equals $B(J^c)$ in both cases if the United States has a divisible sunk demand of positive value that China cannot threaten. The counter-threat then taxes the patron, not the holder. Otherwise Europe's loss can rise or fall.

(b) Policing a hedge. *Stopping a hedge worth y_H to Europe is a reversible concession. If it is the only demand, linkage stops the hedge if and only if $\min\{y_H, \Lambda\} \leq B(J^c)$ and $w_H \geq [y_H - \Lambda]^+$, where w_H is its value to the United States. The United States then pays rent $[y_H - \Lambda]^+$. Linkage is free only if $y_H \leq \min\{\Lambda, B(J^c)\}$, and its free part $\min\{y_H, \Lambda\}$ vanishes as $Q \to 0$. A staked threat to block the hedge by flow jurisdiction, with net ex post cost e_F, is credible if and only if $e_F \leq \bar{m}_U$. It leaves Q unchanged and is never executed. Executing a conditional withdrawal J raises Europe's premium to $Q + d(J)$.*

Part (a) changes who pays for Chinese coercion. A counter-threat makes a concession more expensive for Europe, and by (a)(i) it does so exactly as a sunk cost would. If Washington has other demands that China cannot reach, it fills the budget with them, and Europe's loss stays at $B(J^c)$. What changes is the mix. Washington gives up the demand China has made expensive and takes others it values less, so the counter-threat taxes the patron. China gains only by changing what Washington asks for (part (a)(ii)), and it carries out a threat only when doing so crowds out some other demand (part (a)(iii)). A numerical case shows that this can happen on the equilibrium path. Let $B(J^c) = 2$. Demand 1 is sunk, with $t_1 = 0.3$, $w_1 = 1$ and $z_1 = 5$, and China cannot threaten it. Demand 2 is sunk, with $t_2 = 1.5$ and $w_2 = 20$, and China can attach $\bar{r}_2 = 0.4$ to it at execution cost 0.1. Without counter-threats Washington asks for both. With $r_2 = 0.4$ the pair would cost Europe $2.2 > 2$, so Washington keeps the valuable demand 2 and drops demand 1, and China executes its threat. China's payoff rises from -5 to -0.1, and Europe's loss rises from 1.8 to 1.9. Under complete information, then, an executed Chinese counter-threat needs a demand it crowded out. Section 8.3 reads the Nexperia episode with this requirement in mind; the alternative is that The Hague misjudged China's resolve, which lies outside this complete-information model.

Part (b) yields the prediction that separates the slack logic from simple cost minimization. A coercer that minimizes cost polices partial defection with whatever credible instrument is cheapest,

and sizes sanctions so that punishing the small offence does not erode deterrence of the large one (Stigler, 1970; Shavell, 1992). Holding the costs of instruments fixed, that account predicts no change in Washington's choice of instrument as Europe's outside option improves. The slack principle predicts a change. The free part of linkage, $\min\{y_H, \Lambda\}$, shrinks as the premium rises toward zero, and beyond it Washington must pay the rent $[y_H - \Lambda]^+$ to use linkage at all. Flow jurisdiction does not depend on slack: it is credible whenever $e_F \leq \bar{m}_U$, it leaves Q unchanged, and under complete information it deters without being used. So linkage against hedging should appear when the ally has slack and fade as its premium approaches zero, even where flow jurisdiction is unavailable, while the use of flow jurisdiction should not track the ally's premium at all. Linkage is also a wasting asset once used: a conditional withdrawal that is carried out raises Europe's premium by $d(J)$, so the instrument that punished the hedge no longer deters exit.

The record fits the first half of this prediction. The 2025 National Security Strategy ties favourable treatment to export-control alignment (section 5.3), and in the August 2025 joint statement the United States undertook to facilitate exports of AI chips to the EU once the EU adopted technology-security requirements in line with its own (The White House, 2025d). Both tie the AI instrument and commercial instruments to export-control alignment, a reversible concession. That is what part (b) leads us to expect from a patron dealing with an ally that has slack. The second half, the retreat from linkage as the premium approaches zero, needs variation in premia that the Dutch record alone does not supply. Section 8.5 outlines a comparison across holders (the Netherlands, Japan and South Korea) and over time.

5.3 Conditionality as a choice

So far conditionality has been a parameter. Washington chooses it. Recall from section 4.3 that Washington defends Europe in a crisis if its stake in Europe's security exceeds a crisis cost ξ drawn from a continuous distribution G with density g, that its stake is $s_A = s_X + s_L$ in a loyal Europe and $s_X = s_0 + s_B$ in one that has exited, and that a public stake $m_S \leq \bar{m}_U$ on abandoning an exiter makes the core guarantee conditional by $\kappa = G(s_A) - G(s_X - m_S)$. Choosing m_S is choosing $\kappa \in [\kappa_0, \bar{\kappa}]$, where $\kappa_0 = G(s_A) - G(s_X)$ is the conditionality with no stake and $\bar{\kappa} = G(s_A) - G(s_X - \bar{m}_U)$ the largest Washington can create.

Conditionality is costly to the coercer, and we let it cost $H(\kappa)$, increasing and convex with $H(\kappa_0) = 0$. The cost is one of assurance: other allies discount a guarantee that comes with a broad withdrawal rule, and a guarantee tied to Europe's conduct can be triggered by mistake. With Q_0 the premium at κ_0, the premium is $Q(\kappa) = Q_0 - (\kappa - \kappa_0)S_k$. Washington converts slack into concessions worth $\nu \in [0, 1]$ per unit of European cost; by Corollary 2, ν is the value of its best reversible demands, and we assume $B(J^c) \geq \Lambda$ so that all slack can be converted. Washington chooses κ before the bidding stage to maximize

$$\Pi(\kappa) = -\left[Q(\kappa)\right]^+ + \nu\left[-Q(\kappa)\right]^+ - H(\kappa).$$

Let H' be the right derivative of H. For $x \geq 0$ let $\kappa_x = \inf\{\kappa \in [\kappa_0, \bar{\kappa}] : H'(\kappa) \geq x\}$, with $\inf \emptyset = +\infty$, and write $\kappa_1 = \kappa_{S_k}$ and $\kappa_\nu = \kappa_{\nu S_k}$, so that $\kappa_0 \leq \kappa_\nu \leq \kappa_1$. The conditionality at which the premium is exactly zero is $\kappa_z = \kappa_0 + Q_0/S_k$.

Proposition 4 (Conditionality is the price of concessions)**.** *The conditionality*

$$\kappa^* = \min\left\{\bar{\kappa},\ \max\{\kappa_\nu,\ \min\{\kappa_z, \kappa_1\}\}\right\}$$

maximizes Π*, uniquely if* H *is strictly convex, and* $Q(\kappa^*) = (\kappa_z - \kappa^*)S_k$*.*

(a) If $\nu = 0$*, then* $\kappa^* = \kappa_0$ *when* $Q_0 \leq 0$*, and* $\kappa^* = \min\{\kappa_z, \kappa_1, \bar{\kappa}\}$ *otherwise. The United States never makes its guarantee more conditional than needed to hold the premium at zero. It creates no slack:* $\Lambda^* = [-Q_0]^+$*.*

(b) If $\nu > 0$*,* $\kappa_0 \leq \kappa_z < \bar{\kappa}$ *and* $H'(\kappa_z) < \nu S_k$*, then* $\kappa^* = \min\{\kappa_\nu, \bar{\kappa}\} > \kappa_z$ *and* $\Lambda^* = (\kappa^* - \kappa_z)S_k > 0$*: conditionality is the price of concessions. If* $Q_0 < 0$*, then* $\kappa^* = \min\{\kappa_\nu, \bar{\kappa}\}$ *and* $\Lambda^* = (\kappa^* - \kappa_z)S_k \geq -Q_0$*.*

(c) Let H *be strictly convex and differentiable. Away from regime boundaries:*

(i) if $\kappa^* = \kappa_z$*, then* $d\kappa^*/dQ_0 = 1/S_k$ *and the United States pays no rent: a stronger Chinese bid elicits a more conditional guarantee, not a payment;*

(ii) if $\kappa^* = \kappa_1 < \min\{\kappa_z, \bar{\kappa}\}$ *(assurance is costly at the margin before the premium reaches zero),* κ^* *does not depend on* Q_0 *and the rent* $Q(\kappa_1) > 0$ *rises one for one with* Q_0*;*

(iii) if $\kappa^* = \kappa_\nu \in (\max\{\kappa_z, \kappa_0\}, \bar{\kappa})$*, or if* $\kappa^* = \kappa_0 > \kappa_z$*, then* κ^* *does not depend on* Q_0 *and slack falls one for one with* Q_0*: China's bid is paid for out of the United States' concessions;*

(iv) if $\kappa^* = \bar{\kappa}$*, the United States pays* $[Q(\bar{\kappa})]^+$*, which rises one for one with* Q_0 *when* $Q(\bar{\kappa}) > 0$*. Otherwise slack falls one for one.*

(d) $d\bar{\kappa}/ds_X = g(s_X + s_L) - g(s_X - \bar{m}_U)$*, with* s_L *and* $\bar{m}_U$ *fixed. Hence* $\bar{\kappa}$ *is nonincreasing in* s_X *wherever* g *is nonincreasing on* $[s_X - \bar{m}_U,\ s_X + s_L]$*, for example when Washington would almost surely defend a loyal Europe, and it is then nonincreasing in the access value of European territory and, if* s_B *rises with* V*, in* V*. If instead* $g(s_X + s_L) > g(s_X - \bar{m}_U)$*,* $\bar{\kappa}$ *rises with* s_X*. If* s_A *is held fixed when* s_X *rises,* $\bar{\kappa}$ *is nonincreasing in* s_X *for every* G*.*

If conditionality cost nothing, Washington would set it at $\bar{\kappa}$. With an assurance cost, how conditional the guarantee is depends on what Washington wants from Europe. A patron with no further demands ($\nu = 0$) makes its guarantee just conditional enough to hold the premium at zero, or less if assurance becomes too costly first, and it creates no slack. Conditionality beyond that point is worth buying only because slack can be spent, so the patron that plans to make demands is the one that makes its guarantee more conditional than the rivalry with China requires. That is the sense in which conditionality is the price of concessions: Europe pays for the extra conditionality

with the concessions it then makes, and other allies pay through a weaker assurance. The argument uses only the fact that Washington chooses a conditional part before the bidding stage at a convex cost, so it applies with d_j in place of κS_k to any instrument whose conditional part Washington sets.

Part (c)(i) describes how a patron answers a stronger rival. In the bidding benchmark a rise in China's credible offer raises what Washington must pay once the premium is positive. When assurance is cheap at the margin, Washington instead makes its guarantee more conditional, by exactly enough to keep the premium at zero, and pays nothing. The December 2025 National Security Strategy reads this way to us. It promises "more favorable treatment on commercial matters, technology sharing, and defense procurement" to countries that "align their export controls with ours" (The White House, 2025e). The passage makes treatment depend on alignment, which raises conditional parts relative to the treatment of a country that does not align. In the model's terms this is conditionality. A transfer would be provision above the aligned baseline, which is due only when the premium is positive. Section 8.2 sets out how we code the two in the record. Where assurance is already costly at the margin, or Washington is already spending slack, parts (c)(ii) and (c)(iii) say who pays for China's bid instead: Washington in rent, or Washington in forgone concessions.

Two kinds of US drawdown have opposite places in the model. An untargeted drawdown lowers what Washington provides to a loyal ally, $x_p(A)$, and so lowers the conditional part d_p of posture and enablers along with Europe's slack. The withdrawal of 25,000 or more troops from Europe that the Pentagon was reported in September 2026 to be weighing (De Luce et al., 2026) would be a cut of this kind, to the extent that it is not aimed at particular allies' conduct. A targeted withdrawal after defiance is different. The order of 1 May 2026 to move about 5,000 troops out of Germany followed Chancellor Merz's criticism of US strategy on Iran (Martinez, 2026) (see section 2). It punished an act, so in the model it is an executed threat, and Proposition 3(b) says what execution does: it spends the instrument and raises the target's premium by the instrument's conditional part, since posture already withdrawn cannot be withheld again. Whether the order also raised other allies' beliefs about how conditional US protection is depends on reputation and on uncertainty about κ, which this complete-information model leaves out.

Part (d) gives the sign of the silicon shield, the fourth channel of Corollary 1. If holding the chokepoint raises Washington's own stake in Europe's security whatever Europe's alignment, then s_X rises. In the case the proposition singles out, where Washington would almost surely defend a loyal Europe, the higher s_X lowers the most conditionality Washington can create. The chokepoint then limits how far its holder can be coerced through the guarantee. The same comparative static holds for the access value of European territory, which also raises s_X (Carson et al., 2025). Where defence of a loyal Europe is in real doubt, the sign can reverse, and holding the chokepoint can make the guarantee more conditional. Part (d) therefore makes the silicon shield an empirical question about the distribution of crisis costs.

The assurance literature supplies the cost side of the proposition. Pauly (2024) shows that the steps a coercer takes to make a threat credible undermine its assurance that compliance will

not be punished. In our model those steps are the stakes that raise κ, and part of their cost is H. Gerzhoy (2015) credits US abandonment threats, paired with that kind of assurance, with restraining West Germany; Lanoszka (2018) disputes that they were decisive and finds that allies who doubt the patron build capabilities of their own. Such capabilities raise the ally's premium, as Europe's autonomy investment does in section 6, and the prospect of them is a second component of H. Blankenship (2021) finds that US pressure on allies works when Washington can credibly threaten to reduce protection, and that allies in locations the United States values contribute less. That pattern is what part (d) implies when the density of crisis costs is decreasing: a higher s_X lowers the conditionality Washington can credibly create.

6 Control Rights and Preventive Demands

The static model takes Europe's control of the chokepoint as given and its exit premium as fixed. Yet the premium moves. Chinese credit or market offers raise ω, European rearmament lowers Europe's stake in core defence S_k, and rationing of loyal allies would shrink the conditional part of the AI instrument; Chinese indigenization and stockpiling lower W and push the other way. If Washington expects the premium to turn positive, it expects to pay rents that Europe cannot credibly promise to forgo. A transfer of control can remove them. This section asks when a patron demands control of a chokepoint, what it pays, and when the demand fails.

6.1 Setup

There are two periods, $t \in \{1, 2\}$, and a discount factor $\delta \in (0, 1]$. In each period in which Europe holds control, the bidding contest of Lemma 1 is played at that period's premium Q_t, and Europe receives $[Q_t]^+$. We assume that Washington retains Europe at every contest, including after a cut ($Q_1 + d_1 < V$ and $Q_1 + d_1 \leq \bar{m}_U$, and the same in period 2 at the highest premium Europe can reach), and that payments are made at signing.

The ultimatum. At the start of period 1, before the period-1 contest, Washington may issue an ultimatum at an issuing cost $\iota \geq 0$: Europe transfers control of ASML's customers, output and technology for both periods against a payment $T \geq 0$, or Washington withdraws a bundle J of instruments for period 1. The bundle has unconditional part u_1, conditional part d_1 and direct ex post cost $g_1 \geq 0$ to Washington, which is nonnegative because the baseline is Washington's ex post optimal provision to a loyal ally. If Europe refuses and the threat is carried out, Europe's premium rises to $Q_1 + d_1$, and by Proposition 2(a) the refusal costs Europe the bite

$$B_1 = u_1 + \min\{d_1, \Lambda_1\}, \qquad \Lambda_1 = [-Q_1]^+,$$

net of the rent Washington must then pay. Carrying out the threat costs Washington $e = g_1 + [Q_1 + d_1]^+ - [Q_1]^+$, its direct cost plus the extra rent. Under the commitment technology of

section 4.2 the threat is credible if and only if $e \leq \bar{m}_U$; otherwise $B_1 = 0$. No other period-1 demand draws on the same bite, so the ultimatum has the whole budget of Proposition 2(c) to itself.

The bundle may contain any instrument of section 4.3. We do not expect the core guarantee to back such an ultimatum. On our reading, withdrawing protection from a Europe that stays aligned costs Washington its own stake in defending a loyal ally, so g_1 is large unless s_A is small, and the threat is credible only if Washington can stake more than that. Posture, enablers and the AI instrument cost less to withdraw from one ally for one period.

Control. A transfer lowers Europe's premium at both dates by $\Delta_c \in (0, \infty]$, the enforceability of control. With $\Delta_c = \infty$ Europe can never again choose exit. With Δ_c finite, a Europe whose premium later rises far enough could take back part of what it gave. What is transferred sets Δ_c. Relocating capacity to US soil raises it. So do technical controls held in the United States: Bloomberg reported in 2024, citing anonymous sources, that ASML had told Dutch officials it could disable machines remotely (Baazil et al., 2024), and a disabling capability held in Washington would be hard for The Hague to override. Control exercised through ASML's US subsidiaries, which are US-law entities (ASML Holding N.V., 2026c), sits under US jurisdiction. A controlling vote in a Dutch holding company has low Δ_c, because it stays under Dutch law. In 2025 the Dutch state issued an order under the Goods Availability Act against Nexperia (Gerechtshof Amsterdam, Ondernemingskamer, 2025), and on our reading a vote held by Washington would be exposed to the same kind of emergency power (section 8.3).

Values and costs. Washington places an intrinsic value $v_c \geq 0$ on control, apart from the rents it saves. Europe bears a sovereignty cost $\sigma \geq 0$ if it transfers control. It includes the domestic political cost of surrendering a national champion and governance hurdles that are fixed before any demand is made: ASML's preference-share foundation and screening under the Vifo Act (section 2), and the Foreign Subsidies Regulation. Under that Regulation, a concentration involving an EU target with EU turnover of at least EUR 500 million must be notified when the parties received more than EUR 50 million in third-country financial contributions over three years, and the Commission may prohibit it after an in-depth review (European Parliament and Council of the European Union, 2022). That a US-funded vehicle taking control of ASML would be notifiable is our inference. Because these hurdles are public and set in advance, they raise σ before Washington chooses whether to demand anything. China punishes a completed transfer with $r^c \geq 0$, which Europe bears; Corollary 3(d) asks when that threat is credible.

Autonomy investment. After the period-1 outcome, and before the period-2 contest, Europe can make an observable investment at cost K that raises its period-2 premium by $\bar{I}$, for instance rearmament that lowers S_k. Holding a period-2 premium q, it invests if and only if $\delta(\left[q + \bar{I}\right]^+ - [q]^+) > K$.

Ties. An indifferent Europe accepts; Washington issues only if it strictly gains (the limit $\iota \to 0^+$ when $\iota = 0$); Europe invests only if that is strictly profitable. After a transfer Washington keeps its baseline provision, since withdrawing from a Europe that holds no control would cost it something and gain it nothing. The scope conditions of section 9.1 apply.

6.2 Preventive control

Let $I(q) = \mathbf{1}\{\delta(\left[q + \bar{I}\right]^+ - [q]^+) > K\}$ indicate investment at period-2 premium q. Washington's discounted period-2 rent is $\text{Rent}_U(q) = \delta\left[q + \bar{I}\, I(q)\right]^+$, and Europe's period-2 value net of its investment cost is $\text{Val}_H(q) = \text{Rent}_U(q) - K\, I(q)$.

Proposition 5 (Preventive control)**.** *Let*

$$\Delta R_U = [Q_1]^+ - [Q_1 - \Delta_c]^+ + \text{Rent}_U(Q_2) - \text{Rent}_U(Q_2 - \Delta_c),$$
$$\Delta R_H = [Q_1]^+ - [Q_1 - \Delta_c]^+ + \text{Val}_H(Q_2) - \text{Val}_H(Q_2 - \Delta_c).$$

Let $B_1 = u_1 + \min\{d_1, \Lambda_1\}$ if the threat is credible ($e \le \bar{m}_U$, with $e = g_1 + [Q_1 + d_1]^+ - [Q_1]^+$) and $B_1 = 0$ otherwise. In the subgame-perfect equilibrium, the United States issues the ultimatum, and Europe accepts, if and only if

$$\Gamma \equiv v_c - \iota + \min\{\Delta R_U,\ (\Delta R_U - \Delta R_H) + B_1 - \sigma - r^c\} > 0.$$

The payment is $T^* = [\Delta R_H - B_1 + \sigma + r^c]^+$, the United States gains Γ, and Europe's payoff is $\max\{\Delta R_H - B_1,\ -\sigma - r^c\} + [Q_1 - \Delta_c]^+ + \text{Val}_H(Q_2 - \Delta_c)$. An ultimatum that is issued is accepted. The credibility condition $e \le \bar{m}_U$ does not depend on Δ_c.

The proof is in appendix A. ΔR_U is the rent that control saves Washington and ΔR_H is what Europe gives up by losing it; they differ only by the investment cost that a transfer spares Europe. Washington gains the value of control net of the issuing cost, plus the smaller of the rent saved and what it can extract by force: the bite, less the deadweight cost $\sigma + r^c$, plus the investment cost saved. Europe is paid for the rents it surrenders and for the deadweight cost, less the bite. A refused ultimatum never pays Washington, so the threat is never carried out on the equilibrium path. Its credibility turns on what carrying it out would cost against a Europe that keeps control, which is why Δ_c does not enter.

In the special case $v_c = \iota = 0$, $\Delta_c = \infty$ and $\bar{I} = 0$, the gain is $\Gamma = \min\{[Q_1]^+ + \delta\,[Q_2]^+,\ B_1 - \sigma - r^c\}$, and if $Q_1 < 0$ it is $\Gamma = \min\{\delta\,[Q_2]^+,\ B_1 - \sigma - r^c\}$. Washington then demands control if and only if Europe's future premium is positive and the bite exceeds the deadweight cost of the transfer. The first condition holds only because Washington values control at exactly zero and does not issue when indifferent. With $v_c > \iota$ it disappears, and the general statement is a threshold. Since $B_1 \le u_1 + \Lambda_1$, a purely conditional bundle bites by more than $\sigma + r^c$ only if $Q_1 < -(\sigma + r^c)$: the demand comes while Europe is weak and is expected to grow stronger.

Corollary 3 (Motive, investment, enforceability, China)**.**

(a) Preventive motive. $\Gamma > 0$ *if and only if* $\Delta R_U > \iota - v_c$ *and* $B_1 + (\Delta R_U - \Delta R_H) > \sigma + r^c + \iota - v_c$. *If* $v_c \leq \iota$, *issuance requires* $\Delta R_U > 0$ *and* $B_1 + (\Delta R_U - \Delta R_H) > \sigma + r^c$. *For* $\Delta_c > 0$, $\Delta R_U > 0$ *if and only if* $Q_1 > 0$ *or* $Q_2 + \bar{I}\, I(Q_2) > 0$*: Europe's premium is positive at date 1, or its premium after its own investment is positive at date 2.*

(b) Autonomy investment. $\Delta R_U - \Delta R_H = K\left[I(Q_2) - I(Q_2 - \Delta_c)\right] \geq 0$. *Both arguments of the minimum in* Γ *are weakly larger than when Europe has no investment option (*$\bar{I} = 0$*), so an autonomy investment that Europe cannot commit to forgo weakly raises* Γ. *With investment, the United States can obtain control with no bite at all: if* $B_1 = 0$, *the ultimatum is issued if and only if* $\Delta R_U > \iota - v_c$ *and* $K[I(Q_2) - I(Q_2 - \Delta_c)] > \sigma + r^c + \iota - v_c$, *and the transfer then saves Europe's investment cost.*

(c) Enforceability. *Holding* σ, r^c, v_c *and* ι *fixed,* Γ *and* T^* *are nondecreasing in* Δ_c; B_1 *and the credibility condition do not depend on* Δ_c.

(d) Counter-threats against a transfer. *Before the United States decides, China announces a punishment* $r \in [0, \bar{r}]$ *of a transfer and a stake* $m_C \leq \bar{m}_C$. *Carrying out* r *after a transfer has ex post value* $-c_C(r)$ *to China, where* c_C *is continuous and increasing with* $c_C(0) = 0$*; the threat is credible if and only if* $c_C(r) \leq m_C$. *China gets* $H_C \geq 0$ *if Europe keeps control, and the ex post payoff of its punishment decision (*$-c_C(r)$ *if it punishes,* $-m_C$ *if it does not) if control passes. Let* $\Gamma(r)$ *be* Γ *with* $r^c = r$ *and* $r_{\max} = \max\{r \leq \bar{r} : c_C(r) \leq \bar{m}_C\}$. *In every subgame-perfect equilibrium:*

(i) if control is transferred, China does not punish: on the path $r^c = 0$, *the payment is* $[\Delta R_H - B_1 + \sigma]^+$ *and* $\Gamma(0) > 0$*;*

(ii) a credible punishment $r > 0$ *is announced only if it deters,* $\Gamma(r) \leq 0$*;*

(iii) if $H_C > 0$, *control is transferred if and only if* $\Gamma(r_{\max}) > 0$*; otherwise China deters with a credible* $r \geq v_c - \iota + \Delta R_U - \Delta R_H + B_1 - \sigma$ *whenever* $\Gamma(0) > 0$*;*

(iv) if $H_C = 0$ *and* $\Gamma(0) > 0 \geq \Gamma(r_{\max})$, *both the transfer and its deterrence are equilibrium outcomes.*

If punishing is self-rewarding ($c_C(r) < 0$ *for some* r*), (i) fails: China may let the transfer happen and punish it.*

When $v_c = 0$, the transfer is inefficient. The rents it avoids are transfers between Washington and Europe, while the sovereignty cost, China's punishment and the issuing cost are destroyed. Washington pays $\sigma + r^c + \iota$ in joint surplus to stop paying rents it would otherwise owe. That is the logic of preventive war (Fearon, 1995; Powell, 2006), with a coerced transfer of control in place of a war. Under complete information nothing breaks down, because the object that shapes future bargaining power can itself be transferred, which is close to the case in which bargaining over such an object is efficient in Fearon (1996), an unpublished manuscript. What our setting adds is the

ceiling on coercion. The conditional part of the bite cannot exceed Europe's current slack, so the patron must act while the holder is weak, and the holder's slack is what it spends. Benson and Smith (2023) find that the conditions that make prevention attractive also make it unlikely to be carried out. Our version is starker: the threat behind a successful demand is never carried out, so its cost matters only through its credibility.

Part (b) connects the demand to Europe's own choices. An investment in future bargaining power that Europe cannot commit to forgo invites the demand, as an observable nuclear programme invites prevention or a deal in Debs and Monteiro (2014), Bas and Coe (2016) and Spaniel (2019). The investment raises the rents a transfer saves Washington, and it also creates a surplus that the two can share, because a transfer spares Europe an investment cost it would sink only to secure rents. The second effect permits a purchase with no bite at all. Take $\delta = 1$, $Q_1 = 1$, $Q_2 = -\frac{1}{2}$, $\bar{I} = 3$, $K = 2$, $\Delta_c = \infty$, $\sigma = 1$ and $r^c = v_c = \iota = 0$, with no threat. Europe would invest, so Washington faces a period-2 rent of 2.5 on top of the period-1 rent of 1. It buys control for $T^* = 2.5$ and gains $\Gamma = 1$. Advice that Europe should rearm to raise its premium therefore carries a cost that the static model misses: rearmament Europe cannot commit to forgo can prompt an offer for the chokepoint itself.

Part (c) answers the question of what makes a transfer bind. A transfer that Europe could partly undo removes only $\min\{[Q_t]^+, \Delta_c\}$ of the rent at each date, so the commitment problem returns in period 2 whenever $Q_2 - \Delta_c > 0$. Washington then gains less and pays less. Harstad (2016) reaches a similar conclusion about conservation: leasing beats buying when the seller's consumption value and the buyer's cost of protecting what it owns are large. The arrangements listed in section 6.1 differ mainly in that cost. A demand to relocate production is the high-Δ_c end. Washington floated one for Taiwan in 2025, when the Commerce Secretary proposed that half of the chips the United States uses be made in the United States, and Taipei said it would not agree (Younger, 2025; Focus Taiwan , CNA); section 8.4 compares the cases.

Part (d) concerns China. Its threat to punish a transfer is never carried out on the equilibrium path. If control passes, a credible punishment would cost China something ex post and would have deterred nothing, so China does better by announcing none. The threat therefore protects European ownership only by deterring the demand, which requires that China can stake enough to make r credible and that $\Gamma(r_{\max}) \leq 0$. Sovereignty costs work the same way. A Netherlands that would pay a high price for surrendering ASML is harder to coerce into doing so, which makes the foundation, Vifo screening and the Foreign Subsidies Regulation part of Europe's protection against a demand.

6.3 Windows

Proposition 5 allows one ultimatum and a one-period cut. We now allow both at several dates.

Lemma 2 (Spent and self-limiting cuts)**.** *Let dates $s = 1, \ldots, N$ have premia Q_s, slack $\Lambda_s = [-Q_s]^+$ and discount factor δ. A cut over a set S of dates withdraws at each $s \in S$, before the date-s contest, a bundle with unconditional part u_s, conditional part d_s and direct ex post cost $g_s \geq 0$ to the United*

States. Suppose that at every $s \in S$ the United States still retains Europe after the cut ($Q_s + d_s < V_s$ and $Q_s + d_s \leq \bar{m}_U$), and that the cut does not change the premium at any other date. Then:

(i) the bite of the cut, Europe's loss net of the rents the United States must then pay, is $\sum_{s \in S} \delta^{s-1}(u_s + \min\{d_s, \Lambda_s\})$;

(ii) the cost to the United States of carrying it out is $\sum_{s \in S} \delta^{s-1} e_s$ with $e_s = g_s + [Q_s + d_s]^+ - [Q_s]^+$; with a separate stake at each date, the date-s withdrawal is carried out if and only if $e_s \leq m_s \leq \bar{m}_U$;

(iii) extending a cut into a date with $Q_s \geq 0$ adds $\delta^{s-1} u_s$ to its bite and $\delta^{s-1}(g_s + d_s)$ to its cost to the United States. A purely conditional instrument ($u_s = 0$) extended into such a date adds nothing to the bite.

Both scope conditions matter. If a cut drives Europe out at some date, Europe's loss exceeds the formula. If Europe responds to a cut it learns of in advance, for instance with the autonomy investment of section 6.1, the loss falls short of it. Part (iii) governs threats of permanent exclusion. Whether such a threat is credible depends on how much Washington can stake, but a longer cut adds no bite once Europe's future premium is positive. At those dates a conditional cut hands Europe exit as its alternative and Washington pays the difference as rent.

Corollary 4 (Windows)**.** *Consider N dates with $v_c = 0$ and $\Delta_c = \infty$. At any date t at which Europe still holds control, the United States may issue, at cost $\iota > 0$, an ultimatum: transfer control for all remaining dates against a payment $T \geq 0$, or face a cut at date t only, with bite $b_t = u_t + \min\{d_t, \Lambda_t\}$ and execution cost $e_t = g_t + [Q_t + d_t]^+ - [Q_t]^+$. Call t* eligible *if $e_t \leq \bar{m}_U$, $b_t > \sigma + r^c$, and $Q_{t'} > 0$ for some $t' \geq t$. There is $\bar{\iota} > 0$ such that for every $\iota \in (0, \bar{\iota})$, the United States issues an ultimatum on the equilibrium path if and only if some date is eligible. Every ultimatum issued is accepted, and the first one comes no later than the last eligible date (possibly at a date that is not eligible). The same holds at $\iota = 0$ when the United States issues only if it strictly gains. If $u_t = 0$ at every date, the condition $t' \geq t$ can be replaced by $t' > t$.*

Figure 4 draws the two windows. We say that Washington's threats still bite at date t when the cut is credible and $b_t > \sigma + r^c$. In panel (a) that holds through $t = 5$, and the premium turns positive at $t = 9$. Washington acts at or before $t = 5$, long before Europe's window opens. It can act earlier still: a Europe that refuses remains exposed to a later demand, so an ultimatum at a date where the bite alone would not justify one can pay. Of the 20,199 simulated games with an ultimatum in appendix B, 1,202 issued it at a date that was not itself eligible. In panel (b) the premium also rises, toward -0.4, but never crosses zero. With $v_c = 0$ Washington has nothing to prevent. With $v_c > \iota$ it would demand control anyway whenever the bite covered the deadweight cost net of $v_c - \iota$, as Corollary 3(a) shows for two periods.

The timing differs from the preventive-war models. In Powell (2006) war requires a shift in power that is large and rapid relative to what the declining state can be offered within the period. Krainin (2017) shows that slow but persistent shifts can also produce preventive war, possibly after

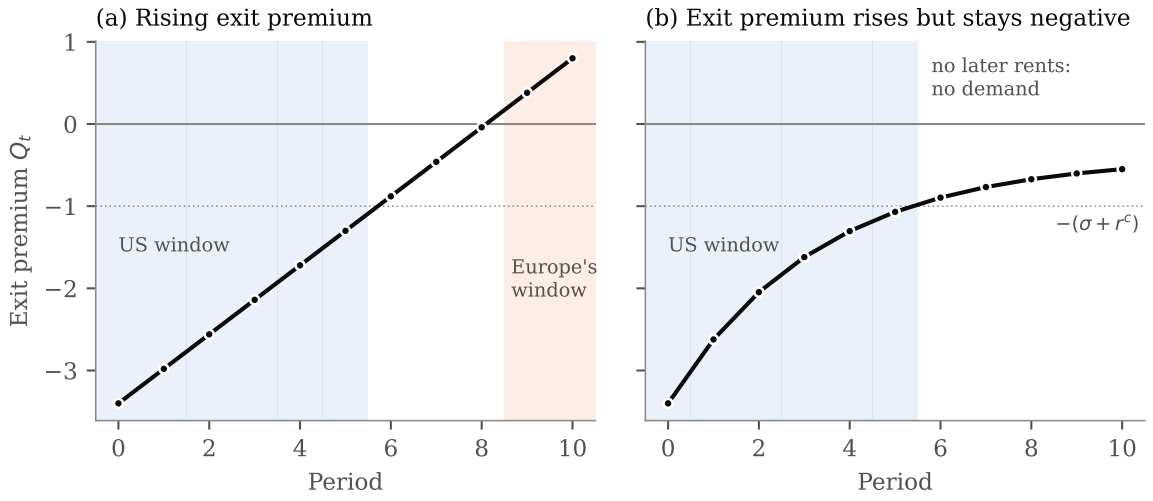


Figure 4: Windows for preventive control (illustrative units). Panel (a): $Q_t = -3.4 + 0.42\,t$. Panel (b): $Q_t = -3.4 + 3(1 - e^{-0.3t})$. In both, $\sigma + r^c = 1$ and Washington holds a conditional instrument with $d_t = 2$ and no unconditional part, so the condition $t' > t$ of Corollary 4 applies; the cut is assumed credible at every date. Blue shading marks dates at which the threat bites by more than $\sigma + r^c$, which here requires $Q_t < -1$. Orange shading marks Europe's window, $Q_t > 0$. In panel (a) Washington's window closes before Europe's opens, and Washington demands control no later than the last blue date. In panel (b) the premium rises but stays negative, there are no rents to remove, and no demand is made.

a delay. Here, under complete information, the speed of the shift affects when Washington acts and what it pays, and not whether it acts; what matters is whether a date at which the threat still bites precedes a date with positive premium. The difference comes from treating control as a commitment device. Transferring it removes every future rent at once, so the patron need not offer within one period an amount that matches the whole future shift. That holds only when $\Delta_c = \infty$. With finite enforceability a transfer removes at most Δ_c of rent per date, a shift far above Δ_c leaves rents that no transfer can reach, and size and speed matter again, much as in Powell's account. Bas and Coe (2016) find that an observer that monitors imperfectly waits until a shift looks imminent. Our patron, under complete information, acts before its own window closes, which may be long before Europe's opens.

The windows also make the direction of drift decisive. A demand requires that Europe's premium be expected to rise. Rearmament and rationing push it up. Chinese indigenization and stockpiling push it down by lowering W, and with them the US motive for a preventive demand. Which effect dominates is an empirical question, and section 8.1 states what each answer implies.

6.4 Promises or threats

Europe could try to remove the commitment problem itself. Allied or Europeanized export controls, or controls entrenched in treaty, would bind Europe against exit without transferring control.

Binding raises Europe's slack and with it the bite of Washington's threats.

Remark 1 (Self-binding and dual key). Before period 1, Europe can adopt a binding rule (Europeanized or treaty-entrenched controls) that lowers its premium at both dates by $\Delta_b \in (0, \infty]$ at cost σ^b, drawing a Chinese punishment r^b; Europe keeps the key. The game of Proposition 5 is then played with premia $Q_t - \Delta_b$, and the bite of the US threat is computed at the new slack.

(a) With $\Delta_b = \infty$, control removes no rents, and the bite becomes $B_1^b = u_1 + d_1$ if $g_1 \leq \bar{m}_U$ (and 0 otherwise). The United States then issues if and only if $\Gamma^b = v_c - \iota + \min\{0,\, B_1^b - \sigma - r^c\} > 0$, so binding pre-empts the ultimatum if and only if $v_c \leq \iota$ or $v_c - \iota + B_1^b \leq \sigma + r^c$. If it does not, the United States demands control and pays $\left[\sigma + r^c - B_1^b\right]^+$.

(b) Let $Q_1 < 0$, $\Delta_c = \infty$, no investment and $\Delta_b = \infty$. Europe's payoff is $-\sigma^b - r^b - \mathbf{1}\{\Gamma^b > 0\} \min\{\sigma + r^c, B_1^b\}$ if it binds, and $\max\{\delta\,[Q_2]^+ - B_1,\, -\sigma - r^c\}$ if it does not bind and $\Gamma > 0$, or $\delta\,[Q_2]^+$ if it does not bind and $\Gamma \leq 0$. In particular, if the United States would issue the ultimatum without binding and binding pre-empts it, Europe strictly prefers to bind if and only if $-\sigma^b - r^b > \max\{\delta\,[Q_2]^+ - B_1,\, -\sigma - r^c\}$. If the United States would not issue without binding, Europe never strictly prefers to bind.

(c) A partial binding ($\Delta_b < \infty$) lowers the rents control would remove but raises the slack, hence the bite, so it can make Γ positive. Example: $\delta = 1$, $Q_1 = -1$, $Q_2 = 10$, $u_1 = 0$, $d_1 = 5$, $g_1 = 0$, $\sigma = 1.5$, $r^c = v_c = \iota = 0$. Without binding $\Gamma = \min\{10,\, 1 - 1.5\} < 0$. With $\Delta_b = 3$, $\Gamma = \min\{7,\, 4 - 1.5\} = 2.5$ and the United States demands control at $T^* = 4.5$.

(d) Dual-key licensing is a partial control transfer with enforceability $\Delta_d < \infty$, sovereignty cost $\sigma_d < \sigma$, value v_d and Chinese punishment r^d. If the United States can demand either arrangement, it demands the one with the larger of Γ_{full} and Γ_{dual} (each given by Proposition 5 with that arrangement's parameters), provided it is positive. Because $\sigma_d < \sigma$, the United States can obtain a dual key when the bite is too small to obtain full control.

Binding pre-empts a demand only if Washington values control little for its own sake. Full binding gives Europe unlimited slack and so maximizes the bite, $B_1^b = u_1 + d_1$, which makes control cheaper to coerce for a patron with $v_c > \iota$. By part (b), binding pays Europe only when it pre-empts a demand that would otherwise come, and only when its own cost is below what the demand would cost Europe. Part (c) runs against intuition. A rule that binds Europe partway removes some of the rents Washington fears and adds to the slack its threats can draw on, and the second effect can turn a demand that would not pay into one that does.

Part (d) separates a dual key from binding. Under binding Europe holds the only key and lowers its own premium. Under a dual key Washington holds a veto over China-bound licences, which is a transfer of a control right over customers, priced by Proposition 5 with its own enforceability and sovereignty cost. The Dutch government's answer to the MATCH Act, that every country is responsible for its own export-control legislation (Sjoerdsma, 2026), suggests to us that σ_d in The Hague is not far below σ.

We read the European steps of 2024 and 2025 (section 2) as partial binding. In September 2024 The Hague took over licensing of the NXT:1970i and NXT:1980i from Washington, a change that ASML said ensures that the Dutch government "is the sole licensing authority" (ASML Holding N.V., 2024). That step lowered Europe's premium and kept the key in Europe, the reverse of a control transfer. The December 2024 exclusion of exporters in the Netherlands and other listed countries from the new SME licence requirements rested on the authority of those governments to "control key SME items of concern" (Bureau of Industry and Security, 2024), and section 8.3 codes it as a reward for self-binding. When Delegated Regulation 2025/2003 put the most advanced immersion tools on the EU list in November 2025, the licence decision stayed in The Hague (European Parliament and Council of the European Union, 2021; ASML Holding N.V., 2026a). On our reading the listing raised Δ_b, because removing the entry now takes an act at the Union level that the Netherlands cannot adopt alone. The August 2025 joint statement, in which the EU said it planned to adopt technology-security requirements in line with those of the United States and Washington said it would then facilitate AI chip exports (The White House, 2025d), is a statement of intent with small Δ_b. None of these steps comes close to $\Delta_b = \infty$. By part (c), they could have strengthened the US motive for a demand as easily as weakened it, depending on how far Washington expected Europe's premium to rise.

6.5 Private sovereignty costs and breakdown

Under complete information an ultimatum is either accepted or never issued. Breakdown requires that Washington be uncertain about what a transfer would cost Europe. We think that uncertainty is real. Since February 2026 the Dutch government has been a coalition of D66, CDA and VVD with minority support in parliament (NL Times, 2026a; Omroep Brabant, 2026), so the domestic price of a transfer would depend on whichever majority the cabinet could find.

Proposition 6 (Breakdown)**.** *Assume the conditions of Proposition 5 with $\Delta_c = \infty$, no investment, $Q_1 < 0$ and $e \le \bar{m}_U$, and let σ be Europe's private information, distributed with CDF F on $[0, \bar{\sigma}]$, density $f > 0$ on $(0, \bar{\sigma})$, and F log-concave. Let $R = \delta\,[Q_2]^+$, $\tilde{b} = B_1 - r^c$ and $\Pi(s; A, x) = F(s)(A - s) - (1 - F(s))x$. For $A + x > 0$ let $s^*(A, x)$ be the unique solution in $(0, \bar{\sigma})$ of $A + x - s = F(s)/f(s)$ if one exists and $\bar{\sigma}$ otherwise; let $s^*(A, x) = 0$ if $A + x \le 0$.*

(a) Threat-backed ultimatum. *With payment T, Europe accepts if and only if $\sigma \le T + \tilde{b} - R$. The optimal cutoff for the United States is $\hat{s} = \max\{\tilde{b} - R, \min\{\bar{\sigma}, s^*\}\}$ with $s^* = s^*(\tilde{b} + v_c, e)$, its payment is $\hat{s} - \tilde{b} + R \ge 0$, and its gain is $\Pi_B = \Pi(\hat{s}; \tilde{b} + v_c, e) - \iota$. Conditional on this ultimatum, Europe refuses and the threat is carried out with probability $P_b = 1 - F(\hat{s})$. For uniform F, $s^* = (\tilde{b} + v_c + e)/2$.*

(b) Purchase offer. *If the United States instead offers to buy control without staking the threat, Europe accepts if and only if $\sigma \le T - r^c - R$, the optimal cutoff is $s_0 = s^*(v_c - r^c, 0)$, and the gain is $\Pi_0 = \Pi(s_0; v_c - r^c, 0) - \iota$. If $v_c \le r^c$, then $\Pi_0 \le -\iota$.*

(c) The United States makes an offer if and only if $\max\{\Pi_B, \Pi_0\} > 0$ and backs it with the

threat if and only if $\Pi_B \geq \Pi_0$. *If* $v_c \leq r^c$ *(in particular if* $v_c = 0$*), it issues if and only if* $F(\hat{s})(\tilde{b} + v_c - \hat{s}) - (1 - F(\hat{s}))e > \iota$, *and every offer is threat-backed.*

(d) Across parameter values at which the United States issues a threat-backed ultimatum, P_b *is nonincreasing in* B_1, e *and* v_c, *nondecreasing in* r^c *and* R, *and independent of* ι. *These comparative statics hold for every continuous* F *(not necessarily log-concave) when* $\hat{s}$ *is the largest, or the smallest, maximizer.*

(e) If $v_c > r^c$ *and the United States chooses between (a) and (b), then across parameter values at which it makes an offer the refusal probability is nonincreasing in* B_1 *and* v_c, *and the breakdown probability (refusal followed by the executed threat) is nonincreasing in* e *and nondecreasing in* R. *The refusal probability is nondecreasing in* r^c *on any range where the threat-backed payment is positive. The other directions can fail: the refusal probability can rise with* e *and fall with* R, *and the breakdown probability can rise with* B_1 *and* v_c *and fall with* r^c.

Breakdown is the outcome in which Europe refuses, loses what the threat withdraws, and keeps the chokepoint. Given a threat-backed ultimatum, it is more likely when China's punishment of a transfer is credible and large, when Europe's future premium is high, and when the threat is cheap for Washington to carry out. The last is the usual risk-return trade-off (Fearon, 1995; Powell, 1999): a patron that loses little from a refusal makes a tougher offer and is refused more often. A larger bite makes refusal costlier for Europe, and a higher value of control makes Washington willing to pay more; both lower the chance of refusal. When $v_c > r^c$, a patron whose threat is expensive to carry out may prefer to offer money alone, since a refused purchase offer costs it nothing, and a rise in the bite can then move it to the threat and raise the probability of breakdown (part (e)). Unlike the complete-information case, a credible Chinese punishment is now carried out on the path with positive probability, because the types that accept are punished.

The result rests on a one-shot ultimatum, which is a commitment assumption. If Washington could not commit to walk away after a refusal and could renew its offer quickly, it would compete with its own later offers. With one-sided private information and offers made by the uninformed side, that side loses most of its power to screen as the time between offers shrinks (Fudenberg et al., 1985; Gul et al., 1986). A patron that cannot commit to a single demand would extract less, and we expect delay to replace some of the refusals that Proposition 6 counts as breakdown. Each refusal would also execute a cut, which brings the repeated game closer to a war of attrition; we do not solve it. The probabilities in Proposition 6 are best read as those of a patron that can commit to one public demand, a commitment that a staked promise to make no further offer could supply (section 5.1).

7 Who Decides, and the Power to Hurt

So far Europe has been one actor. In practice the Dutch licensing authority decides whether ASML may ship to China, the firm decides whether it will, and the European Union decides whether

ASML's tools can be turned against the United States (section 2). This section separates the three. It then gives Europe a way to impose costs on Washington while the two bargain, which is where Washington's dependence on ASML does its most direct work.

7.1 The gatekeeper

Exit toward China is a Dutch decision. Under Article 12(2) of the dual-use regulation the licence is granted by the competent authority of the member state where the exporter is established (European Parliament and Council of the European Union, 2021), and other member states may only request consultations that do not bind. The Commission has no licensing power and no veto. Let $Q_{NL} = \omega_{NL} - c_{NL}$ be the gatekeeper's own exit premium: the value to The Hague of China's credible offer, net of the credible costs of exit that fall on the Netherlands. By Lemma 1, Washington keeps the Netherlands aligned at price $[Q_{NL}]^+$ when $Q_{NL} < V$ and $Q_{NL} \leq \bar{m}_U$.

Remark 2 (Gatekeeper). Let $x_h \geq 0$ be the value to the Netherlands of a benefit that the United States provides to a loyal Netherlands in the baseline and credibly withdraws if the Netherlands licenses exit. Then the gatekeeper's premium becomes $Q'_{NL} = Q_{NL} - x_h$. If $Q'_{NL} < V$ and $Q'_{NL} \leq \bar{m}_U$, the United States' payment falls from $[Q_{NL}]^+$ to $[Q_{NL} - x_h]^+$, that is, by $\min\{x_h, [Q_{NL}]^+\}$. The gatekeeper's slack rises from $[-Q_{NL}]^+$ to $[x_h - Q_{NL}]^+$, that is, by $\left[x_h - [Q_{NL}]^+\right]^+$. If $Q_{NL} > \min\{V, \bar{m}_U\}$, the benefit restores alignment if and only if $Q_{NL} - x_h \leq \bar{m}_U$ (and $Q_{NL} - x_h < V$). A benefit paid as a promised increment above the baseline lowers no payment, since the total outlay is then $\max\{x_h, Q_{NL}\} \geq [Q_{NL}]^+$.

The remark is divide and rule. A benefit aimed at the gatekeeper alone, whose withdrawal on exit is credible (section 4.2), buys alignment at a discount, and when the gatekeeper already has slack every unit of it adds to the stock Washington can spend on other demands (Proposition 2). Only provision that Washington would make to a loyal Netherlands anyway does this work. A side payment promised on top of the baseline is a transfer like any other and saves Washington nothing.

Partners inside Europe affect the gatekeeper's premium only through transfers. Germany, France or the Commission cannot refuse a Dutch licence, so solidarity reaches The Hague only as compensation, coordinated retaliation or cover against Chinese counter-coercion, each of which moves ω_{NL} or c_{NL}. Washington can tier member states directly. The AI Diffusion Rule of January 2025 placed ten EU members, the Netherlands among them, in its favoured group and the other seventeen in the middle tier (Bureau of Industry and Security, 2025). By our computation from the 2026 Council population figures (Council of the European Union, 2025), the ten held 70.46 percent of EU population, too few members to form a qualified majority by themselves, while the seventeen held 29.54 percent and enough members to block by number. The rule was rescinded before it took effect (section 2), so tiering is a latent instrument, but it shows how one rule can reward the gatekeeper and split the Council at once (section 7.3).

7.2 The firm between two jurisdictions

A licence permits a shipment and orders nothing. ASML decides whether to ship, and it answers to two sovereigns besides its own.

Remark 3 (Firm between jurisdictions). Suppose the Netherlands licenses exit. The firm earns y from its Chinese business and faces the credible US penalty $f^F = \chi_f f$ if it ships. If it complies with US measures instead, it faces a Chinese penalty $p_C \geq 0$ for that compliance and a penalty $p_B \geq 0$ under the EU Blocking Statute, which is positive only if the US measures are listed in the statute's annex. The firm ships if and only if $y + p_C + p_B \geq f^F$ (ties ship). If it would not ship, exit requires the state to compel it at cost $\zeta \geq 0$, where ζ excludes the firm's own net loss (already counted in ω and c through y and d_f). Europe's effective exit premium is $Q - \zeta\,\mathbf{1}\{y + p_C + p_B < f^F\}$, and Europe compels the firm only if $Q - \zeta > 0$.

Since April 2026, p_C has a legal basis. Decree 835 lets Beijing put those who implement foreign extraterritorial measures on a Malicious Entity List, issue prohibition orders, and expose them to private damages suits and fines (Li et al., 2026). ASML runs a repair centre in Beijing and services a large installed base in China (section 2), so its staff in China who implemented a US servicing ban would be exposed to these tools (ASML Holding N.V., 2026a). A higher p_C makes a licensed firm more willing to ship, which shrinks the range in which compulsion is needed and weakly raises Europe's effective premium. The decree leaves the credibility of Washington's jurisdiction over China-bound flows unchanged, since that turns on Washington's own ex post cost. What it raises is the penalty the firm must expect before it prefers compliance. For a US ban on servicing in China the same penalty enters Proposition 3(a) as a counter-threat r_k only if Europe internalizes the firm's loss.

Europe's own instrument for this margin is weak. The Blocking Statute bars EU persons from complying with listed foreign laws, denies recognition to foreign judgments based on them and allows damages claims (Council of the European Union, 1996). Its annex lists US sanctions on Cuba and Iran, updated in 2018 after Washington reimposed Iran sanctions (European Commission, 2018), and no US export-control law. So $p_B = 0$ today, and making it positive would take a further delegated act. Even then the penalty would be bounded. In *Bank Melli* the Court of Justice held that the prohibition applies without any US order, but that a national court may annul a termination made to comply only if annulment does not have disproportionate effects, weighing the probability and extent of the firm's economic loss (Court of Justice of the European Union, 2021). Enforcement runs through national courts and is bounded by proportionality toward firms with a large US business. We read this as keeping p_B small for ASML, whose US subsidiaries and staff (section 2) give it such a business.

In the Nexperia case the Dutch state used the Goods Availability Act to compel a firm and suspended the order seven weeks later, after China had blocked exports from Nexperia's Chinese plants; we read the suspension as evidence that ζ is high when China retaliates against the compulsion itself (section 8.3).

7.3 The Council

Turning ASML against Washington is collective, because EUV and the most advanced immersion tools ship to US fabs under the Union general export authorisation EU001 (section 2). Two routes could stop those shipments.

The Anti-Coercion Instrument has two stages (European Parliament and Council of the European Union, 2023). The Commission examines the case, normally within four months. The Council then determines by qualified majority, on a Commission proposal, that coercion exists; it acts within eight weeks, and normally within ten. Response measures are Commission implementing acts under the examination procedure, with a clause that bars adoption when the committee delivers no opinion. In the appeal committee, however, the Commission may adopt unless a qualified majority votes against (European Parliament and Council of the European Union, 2011), and the instrument's recital 85 asks it to avoid going against any predominant position among member states. The second route is a delegated act removing the United States from EU001 (European Parliament and Council of the European Union, 2021). Under the urgency procedure it enters into force at once and stands unless the European Parliament objects by a majority of its component members or the Council objects by a reinforced qualified majority of 20 members representing 65 percent of the population (European Union, 2016). Eight members, or members holding more than 35 percent of the population, keep the act in force. The Netherlands would then still have to deny the individual licences, so the delegated-act route hands the decision back to the gatekeeper of section 7.1.

Let q_i be member i's net gain from adopting an anti-US countermeasure. A qualified majority needs at least 15 of 27 members representing at least 65 percent of the population, and abstention counts against.

Remark 4 (Blocking minorities). Let $\mathcal{B}$ be the family of coalitions that block the Council's determination at the first stage: at least 13 members, or at least four members with more than 35 percent of the population. With enforceable vote-contingent payments and no rival bidder, the United States defeats the determination at cost $\Phi(0) = \min_{J \in \mathcal{B}} \sum_{i \in J} [q_i]^+$. Members with $q_i \leq 0$ join any blocking coalition at zero cost, so $\Phi(0)$ is the cheapest completion of $F = \{i : q_i \leq 0\}$ into a blocking coalition, and $\Phi(0) = 0$ if and only if $F \in \mathcal{B}$. The six frontline states (15.09 percent of the population) do not block. Thirteen eastern and Nordic members (26.77 percent) do, by number. Defeating a response measure in the appeal committee, or a delegated act by Council objection, needs a qualified majority against, which is itself in $\mathcal{B}$, so it costs at least $\Phi(0)$.

The cost function is the one-voter logic of Lemma 1 applied to a weighted body, as in Groseclose and Snyder (1996). We assume no one bids for adoption; with a counter-bidder that moves last, Washington would have to buy more than a minimal blocking coalition (Groseclose and Snyder, 1996; Banks, 2000). The remark puts the veto at the front. Every later Council stage of either route requires a coalition at least as large as a stage-one block, so a Washington that wants to stop a countermeasure buys its blocking minority before the determination. Table 2 gives the arithmetic for the coalitions that matter.

The table yields a prediction. Washington can make posture and enablers conditional member

Table 2: Blocking arithmetic in the Council, 2026 population weights

Coalition	Members	Population (%)	Blocks determination	Keeps delegated act in force
Frontline core: PL, LT, LV, EE, FI, RO	6	15.09	no	no
Core plus SE, DK	8	18.75	no	yes (members)
Bucharest Nine plus FI, SE, DK	12	25.91	no	yes
Core plus SE, DK, CZ, SK, HU, BG, HR	13	26.77	yes (members)	yes
the same with SI for HU	13	25.13	yes (members)	yes
the same with SI and GR for HU and HR	13	26.57	yes (members)	yes
Eight smallest members	8	3.18	no	yes (members)
DE, FR, IT	3	46.84	no	yes (population)

Note: A coalition blocks the determination (first stage of the Anti-Coercion Instrument) if it has at least 13 members, or at least four members with more than 35 percent of the population. It keeps a delegated act removing the United States from EU001 in force if it has at least 8 members or more than 35 percent. No coalition in the table can defeat a response measure in the appeal committee, which takes a qualified majority against. Population shares from Council Decision 2025/2427 (Council of the European Union, 2025); blocking status is our computation from the treaty rules, checked by enumeration (appendix B).

by member (section 4.3), so the members whose defence depends most on US forward presence have the lowest q_i and are the natural blockers. They are not enough. The frontline core falls short on both counts, and a population block without Germany, France, Italy and Spain needs at least twelve members. If all thirteen eastern and Nordic members in the fourth row had $q_i \leq 0$, Washington would defeat a determination for free. Hungary's support is doubtful, but Slovenia or Greece can replace it and the group still has thirteen members. We therefore expect opposition to an anti-US measure to start on the frontline and to block only once it spreads to central Europe and the Nordics, or once one large state joins.

Informal norms and procedural defaults pull the outcome away from the formal arithmetic, in opposite directions. Council practice is consensual. About four in five legislative acts passed with no recorded negative vote or abstention in 1994–2002 (Heisenberg, 2005), and Häge (2013) models that consensus as a by-product of negotiators organizing around blocking minorities before any vote is taken. Recorded votes also understate opposition, because a vote would disclose who lost (Novak, 2013). The data cover legislation before the Lisbon voting rules, so we treat them as indicative, but they suggest that a measure large states oppose rarely reaches a vote. Reversed defaults pull the other way. On 4 October 2024 the trade-defence committee voted on countervailing duties on battery electric vehicles from China: ten member states in favour, five against (Germany among them) and twelve abstaining (Agence France-Presse, 2024). The committee delivered no opinion, no qualified majority opposed, and the Commission adopted the duties on 29 October (European Commission, 2024a; European Parliament and Council of the European Union, 2011). That vote is not evidence

that an ACI measure could pass with a minority, since the ACI's no-opinion clause reverses this default at the committee stage. It shows what the Commission does when the procedure puts the burden on opponents, as the ACI's appeal stage and the delegated-act route both do.

The same arithmetic applies to the Council vote in *Europe 2031*. The seven states the scenario names as opposing or abstaining on its 2029 countermeasure (the Netherlands, Ireland, Poland, the three Baltic states and Italy) hold 28.06 percent of EU population (Juijn et al., 2026; Council of the European Union, 2025). They block neither by number nor by population. For the vote to fail, Germany, France or Spain would also have to withhold support, or six more members of any size. Appendix C sets this beside the scenario's other episodes.

7.4 Hold-up

Proposition 1 lets the status quo run unchanged while Europe and Washington bargain. A dispute over control of ASML or over Europe's access to frontier AI would not leave it unchanged. Washington could cut Europe's AI allocation while talks continue, and Europe could interrupt ASML's servicing of US-aligned fabs. Let disagreement carry flow costs $c_E(t)$ to Europe and $c_U(t)$ to the United States, and write $\hat{C}_i(t) = (r+\lambda)\int_0^\infty e^{-(r+\lambda)s} c_i(t+s)\,ds$ for the average discounted cost from date t on.

Proposition 7 (Pain and rupture)**.** *Consider the game of Proposition 1 with $r > 0$ and disagreement flow costs $c_E(t), c_U(t) \geq 0$ that are bounded and piecewise uniformly continuous, with finitely many discontinuities in every bounded interval. Let $M = Q + \beta(V-Q)$, $n(t) = \beta c_U(t) - (1-\beta)c_E(t)$, $N(t) = \beta\hat{C}_U(t) - (1-\beta)\hat{C}_E(t)$ and*

$$x^\circ(t) = \psi M + (1-\psi)N(t).$$

(i) For every $\Delta > 0$, Europe's Markov perfect equilibrium payoff is unique. In every round in which $\lambda > 0$ or $c_E + c_U > 0$, every Markov perfect equilibrium has immediate agreement. If $\lambda = 0$ and $c_E(t) + c_U(t) = 0$ on an interval, there are also payoff-equivalent equilibria with delay during that interval.

(ii) As $\Delta \to 0$, Europe's equilibrium payoff converges to

$$\begin{aligned} P^* &= \sup_{\tau\in[0,\infty]} \Big\{ \int_0^\tau e^{-(r+\lambda)s}\big(\lambda M + r\,n(s)\big)\,ds + e^{-(r+\lambda)\tau} Q \Big\} \\ &= x^\circ(0) + \left[\sup_{t\geq 0} e^{-(r+\lambda)t}\big(Q - x^\circ(t)\big)\right]^+. \end{aligned}$$

(iii) $P^* = x^\circ(0) = \psi\big[Q + \beta(V-Q)\big] + (1-\psi)\big[\beta\hat{C}_U(0) - (1-\beta)\hat{C}_E(0)\big]$ if and only if

$$Q \leq \inf_{t\geq 0} x^\circ(t) \quad\Longleftrightarrow\quad Q \leq \bar{Q} \equiv \inf_{t\geq 0} \frac{\lambda\beta V + r\,N(t)}{r+\lambda\beta}.$$

Under this condition the limiting equilibrium value at every date t is $x^\circ(t)$, so the exit option is never used. The condition $Q \le x^\circ(0)$ alone is not sufficient. With constant costs, $P^* = \max\{Q, x^\circ\}$.

When exit does not bind, Europe's transfer is a weighted average of two terms. With weight ψ it gets the rupture value M of Proposition 1. With weight $1-\psi$ it gets the hold-up term N: Washington's average pain weighted by Europe's agenda power, minus Europe's own pain weighted by Washington's. With zero costs the proposition returns Proposition 1. The hold-up term needs no rupture risk. At $\lambda = 0$, Proposition 1 gives $[Q]^+$ whatever V is, while here $P^* = \max\{Q, \beta\hat{C}_U - (1-\beta)\hat{C}_E\}$ with constant costs. Hold-up is the fifth channel of Corollary 1(e).

Keohane and Nye (1977) distinguished sensitivity, how quickly and how sharply a cutoff hurts before anyone adjusts, from vulnerability, the cost that remains after adjustment. The proposition prices both. A buffer of stockpiled parts and transferred know-how delays the onset of Washington's cost and so shrinks $\hat{C}_U$; that is sensitivity. The flow cost once the buffer is spent, net of whatever servicing Washington can compel, is vulnerability. The next result makes the split explicit for servicing. Let baseline disagreement flows be c_E^0 and c_U^0, which may include Washington's own cost of cutting a loyal Europe's AI access from date zero, and let x_0° be $x^\circ(0)$ under these baseline costs.

Corollary 5 (Interruption)**.** *Let Europe choose, at the start of each round and before the proposer is drawn, whether servicing of US-aligned fabs is interrupted. Interruption costs Europe a flow ℓ, and costs the United States a flow $(1-h)c_S$ once its buffer is exhausted, where h is the share of that servicing Washington can compel. The buffer is a stock of t_U units of interruption time, depleted only in rounds with interruption. Let $D = \beta(1-h)c_S e^{-(r+\lambda)t_U} - (1-\beta)\ell$. In the frequent-offer limit:*

(i) If exit never binds without interruption (Proposition 7(iii) with the baseline costs), allowing interruption raises P^* by $(1-\psi)\,[D]^+$. Interruption raises P^* if and only if $\beta(1-h)c_S e^{-(r+\lambda)t_U} > (1-\beta)\ell$, that is, if and only if $t_U < \bar{t} \equiv \ln[\beta(1-h)c_S/((1-\beta)\ell)]/(r+\lambda)$ (with $\bar{t} = \infty$ if $\ell = 0 < \beta(1-h)c_S$).

(ii) With constant baseline costs and the exit option, $P^* = \max\{Q,\, x_0^\circ,\, x_0^\circ + (1-\psi)D\}$. Interruption raises P^* if and only if $(1-\psi)D > [Q - x_0^\circ]^+$. If exit binds without interruption, the condition of part (i) is necessary but not sufficient.

(iii) The decision is time consistent. At stock b, Europe interrupts if and only if $\beta(1-h)c_S e^{-(r+\lambda)b} > (1-\beta)\ell$. Along an interruption path b falls, so the condition keeps holding. If it fails at $t = 0$, the stock never moves and it fails in every round.

The result depends on the stock buffer and on the timing. (a) If the buffer runs out at calendar date t_U whatever Europe does, Europe never interrupts before t_U; interruption raises P^* by $(1-\psi)e^{-(r+\lambda)t_U}\,[\beta(1-h)c_S - (1-\beta)\ell]^+$, so it helps if and only if $\beta(1-h)c_S > (1-\beta)\ell$, for any finite t_U. (b) If Europe chooses interruption after an offer is rejected and $\ell > 0$, the unique Markov equilibrium has no interruption and P^* is unchanged. (c) If Europe chooses after the proposer

is drawn but before offers, it interrupts in the rounds in which it proposes, at no cost on the equilibrium path; interruption then raises P^* whenever $\beta(1-h)c_S > 0$ and $t_U < \infty$, by at least $(1-\psi)\beta(1-h)c_S e^{-(r+\lambda)t_U/\beta}$, whatever the value of ℓ.

In D, vulnerability is the comparison of $(1-h)c_S$ with ℓ, and sensitivity is the discount $e^{-(r+\lambda)t_U}$, which rupture risk and impatience make steep. Variant (b) is the substantive restriction. Interruption helps only if a rule in place before offers decides it. Decided after a rejection, it is never used, because pausing it for one round saves Europe ℓ and changes nothing that determines the outcome. A government that wants this option must commit in advance to interrupt for as long as the dispute lasts. When one side chooses its disagreement action every round, complete-information bargaining generally has many subgame-perfect equilibria, some with costly delay (Fernandez and Glazer, 1991; Haller and Holden, 1990; Busch and Wen, 1995). We characterize Markov perfect equilibria, in which strategies depend only on calendar time, the buffer stock and the current offer. Proofs are in appendix A.

Hold-up is nested too. ASML's US subsidiaries and the people they employ in the United States are US persons (Bureau of Industry and Security, 2026; ASML Holding N.V., 2026c), and ASML keeps offices next to US customer fabs (ASML, 2026). No source we found names the entity that employs US field-service staff, but on this structure Washington could compel most servicing of US-sited fabs, so h is close to one there. The scenario's 2029 measure, which restricts ASML's exports and servicing to US fabs in Arizona (Juijn et al., 2026), is a game of this kind. ASML's entire US-sited revenue in 2025 was EUR 4.09 billion (ASML Holding N.V., 2026a), so Europe's own cost ℓ would be at most about EUR 0.34 billion a month, and its means of hurting those fabs would be limited to new optics, modules and software. A broad interruption covering the EUV fleet in Taiwan and Korea is a different game. There h depends on Taipei and Seoul as well as on Washington, and Europe's own cost is larger, about EUR 1.9 billion a month of ASML revenue on a 2025 basis (ASML Holding N.V., 2026a). The flows that cannot be commandeered abroad include Zeiss optics and Trumpf drive lasers (ASML Holding N.V., 2026a; Varas et al., 2021), so on our reading Germany would have to take part in any credible interruption.

The other parameters are poorly measured. The US buffer t_U runs from about one month to two years or more. The short end takes at face value an anonymously sourced report that EUV tools need such frequent upkeep that "without ASML's spare parts they quickly stop working" (Bloomberg News, 2024), together with a collector lifetime of over 100 billion pulses demonstrated in 2017 (Fomenkov et al., 2017). The long end allows for stockpiled consumables and for parts and repair capacity seized locally. Because collector reflectivity declines by fractions of a percent per billion pulses (Fomenkov et al., 2017), $c_U(t)$ should rise smoothly, which Proposition 7 allows. For the US cost of a broad interruption, TSMC's 2025 filing implies about US$3.9 to 4.8 billion a month of wafer revenue from EUV nodes and US-headquartered customers (Taiwan Semiconductor Manufacturing Company Limited, 2026). With a downstream multiplier of 5 to 12, the upper end from the one-year Taiwan disruption scenario of Varas et al. (2021), and a post-buffer output loss that we assume lies between 10 and 70 percent, c_S runs from about US$2 billion to US$40 billion a

month, with a central value near US$13 billion. These are revenues at risk, not welfare losses, and they omit Korean memory. With $\beta = 1/2$ and central costs, interruption pays at $t_U = 0$ only if h is below about 0.85. With an illustrative $r + \lambda$ of 0.55 a year, $\bar{t}$ is about 26 months at $h = 0.5$ and about 15 months at $h = 0.7$. The answer turns more on h than on t_U, and h is the parameter with the least evidence behind it. Since $\bar{t}$ falls as h rises, ASML's own plan to have 50 to 70 percent of service parts repaired locally (ASML Holding N.V., 2026a) erodes Europe's hold-up option.

Servicing is not Europe's only hold-up instrument. In March 2026 Spain refused US use of jointly operated bases for the Iran war and closed its airspace to the aircraft involved (Roque, 2026). Denial of access raises $c_U(t)$ during a dispute, and Washington cannot compel access to another state's territory the way it can compel US persons, so on our reading its h is near zero. It is decided by one member state, outside the Council arithmetic, and its cost to that state includes the conditional part of US posture (section 5.3). The timing condition applies as well: by variant (b), a denial improvised after a rejected offer would never be carried out.

Washington's dependence on ASML matters most here. In Proposition 1, V is the loss from China gaining the tools, and it reaches Europe's transfer only through agenda power under rupture risk and through Q. In Proposition 7 the relevant loss is c_S, what Washington's own fabs lose without ASML, and it raises Europe's transfer without any rupture risk. The scenario's claim that Washington could outlast Europe through its Taiwanese partners is a claim that t_U exceeds $\bar{t}$ (appendix C). For a measure confined to US soil the claim is plausible, since h is near one there. For a broad interruption it depends on h, which the record does not pin down.

8 The Record as a Plausibility Probe

The parameters of the model are not measured, and a negative exit premium can rationalize almost any act of compliance after the fact. Compliance alone also cannot separate coercion from shared interest, because successful coercion ends at the threat stage (Drezner, 2003). We therefore ask what the model implies that its rivals do not, fix rules for coding what Washington gave and threatened, and read the Dutch record and nine comparison cases with those rules. None of this is a test. Section 8.5 describes the designs that would be.

8.1 Observable implications

Table 3 sets seven implications of the model beside five rival accounts of allied compliance. Under shared threat perception the Netherlands aligns because it shares Washington's view of China, which in the model is a large intrinsic cost of exit η. In Lake (2009) compliance is owed to a patron whose authority rests on protection already provided and partly on legitimacy; in Morrow (1991) the protégé trades autonomy for security, so protection is itself the payment. Drezner (1999) argues that adversaries, who expect future conflict, rarely win concessions by coercion, while allies win them when they try. A cost-minimizing coercer polices each demand with the cheapest credible instrument and sizes sanctions so that punishing a small offence does not erode deterrence of a large

one (Stigler, 1970; Shavell, 1992).

The threshold separates the model from hierarchy and the autonomy–security trade-off only through holders with a positive premium, which should be paid, as Kyrgyzstan was for Manas when a Russian aid package raised the rent Washington paid (Cooley, 2012). The second and third rows discriminate most. In both, what Washington can obtain shrinks as China's credible offer rises, whatever its instruments cost. Cost minimization implies no such change, and none of the alliance accounts lets a rival's offer enter the price of compliance. In the fourth and fifth rows, hierarchy implies a patron that shields its subordinate, while our model implies that a Chinese counter-threat changes what Washington asks for and that a stronger Chinese bid makes the guarantee more conditional. The frontline row discriminates least: every account in which security dependence produces deference shares it.

8.2 Coding transfers, threats and leverage

Reading the record through Proposition 1 requires a line between a payment and a threat that was never carried out. We draw it at Washington's ex post optimal provision to an aligned ally, the baseline $x_j(A)$ of each instrument (section 4.3).

A *transfer* is a benefit conditional on alignment that exceeds what Washington would provide to an aligned ally if it optimized ex post. Only transfers enter P^*. A *threat left unexercised* is either a baseline benefit that Washington would withdraw after exit or a sanction it could impose and does not. It enters the exit cost c as a conditional part d_j, however valuable it is to the holder. Remark 2 draws the same line: a benefit Washington provides to a loyal Netherlands anyway, and withdraws on exit, lowers the gatekeeper's premium, while a promise above the baseline is a payment like any other.

The December 2024 exclusion of the Netherlands and the other countries on Supplement No. 4 from the new SME licence requirements (section 2) is the case that matters. BIS conditioned it on the listed governments' own capacity: they "have the authority to control key SME items of concern" (Bureau of Industry and Security, 2024). The exclusion lasts as long as The Hague controls the items itself, and BIS can withdraw it by amending the rule. We therefore code it as a threat left unexercised, conditioned on Europe's own capacity to control, or equivalently as a reward for self-binding (remark 1). It is not a transfer. With this coding the exclusion is consistent with a nonpositive premium: it adds to c and lowers Q, and under Corollary 1 a holder with $Q \leq 0$ and little risk of rupture receives no transfer. The same rule codes the "more favorable treatment" that the National Security Strategy offers countries aligning their export controls as conditionality (section 5.3).

We code the instrument a US demand invokes in three classes. *Flow jurisdiction* reaches China-bound flows directly (de minimis and foreign direct product rules, entity listings, licences for US-origin items); it leaves Q unchanged and needs no concession from the holder (Proposition 3(b)). *Conditional linkage* threatens to withdraw from an aligned holder something in another domain that an exited holder would lose anyway, such as posture, AI access or trade terms. An *unconditional*

instrument hurts the holder whatever it does with the chokepoint. Chinese responses are coded as counter-threats if announced before the holder concedes and as executed if carried out.

Transfers are a narrow measure of leverage, on which the Netherlands has little to show almost by construction. Four broader measures also have meaning in the model: *delay* between a US move and the holder's step, the *scope* of the holder's controls relative to the demand, *retained jurisdiction* over licensing, and *tolerated hedges*, flows Washington could reach but leaves alone. None is a payment. On our reading they measure how much of the holder's slack Washington chose to spend: by Corollary 2 a patron leaves a hedge alone when its value per unit of the holder's cost is low, and retained jurisdiction is self-binding with the key kept in Europe (remark 1).

8.3 The Dutch record and Nexperia

Alignment without a price. Since 2019 The Hague has aligned in steps, and from 2023 each step followed a US move (section 2). Under our rules it received no transfer: its one benefit, Supplement No. 4, is a threat left unexercised. If talks carry no risk of rupture, or Washington makes the offers, Corollary 1(b) gives $P^* \leq 0$ whenever $Q \leq 0$, and the record is consistent with a holder whose premium has been negative throughout, even though customers in China took between 26 and 36 percent of ASML's sales from 2023 to 2025 (section 2).

The negative premium needs no security term. On our reading, US flow jurisdiction over China-bound tools blocks most of what China would gain from exit, which makes θ large and ω small (eq. (2)). ASML's US entities and staff are US persons (section 2), so the firm would not ship against a credible US penalty even if The Hague licensed exit, and Europe would then have to compel it (remark 3). With $\kappa = 0$ these terms alone can make Q negative. The record therefore supports the jurisdictional channel and leaves the security channel untested: we have found no documented case that links US security commitments to ASML. Cross-domain linkage appears elsewhere (section 8.4), never on Dutch chip controls. The disputes since 2023 concern legacy sales and servicing, a hedge that by Proposition 3(b) Washington polices with flow jurisdiction whenever that is credible, so they could not reveal κ even if it were large.

Nor do the security signals of 2025 and 2026 settle it. An untargeted drawdown should raise Europe's premium, and a withdrawal aimed at an ally's conduct is an executed threat (section 5.3), yet the Dutch response to US pressure in 2026 was voice without exit. Either jurisdiction alone kept the premium negative, or perceived conditionality rose with the targeted moves. The record cannot tell these apart.

Slack spent and unspent. The broader measures show what The Hague kept. Licensing of the older immersion tools came a year after the advanced ones, The Hague took the licensing pen from Washington in September 2024, and servicing in China continued at least into mid-2026 (section 2). We read this as rationing: Washington spent Europe's slack first on the sales controls it valued most and left the servicing hedge, costly to ASML, until 2026. A shared-interest account fits the substance of Dutch alignment but not its timing, since a holder aligning for its own reasons need

not wait for US rules.

Nexperia. The Nexperia dispute of 2025 is the one episode in which a Chinese counter-threat was executed against a Dutch step taken under US pressure. On 5 June 2025 the Dutch economy ministry told Nexperia that Washington might soon introduce a "50% rule" and that, if the Nijmegen firm sought an exemption, Washington would look at measures to stop transfers of US intellectual property and technology. On 12 June the Dutch Foreign Ministry met the State Department's Bureau of International Security and Nonproliferation. The Dutch minutes record that the US side called it problematic that the chief executive was still the same Chinese owner, and conclude that it was "almost certain the CEO will have to be replaced to qualify for the exemption from the entity list" (Gerechtshof Amsterdam, Ondernemingskamer, 2025, paras. 2.19–2.20; our translation). The minutes, reported in October 2025 (DutchNews.nl, 2025), are a Dutch record of US views, not a formal US condition. On 30 September, after a US rule threatened Nexperia's operations, the economy minister used the Goods Availability Act to take control of the firm, and the Enterprise Chamber suspended the chief executive in early October (Taylor Wessing, 2025; Gerechtshof Amsterdam, Ondernemingskamer, 2025). China then blocked exports from Nexperia's Chinese plants, which threatened production at European carmakers (Taylor Wessing, 2025). At Busan on 30 October, Washington agreed to suspend the Affiliates Rule for one year from 10 November, and China agreed to "take appropriate measures to ensure the resumption of trade from Nexperia's facilities in China" (Ministry of Foreign Affairs of the People's Republic of China, 2025; The White House, 2025b). On 19 November the Dutch minister suspended his order (Taylor Wessing, 2025). The Enterprise Chamber's measures were separate from the order and outlasted it.

We read the episode as a US-backed demand on the Netherlands (an entity-list exemption in exchange for removing Chinese control of the firm), met by a Chinese counter-threat that was executed, with the Dutch suspension following Washington's suspension of the rule behind the demand. Under complete information, Proposition 3(a)(iii) allows execution on the equilibrium path only when the counter-threat crowds out another demand. That reading requires three things: a Chinese threat known in The Hague and Washington before 30 September; a US demand on the Netherlands that Washington shelved because the Nexperia concession had become expensive; and a Dutch government that conceded while expecting to bear the cost. The alternative, outside our complete-information model, is that The Hague, and perhaps Washington, misjudged China's resolve, which requires that the Chinese response surprised them. The reversal does not discriminate, since by then the rule behind the US demand had been suspended. Neither reading is established by the public record, and the minutes that revealed the US demand exist only because a court published them.

Europe bore the cost and was not compensated. By Proposition 3(a)(v) a counter-threat leaves the holder's loss unchanged when Washington has a divisible sunk demand that China cannot reach, and otherwise the holder's loss can rise. The relief came through a US concession to China at Busan. US–China bargaining is outside the model, so we do not read Busan as a transfer to Europe.

The reversal under Chinese pressure is also evidence that compelling a firm is costly when China retaliates against the compulsion itself (remark 3). And the order shows that the Dutch state can override control of a firm under its own law, which on our reading lowers the enforceability of any control right that Washington might hold through a Dutch entity (Corollary 3(c)).

Servicing and the MATCH Act. Servicing of China's installed base is the hedge in our terms, and Washington tolerated it into 2026. The MATCH Act would police it by flow jurisdiction: allies that do not align within 150 days would face US controls on sales and servicing (section 2). In the model a statute is a stake that makes the threat credible if $e_F \leq \bar{m}_U$, and a credible threat to block the hedged flows leaves Q unchanged and need not be carried out (Proposition 3(b)). The design also moves the burden. Washington would act on the flows itself if The Hague did not concede, and on our reading an order to stop servicing carries a Chinese counter-threat after Decrees 834 and 835 (State Council of the People's Republic of China, 2026; Li et al., 2026), so the concession has become more expensive for Europe (Proposition 3(a)(i)). The Dutch answer was voice: the prime minister and the trade minister took objections to Washington in April and June 2026, and the cabinet told parliament it opposed the bill's extraterritorial reach (NL Times, 2026c; van Gerven, 2026; Sjoerdsma, 2026). The Hague did not threaten to stop aligning. With $Q < 0$, voice and the hold-up of section 7.4 are all a holder has.

China's share of ASML's system sales rose from 14 percent in the second quarter of 2026 to 23 percent in the third (ASML Holding N.V., 2026d,b). The model has no inventories, so a stockpiling reading is not one of its implications. A rise in Dutch licensing of legacy tools, which would be a larger tolerated hedge, is. Licensing decisions and order dates would separate the two.

8.4 Comparison cases

Table 4 sets the Dutch record beside six contemporary and three historical cases, coded with the rules of section 8.2. The equipment holders in Japan, South Korea and Germany face the structure of the model. Taiwan and the United Arab Emirates fall outside the scope conditions of section 9.1: China threatens Taiwan rather than bidding for it, and the UAE has no treaty guarantee. The historical cases add variation in linkage and compensation.

What the comparison shows. No holder in the table was paid in cash for aligning on a chokepoint. The currency is status (Supplement No. 4, validated end-user status, Country Group A:5), which BIS grants and withdraws by rule. Under our coding such statuses are exit costs, the pattern Corollary 1 leads us to expect from holders with negative premia. Where money moved, it moved from the client to the patron (the offsets, the UAE) or to a third party: in 1967 Washington placed orders in Britain to close the trilateral deal, and in 1977 it offered Brazil, not Bonn, a fuel guarantee (Sampson, ed, 1995; Tudda, ed, 2015).

Resistance clusters on the installed base and on deals already signed. Tokyo and The Hague kept servicing Chinese fabs, Bonn kept its contract with Brazil while offering restraint on future transfers,

and the Korean fabs kept operating under licence. In the model these concessions cost the holder much relative to the bite that remains, and in Japan a Chinese counter-threat sat on exactly the concession Washington sought. A cost-minimizing coercer that found enforcement against servicing expensive would also tolerate it, so this pattern fits our account without separating it from that rival.

Explicit links between an ally's conduct on one issue and US treatment on another appear in three places: Johnson's link between troop levels and offset payments in 1966, the posture and trade measures against Germany and Spain in 2026, and the BIS rule of July 2026 that cited the UAE's role in Operation Epic Fury. None of them concerns a European holder's chip controls. Washington uses the security channel, but the record does not show it using that channel on ASML.

Where the comparison strains the model. The offsets fit only partly. Washington linked troop levels to German payments, and the payments were sunk concessions, which escape the slack cap (Proposition 2(b)). Renegotiation recurred from the 1961 memorandum through the informal settlement of 1964 to the trilateral talks of 1966–67 (Gerakas et al., eds, 1995; Miller, ed, 1999; Sampson, ed, 1995), as Proposition 2(d) leads us to expect of a patron that cannot commit to stop demanding, though FRUS does not show Bonn reasoning that way. The redeployment of up to 35,000 personnel agreed in 1967 strains the model, since a complete-information patron does not both collect and cut. We see two readings. If the troops were at Washington's ex post optimum, the linkage was a costly conditional threat and the partial cut is unexplained. If they were above it, as balance-of-payments pressure and the Mansfield resolution suggest, keeping them was a transfer to Bonn, the offsets were its price, and Washington cut where cutting saved money (Sampson, ed, 1995; Gavin, 2004; Zimmermann, 2002). FRUS does not let us choose. Erhard's fall is outside the model. McCloy recorded Brandt calling the package approach "a major mistake" that hastened it (Miller, ed, 1999). Our holder bears a sovereignty cost only when it transfers control, and the case suggests that other concessions to the patron carry a domestic cost too.

The United Kingdom in 1957 shows a secondary demand that Washington chose not to back with linkage. Eisenhower conceded privately that there was "very little of profit in the matter", Macmillan answered that "We live by exports", and in September the National Security Council decided that no sanctions would be applied against allies for eliminating the special China controls (McMahon et al., eds, 1989; Mastanduno, 1992). A demand worth little to Washington and costly to the holder is the first to go under Corollary 2. The strain is that the holder did not act alone: twelve of sixteen members backed abolition (McMahon et al., eds, 1989), a collective exit that our single-holder model does not represent. The Brazil deal adds a different strain. German officials feared for the US security commitment, although in the FRUS record Washington did not tie its pressure to it (Rasmussen, ed, 2014; Tudda, ed, 2015). Conditionality that exists only in the holder's beliefs has no place in our complete-information model. And Gray (2012) reads Bonn's resistance as commercial rivalry, against the finding of Kroenig (2009) that sensitive nuclear transfers follow strategic motives.

8.5 Designs that could test the model

An event dataset. The unit would be a US demand on a holder of a chipmaking-equipment chokepoint, 2019–2026: the Netherlands, Japan, South Korea and Germany. Each demand would be coded, from the Federal Register, national gazettes, parliamentary records and firm filings, for the instrument threatened (section 8.2), the holder's response, any compensation and any Chinese response. China's share of the firm's sales and credible Chinese counter-threats proxy for the premium. The model implies that conditional linkage against secondary demands appears when a holder has slack and fades as its premium rises toward zero, while the use of flow jurisdiction does not track the premium (Proposition 3(b)). It also implies that concessions in one issue area crowd out demands in another within the same period (Corollary 2). Cost minimization implies neither.

An elite survey experiment. Conditionality is a property of US behaviour, so the respondents should be US foreign-policy elites. A vignette would describe an ally holding a chokepoint and vary whether it complied with US export controls on China, refused, or complied on sales while servicing Chinese fabs. The gap in support for defending the ally, and for keeping US forces and intelligence sharing there, between compliant and noncompliant allies estimates the conditionality of the core guarantee and of posture. Further arms would vary whether the president had publicly tied protection to compliance (the stake of Proposition 4) and whether the ally hosts bases Washington uses (the sign in part (d)).

Process tracing. Requests under the Dutch Open Government Act (Woo) for ministerial records of export-control talks with US counterparts from 2019 to 2026 would show whether security commitments, posture or the F-35 programme were ever raised, which is the evidence the security channel needs. The same records would date Dutch drafting against US threats, which separates coercion from shared interest better than outcomes do, and show what The Hague expected from China before 30 September 2025.

The Busan expiry. The US suspension of the Affiliates Rule ran for one year from 10 November 2025, and China's suspension of its October 2025 controls ran to 10 November 2026 (The White House, 2025b; Ministry of Commerce of the People's Republic of China and General Administration of Customs of the People's Republic of China, 2025b). In September 2026 the two governments agreed to extend the arrangement that paused them to 10 January 2027, but no legal act extending the suspensions themselves had appeared by late October (Ministry of Commerce of the People's Republic of China, Department of American and Oceanian Affairs, 2026). If both lapse, the rule behind the Nexperia demand and China's extraterritorial rare-earth controls return together (Mayer Brown, 2025), and Proposition 3(a) leads us to expect Washington to move its demands on The Hague away from concessions that carry credible Chinese counter-threats, toward demands China cannot reach or flow jurisdiction it exercises itself. A second executed Chinese counter-threat against a Dutch step would be hard to square with misjudgment, which learning should correct, and the

crowding-out reading would then require us to find the demand it displaced.

9 Implications and Scope

9.1 Scope

The results hold under seven conditions.

(i) *Security dependence.* The holder relies on its patron for protection it cannot replace on the relevant horizon ($S_k > 0$).

(ii) *A partly conditional guarantee.* Some of that protection can be withheld after exit at a cost the patron can stake, and making all of it conditional carries an assurance cost ($\kappa S_k + d_p > 0$; Proposition 4).

(iii) *A rival that bids.* A second great power can credibly promise something for the chokepoint ($\omega > 0$).

(iv) *A nested chokepoint.* The chokepoint sits partly inside the patron's jurisdiction, so flow jurisdiction is cheap ($\theta > 0$), but the holder's licence still decides whether exit happens ($\theta < 1$).

(v) *Complete information* about conditionality.

(vi) *Unitary great powers.*

(vii) *One rival*, with holders bargaining one at a time.

The Netherlands meets the first four (section 2), and section 7 relaxes the unitary holder. On our reading Japan and South Korea meet them too. Both are treaty allies; Beijing threatened retaliation when Washington pressed Tokyo to restrict its toolmakers' sales and servicing in China (Bloomberg, 2024), and Korea's fabs in China run on US-origin tools whose validated end-user status BIS withdrew in 2025 (U.S. Department of Commerce, Bureau of Industry and Security, 2025). Taiwan fails the third condition: China threatens its security and does not bid for its alignment, so Washington's 2025 proposal that half the chips it uses be made in the United States, which Taipei rejected (Younger, 2025; Focus Taiwan , CNA), is a demand for control without a bidder. The Gulf partners that traded security cooperation and investment for chip access (The White House, 2025a; U.S. Department of Commerce, 2025; U.S. Department of Commerce, Bureau of Industry and Security, 2026) have no treaty guarantee and fail the first. Kyrgyzstan, which used a Russian aid package to raise the rent for the Manas base (Cooley, 2012), fits only the bidding benchmark of Lemma 1. Section 8.4 compares the cases. Because conditionality is a choice (section 5.3), a less transactional administration would move Europe to another point in the parameter space and leave it within the scope.

9.2 Implications for Europe

Each implication holds in a region of the parameter space. On our reading Europe has a nonpositive premium and slack to spend: The Hague has received revocable statuses, which we do not code as transfers (section 8.2), and Washington has tied favourable treatment and AI chip exports to export-control alignment, the linkage Proposition 3(b) leads us to expect against an ally with slack (section 5.2). That reading rests on the jurisdictional channel. How much of the slack comes from the conditional part of the guarantee, the record leaves untested (section 8).

Reduce the conditional part of dependence. At The Hague in June 2025 NATO allies committed to spend 5 percent of GDP a year by 2035, at least 3.5 percent on core defence (North Atlantic Treaty Organization, 2025). The target counts spending. In the model a euro raises Europe's premium only insofar as it replaces something Washington could withhold after exit, which points to substitutes for posture and enablers (intelligence, munitions, sustainment, forces that can be moved one ally at a time) more than to duplicates of the statutory core (section 4.3). Below zero, each unit of premium gained removes a unit of the slack Washington can spend (Corollary 2), though Europe still receives no transfer. Washington can compete the gain away. Where assurance is cheap at the margin, it answers a higher premium with a more conditional guarantee and pays nothing (Proposition 4(c)(i)); Europe gains only where assurance is already costly, where Washington is already buying slack to spend, or where conditionality is at its ceiling. If public US statements tying protection to European conduct multiply as European capability grows, Europe is in the first regime.

Reduce the nesting. Moving US-origin content and US-person servicing out of US jurisdiction lowers θ and raises China's credible offer, but only while China's commitment cap does not bind (eq. (2)). Jurisdiction is a choice variable (Kalyanpur and Newman, 2019), but changing it is slow and visible. On the servicing side, de-nesting lowers the share h that Washington can compel, which is what gives hold-up its value (section 7.4). We do not know whether the cap binds; Chinese offers limited by what Beijing can credibly deliver, and not by US reach, would show that it does, and then de-nesting buys nothing.

Prepare an interruption only as a standing rule. Interrupting servicing raises Europe's transfer only if a rule fixed before offers are made decides it and keeps it in force for the dispute; decided after a rejected offer, it is never used (Corollary 5). A response measure under the Anti-Coercion Instrument (European Parliament and Council of the European Union, 2023), once adopted through the stages of section 7.3, could be such a rule. It helps only for servicing Washington cannot compel, which for US-sited fabs is a small share on our reading of ASML's US structure (section 7.4).

Expect a preventive demand. Autonomy investments that Europe cannot commit to forgo strengthen Washington's motive to demand control while its threats still bite, and with investment Washington can buy control with no bite at all (Corollary 3(a)–(b), Corollary 4). Whether the window opens depends on the net drift of the premium, which rearmament raises and Chinese indigenization lowers (section 6.3). Sovereignty costs fixed in advance (ASML's preference-share foundation, Vifo screening, the Foreign Subsidies Regulation) deter the demand (section 6.1). Demands for relocation, board seats or a licensing veto that follow visible European investment would show that Europe is in the window (section 8.1).

Bind, or keep the key. Binding against exit pays only when it pre-empts a demand that would otherwise come, at lower cost, and it pre-empts only if Washington values control little for its own sake; a partial binding can invite a demand by raising the slack Washington's threats draw on (remark 1). The EU has moved partway. The 2023 Economic Security Strategy flagged advanced semiconductors for possible new export tools (European Commission and High Representative of the Union for Foreign Affairs and Security Policy, 2023), the 2024 White Paper on Export Controls, COM(2024) 25, proposed uniform EU controls and a senior-level coordination forum (European Commission, 2024b), and Delegated Regulation 2025/2003 put the most advanced immersion tools on the EU list (section 6.4). On our reading these raise Δ_b without approaching $\Delta_b = \infty$, and whether that weakened the US motive turns on v_c. A shift in US demands from alignment of controls toward ownership or a licensing veto would indicate a high v_c.

Chinese counter-threats. The Critical Raw Materials Act sets 2030 benchmarks of 10 percent EU extraction, 40 percent processing and 25 percent recycling of strategic raw materials, with at most 65 percent from any one third country (European Parliament and Council of the European Union, 2024). In the model, removing China's counter-threats weakly raises Washington's payoff and leaves Europe's loss at $B(J^c)$ when Washington has demands China cannot reach; otherwise Europe's loss can rise or fall (Proposition 3(a)). It also weakens the Chinese threat that protects European ownership against a transfer (Corollary 3(d)). Europe's gain from de-risking therefore lies mostly in disputes over its own measures, such as the Nuctech investigation (section 5.2) and Nexperia (section 8.3). The Blocking Statute (Council of the European Union, 1996) lists no US export-control law (European Commission, 2018), so its penalty is zero today (remark 3); listing such laws would matter only once the premium is positive and Europe licenses exit. We treat China's coercive reach and the credibility of its promises as separate. If Europe discounts the promises of a China able to cut its supplies, de-risking also raises λ_C and with it ω.

9.3 Lessons for Washington

Conditionality is costly. A patron with no further demands makes its guarantee just conditional enough to hold the premium at zero, and more conditionality pays only if the slack is spent, while other allies pay through weaker assurance (Proposition 4). Demands across domains (tariffs, the

spending target, procurement, servicing in China) share one budget, so a stronger Chinese bid forces Washington to drop its lowest-value demands first (Corollary 2). Assurance raises what coercion yields. Conditional parts are measured from what a loyal ally receives, so untargeted drawdowns and rationing of loyal allies lower d_p and d_{ai} and with them Europe's slack (section 5.3), and a targeted cut that is carried out spends the instrument (Proposition 3(b)).

9.4 Limitations

Complete information about conditionality is the strongest assumption. If κ were private, Washington would gain from overstating it, European beliefs would set Q, and documents such as the 2025 National Security Strategy would be partly cheap talk. Refusals and executed counter-threats could then come from misjudgment, the reading of Nexperia that lies outside the model (section 8.3). The assurance cost H is a reduced form, and who pays for China's bid depends on its shape (Proposition 4(c)); if it reflects Washington's reputation with Tokyo and Seoul, conditionality toward Europe depends on what other holders observe. The great powers are unitary, although Congress can set stakes the executive would not, as the MATCH Act's 150-day deadline for allies would (section 2), and member-specific exposure enters only the Council remark (remark 4). With one rival and holders bargaining alone, the model is silent on whether a coalition of the Netherlands, Japan and South Korea would help; that would require assets that are complements for China and members able to compensate one another.

10 Conclusion

A chokepoint holder's leverage over its patron is bounded by its exit premium, and the patron's demands are bounded by the holder's slack. Because the conditional parts of the patron's threats draw on that one budget, what Europe can obtain for ASML, and what it can be made to give up, turn on the conditional part of its dependence on Washington and on what China can credibly promise.

The result adds a condition to the theory of weaponized interdependence. Farrell and Newman (2019) require that a state have jurisdiction over a hub and the domestic institutions to use it. When the hub's holder bargains with a state that depends on the hub, a third condition applies: the hub confers power on its holder only if the holder's exit premium is positive. The Netherlands licenses ASML's exports, so it has the institutions. It has jurisdiction only in part, because Washington reaches the firm through US-origin inputs and US persons (section 2). Whether it meets the third condition depends on how conditional the US guarantee is on chip policy, and that is the quantity our evidence measures least well.

The limitation we would worry about most is information. The model assumes that all three powers know how conditional the guarantee is, and it models the cost of making the guarantee conditional as a reduced-form assurance cost (Proposition 4). If conditionality were private information, Washington would gain from overstating it, its public statements would be partly cheap

talk, and European beliefs would set the exit premium. Our results then describe the case in which those beliefs are correct. A model of where the assurance cost comes from (a reputation with other holders, say, or a Congress unwilling to defend allies who help China) could change how much conditionality Washington chooses.

How conditional the guarantee is can be measured. A survey experiment on US foreign-policy elites could describe an ally that has or has not complied with a US export-control demand, hold the ally and the crisis fixed, and ask whether respondents would support defending it. The difference estimates the issue-specific conditionality that the public record does not reveal. The event dataset described in section 8.5 would test the rationing logic of Corollary 2 directly, by coding each US demand on a holder of a chipmaking-equipment chokepoint for the instrument threatened, the holder's response, any compensation and any Chinese response.

What the record shows so far fits the jurisdictional half of the argument. What The Hague obtained in December 2024 was a place on Supplement No. 4, which BIS justified on the ground that governments on the list "have the authority to control key SME items of concern" and which it can revoke by amending a rule (Bureau of Industry and Security, 2024). What The Hague opposed in May 2026 was the extraterritorial reach of the MATCH Act. Its trade minister told parliament that the cabinet was against the proposal's extraterritorial effect, since "every country is responsible for its own export control legislation" (Sjoerdsma, 2026, our translation).

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A Proofs

This appendix proves the results in the order in which they appear in the text. Where a statement leaves a modelling point open, the proof begins by fixing it. Symbols introduced inside a proof are local to that proof. Throughout, $[x]^+ = \max\{x, 0\}$ and $\mathbf{1}\{\cdot\}$ is the indicator function. Appendix B describes the brute-force check of each result.

A.1 Proof of Lemma 1

Assumptions. We fix four points. (1) A winning Chinese promise of τ_C costs China τ_C, Europe values it at $\lambda_C \tau_C$ with $\lambda_C \in (0, 1]$, China values Europe's exit at W_C, and a losing bid gives China zero. This is the reading under which $\omega = \lambda_C \min\{W_C, \bar{m}_C\}$. If λ_C were instead the probability that China honours a promise, and China paid only when it did, China would bid up to $\min\{W_C/\lambda_C, \bar{m}_C\}$ and its credible offer would be $\min\{W_C, \lambda_C \bar{m}_C\}$. (2) Each announced conditional action carries its own stake of at most $\bar{m}_U$, so a promise and a threat are separate commitments (section 4.2). If $\bar{m}_U$ were a budget shared by the bid and the threats, the two would compete and the lemma would change. (3) Washington's ex post payoff from its instruments does not depend on whether it has paid its promise, and payoffs are additive in transfers and instruments. (4) An indifferent Europe aligns, and a power that is indifferent between paying a staked promise and forfeiting the stake pays.

Proof of Lemma 1. (i) Stakes on threats are attached before Europe moves and are separate from stakes on promises. Once Europe's alignment a is fixed, Washington sets each instrument to maximize its ex post payoff given these stakes, and by assumption (3) this choice does not depend on whether it pays a promise. So it provides $x_j(a)$, and Europe's payoff before transfers is $\sum_j v_j(x_j(a)) - \eta\, \mathbf{1}\{a = X\}$. The difference between $a = A$ and $a = X$ is $\sum_j d_j + \eta = c$.

(ii) *Credibility.* At the payment stage, paying a promise of τ costs τ and has no ex post value, while not paying forfeits the stake $m \le \bar{m}_i$. The promiser pays if and only if $\tau \le m$, so the largest promise that is honoured is $\bar{m}_i$. Europe anticipates this and assigns zero value to larger promises.

The clock auction. Measure the clock $z \ge 0$ in units of Europe's payoff relative to the aligned baseline. While the United States is active at z it stands behind the promise $\tau_U = z$. While China is active at z it stands behind $\tau_C = (z + c)/\lambda_C$, which gives Europe $\lambda_C \tau_C - c = z$ if it exits. A bidder that drops cannot re-enter. When one bidder drops at z (or never enters, in which case $z = 0$), the other wins at z. If both drop at the same level, Europe is indifferent and aligns. A strategy is a drop level z_i.

Washington's drop level. Washington's maximal credible bid is $\bar{z}_U = \min\{V, \bar{m}_U\}$: above $\bar{m}_U$ its promise is worthless, and above V winning is worse than losing. Fix China's drop level z_C. If $z_C \le \bar{z}_U$, dropping at $\bar{z}_U$ wins (ties go to alignment) at the price $\max\{z_C, 0\} \le V$. Every strategy that wins pays at least this price, and losing yields $-V \le -\max\{z_C, 0\}$, so dropping at $\bar{z}_U$ is a best response. If $z_C > \bar{z}_U$, dropping at $\bar{z}_U$ loses and yields $-V$. To win, Washington would have to stand behind a promise of at least z_C. If $\bar{z}_U = \bar{m}_U$, such a promise exceeds $\bar{m}_U$ and is worthless, so Washington cannot win. If $\bar{z}_U = V$, winning costs at least $z_C > V$, which is worse than losing. Hence dropping at $\bar{z}_U$ is a best response to every z_C, that is, weakly dominant.

China's drop level. China wins at level z only if it stays strictly longer than the United States. It then earns $W_C - (z + c)/\lambda_C$, which is nonnegative if and only if $z \le \lambda_C W_C - c$, and its promise is credible if and only if $z \le \lambda_C \bar{m}_C - c$. Its maximal credible bid is therefore $\lambda_C \min\{W_C, \bar{m}_C\} - c = Q$. Fix Washington's drop level z_U. If $z_U < Q$, dropping at Q wins at the price z_U, with a credible promise and a positive payoff. If $z_U \ge Q$, dropping at Q loses and yields zero, while winning would need a level above Q, at which China's promise is either not credible or unprofitable. So dropping at Q is weakly dominant. If $Q < 0$, China cannot offer Europe even the baseline and stays out.

Outcome. With both bidders at their dominant drop levels, Washington wins if and only if $\bar{z}_U \ge Q$ (ties go to alignment), and it pays the level at which China drops, $\max\{Q, 0\} = [Q]^+$. If $Q < V$ and $Q \le \bar{m}_U$, then $Q \le \bar{z}_U$, which gives (a). If $Q > \bar{z}_U$, China wins at the level $\bar{z}_U \ge 0$ at which Washington drops. Europe receives $\lambda_C \tau_C - c = \bar{z}_U$, and China's payment $\tau_C = (\bar{z}_U + c)/\lambda_C$ is credible and profitable because $\bar{z}_U < Q$. This gives (b). In case (a) Europe's payoff is $[Q]^+ = [\min\{Q, \bar{z}_U\}]^+$, and in case (b) it is $\bar{z}_U = [\min\{Q, \bar{z}_U\}]^+$; in both, $\min\{Q, \bar{z}_U\} = \min\{Q, V, \bar{m}_U\}$. The case $Q = V \le \bar{m}_U$, which the statement excludes, leaves Washington indifferent between winning and losing, and both outcomes occur.

(iii) If $\bar{m}_U < Q < V$, then $\bar{z}_U = \bar{m}_U$ and by (ii)(b) Europe exits with payoff $\bar{m}_U$ while Washington receives $-V$. A binding transfer $\tau \in [Q, V)$ with alignment gives Europe $\tau \ge Q > \bar{m}_U$ and Washington $-\tau > -V$. □

The alternating-raise auction. The clock auction builds in the assumption that the losing bidder stays in until its maximal credible bid. We now drop that assumption. Bidders alternate. At its turn a bidder may raise its standing bid to a point on a grid or pass, and two consecutive passes

end the auction, after which Europe chooses. Washington's bids lie on a grid of mesh ε in τ_U, so their value to Europe lies on a grid of mesh ε. China's lie on a grid of mesh ε in τ_C, so their value to Europe, $\lambda_C\tau_C - c$, lies on a grid of mesh $\lambda_C\varepsilon$. Bids worth more to Europe than the bidder's maximal credible bid ($\bar{z}_U$ for Washington, Q for China) are excluded; they are weakly dominated. We claim the following. (a) Every subgame-perfect equilibrium has Europe aligned when $Q < \bar{z}_U - \varepsilon$ and exiting when $Q > \bar{z}_U + \lambda_C\varepsilon$. (b) Subgame perfection does not pin down the price: when $0 < Q < \bar{z}_U - \varepsilon$, every price in $[0, Q]$, up to one grid step, is the outcome of some subgame-perfect equilibrium. (c) If a losing bidder that is indifferent between raising and passing raises, the price is within one grid step of $[Q]^+$. (d) Part (iii) of Lemma 1 holds in every subgame-perfect equilibrium for ε small.

Proof of the claim. The game is finite. (a) Let $Q < \bar{z}_U - \varepsilon$ and suppose that some equilibrium path ends with China leading. China's last raise was followed by a move of Washington, which passed. At that node Washington could instead raise to the smallest grid bid $b \geq Q$, which satisfies $b < Q + \varepsilon < \bar{z}_U$, so $b < V$ and $b \leq \bar{m}_U$. No Chinese bid can beat b, because China's bids are worth at most $Q \leq b$ to Europe and ties go to alignment. The raise therefore guarantees Washington $-b > -V$, while passing gave it $-V$, a contradiction. Now let $Q > \bar{z}_U + \lambda_C\varepsilon$ and suppose that some path ends with Washington leading, at a bid worth at most $\bar{z}_U$ to Europe. At China's last move China could raise to a grid bid whose value to Europe lies in $(\bar{z}_U, Q)$, an interval longer than $\lambda_C\varepsilon$. Washington cannot top it, and China's payoff from it is strictly positive because its value to Europe is below Q, whereas losing gives zero. Again a contradiction.

(b) Let p be a point of China's grid with value to Europe in $[0, Q]$. Consider the following profile. China, as long as it has not yet bid, raises to a bid worth p whenever no bid worth p or more stands, and it passes at every other node. Washington answers any Chinese lead with its smallest grid bid worth at least as much to Europe, and passes whenever it leads or no bid stands. Since China's bids are worth at most Q and $Q + \varepsilon < \bar{z}_U$, Washington's answer is credible and below V. China wins in no subgame, so it earns zero whatever it does and its strategy is optimal. When China leads, answering is optimal for Washington: China then passes and the auction ends with Washington paying the least it can pay to win, while passing would hand the win to China and give Washington $-V$. When Washington leads, or no bid stands, passing is optimal. If China has already bid, it passes next and the auction ends. If it has not, a US bid worth less than p would be met by China's bid worth p and answered again, at a price no lower than on the path, and a US bid worth p or more costs at least as much. On the path Washington pays its smallest grid bid worth at least p, which is within ε of p. The price 0 arises from the same profile with China never bidding: no bid is made, and Europe aligns, since without a Chinese promise its exit payoff is $-c \leq 0$.

(c) Let $0 < Q < \bar{z}_U - \varepsilon$. By (a) China loses and earns zero whatever it does, so it is indifferent between raising and passing whenever a raise to a bid worth at most Q is available. If it raises in that case, the auction cannot end while Washington's standing bid is worth less than $Q - \lambda_C\varepsilon$, so the price is at least $Q - \lambda_C\varepsilon$. Washington never needs to bid more than the smallest grid bid worth at least Q, which is below $Q + \varepsilon$. If $Q \leq 0$, China never bids and the price is 0.

(d) Let $\bar{m}_U < Q < V$, so $\bar{z}_U = \bar{m}_U$, and let ε be small enough that $Q > \bar{z}_U + \lambda_C \varepsilon$. By (a) Europe exits. China's last raise is optimal only if no cheaper bid would also win for sure. A bid worth to Europe a value in $(\bar{z}_U, \bar{z}_U + \lambda_C \varepsilon]$ does, since Washington's bids are worth at most $\bar{z}_U$, and China's payoff falls with its bid. So Europe receives at most $\bar{m}_U + \lambda_C \varepsilon < Q$, and every binding transfer $\tau \in [Q, V)$ makes both Europe and Washington strictly better off. □

A.2 Proof of Proposition 1

Discrete closed forms. Let $\mu = 1 - e^{-\lambda\Delta}$, $\delta = e^{-r\Delta}$, $\upsilon = \delta(1-\mu) = e^{-(r+\lambda)\Delta}$ and $\psi_\Delta = \mu/(1-\upsilon)$, and define

$$x_A = \psi_\Delta(Q + \beta(V - Q)), \qquad x_B = \frac{(1-\beta)Q + \beta\mu V}{1 - \beta\upsilon}, \qquad x^*(\Delta) = \max\{x_A, x_B\}.$$

Part (i) of Proposition 1 holds with this $x^*(\Delta)$, and there is a stationary equilibrium in which Washington offers $P_U = \max\{Q, \mu Q + \upsilon x^*\}$, Europe demands $P_E = \mu V + \upsilon x^*$, each offer is accepted at once, and Europe would exit after rejecting a US offer if and only if $Q > \delta x^*$. When $\lambda > 0$ this equilibrium is the unique one (part (ii)). Part (iii) refers to P_U and P_E as the equilibrium offer and demand.

Assumptions. After a rejection without voluntary exit, rupture is realized at once with probability μ, and exit flows start immediately. Otherwise the round's status-quo flows $(0, 0)$ accrue and the next round starts after discounting by δ. Offers are real numbers with no sign restriction. We assume $Q < V$, $\beta \in [0, 1)$, $r, \lambda \geq 0$ and $r + \lambda > 0$, so that $\upsilon < 1$.

Proof of Proposition 1. We measure payoffs as flows, so that a constant flow z forever is worth z. Agreement on P gives $(P, -P)$, exit gives $(Q, -V)$, and the status quo gives $(0, 0)$ per period.

Step 1 (feasible payoffs). Every outcome gives a total payoff of 0 (agreement or perpetual status quo) or $Q - V < 0$ (exit), and a mixture of these, so in every subgame the two players' payoffs sum to at most zero. Payoffs are bounded. Washington can reject every demand and make offers Europe may accept or reject, which gives it at least $\min\{0, -V\} = -V$, and Europe can reject every offer without exiting and demand 0, which gives it at least $\min\{Q, 0\}$. The upper bounds follow from the sum. Because $\upsilon < 1$, payoffs are continuous at infinity, and the one-shot deviation principle applies.

Step 2 (the value map). Let $\mathcal{R}(v) = \max\{Q, \mu Q + \upsilon v\}$ be Europe's value of rejecting a US offer when the continuation value is v, and let $\mathcal{T}(v) = \beta(\mu V + \upsilon v) + (1-\beta)\mathcal{R}(v)$. $\mathcal{T}$ is nondecreasing with Lipschitz constant $\upsilon < 1$, so $\mathcal{T}(v) - v$ is strictly decreasing and $\mathcal{T}$ has a unique fixed point. Write $\mathcal{T} = \max\{\mathcal{T}_A, \mathcal{T}_B\}$ with $\mathcal{T}_A(v) = \mu(Q + \beta(V - Q)) + \upsilon v$ and $\mathcal{T}_B(v) = \beta\mu V + (1-\beta)Q + \beta\upsilon v$, whose fixed points are x_A and x_B. If $x_A \geq x_B$, then $\mathcal{T}_B(x_A) - x_A \leq \mathcal{T}_B(x_B) - x_B = 0$ because $\mathcal{T}_B(v) - v$ is strictly decreasing, so $\mathcal{T}(x_A) = x_A$. The case $x_B \geq x_A$ is symmetric. Hence the fixed point is $x^* = \max\{x_A, x_B\}$. Finally, $\mathcal{R}(x^*) = Q$ if and only if $Q \geq \mu Q + \upsilon x^*$, that is, $(1-\mu)Q \geq \delta(1-\mu)x^*$, that is, $Q \geq \delta x^*$.

Step 3 (a stationary equilibrium). Consider the profile in which Europe accepts a US offer P if and only if $P \geq \mathcal{R}(x^*)$ and, after rejecting, exits if and only if $Q > \mu Q + \upsilon x^*$; Washington accepts a demand P if and only if $P \leq P_E = \mu V + \upsilon x^*$; Washington offers $P_U = \mathcal{R}(x^*)$; and Europe demands P_E. Under this profile Europe's value is $\beta P_E + (1-\beta)P_U = \mathcal{T}(x^*) = x^*$ and Washington's is $-x^*$. A responding Europe gets P by accepting and $\mathcal{R}(x^*)$ by rejecting (with its exit choice optimal), and a responding Washington gets $-P$ by accepting and $-\mu V - \upsilon x^* = -P_E$ by rejecting, so the acceptance rules are optimal. A proposing Washington gets $-\mathcal{R}(x^*)$ from P_U. A rejected offer gives it $-V$ if Europe then exits, which happens only when $\mathcal{R}(x^*) = Q < V$, and $-\mu V - \upsilon x^* \leq -\mu Q - \upsilon x^*$ if Europe does not exit, which happens only when $\mathcal{R}(x^*) = \mu Q + \upsilon x^*$. In both cases a rejected offer gives at most $-\mathcal{R}(x^*)$. A proposing Europe gets P_E, while a rejected demand gives it $\mu Q + \upsilon x^* \leq P_E$. Both comparisons are strict when $\mu > 0$. By the one-shot deviation principle the profile is a subgame-perfect equilibrium.

Step 4 (all equilibria have the same payoffs). Every subgame that starts a round is a copy of the game. Let $\bar{x}$ and $\underline{x}$ be the supremum and infimum of Europe's equilibrium payoffs, and $\bar{\pi}$ and $\underline{\pi}$ those of Washington. They are finite by Step 1. (a) By Step 1, $\bar{\pi} \leq -\underline{x}$. (b) When Europe proposes, every demand $P < \mu V - \upsilon\bar{\pi}$ is accepted in every equilibrium, because rejecting it gives Washington at most $-\mu V + \upsilon\bar{\pi} < -P$. So Europe's payoff at such a node is at least $\mu V - \upsilon\bar{\pi} \geq \mu V + \upsilon\underline{x}$. (c) When Washington proposes, Europe can reject and obtain at least $\mathcal{R}(\underline{x})$. Hence $\underline{x} \geq \beta(\mu V + \upsilon\underline{x}) + (1-\beta)\mathcal{R}(\underline{x}) = \mathcal{T}(\underline{x})$, and since $\mathcal{T}(v) - v$ is strictly decreasing with root x^*, $\underline{x} \geq x^*$. (d) When Washington proposes, every offer $P > \mathcal{R}(\bar{x})$ is accepted in every equilibrium, because rejecting it gives Europe at most $\mathcal{R}(\bar{x})$. So Washington's payoff at such a node is at least $-\mathcal{R}(\bar{x})$. When Europe proposes, Washington can reject and obtain at least $-\mu V + \upsilon\underline{\pi}$. Hence $\underline{\pi} \geq \beta(-\mu V + \upsilon\underline{\pi}) - (1-\beta)\mathcal{R}(\bar{x})$, that is, $-\underline{\pi} \leq [\beta\mu V + (1-\beta)\mathcal{R}(\bar{x})]/(1-\beta\upsilon)$. (e) At every node Europe's payoff is at most minus Washington's (Step 1). So Europe gets at most $\mu V - \upsilon\underline{\pi}$ when it proposes and at most $\mathcal{R}(\bar{x})$ when Washington proposes. Using (d),

$$\bar{x} \leq \beta\mu V + (1-\beta)\mathcal{R}(\bar{x}) + \beta\upsilon\,\frac{\beta\mu V + (1-\beta)\mathcal{R}(\bar{x})}{1-\beta\upsilon} = \frac{\beta\mu V + (1-\beta)\mathcal{R}(\bar{x})}{1-\beta\upsilon} \equiv \mathcal{S}(\bar{x}).$$

$\mathcal{S}$ has Lipschitz constant $(1-\beta)\upsilon/(1-\beta\upsilon) < 1$, and $\mathcal{S}(v) = v$ if and only if $\mathcal{T}(v) = v$. So $\mathcal{S}(v) - v$ is strictly decreasing with root x^*, and $\bar{x} \leq x^*$. Therefore $\underline{x} = \bar{x} = x^*$. Then $\bar{\pi} \leq -x^*$ by (a) and $\underline{\pi} \geq -\mathcal{S}(x^*) = -x^*$ by (d), so Washington's payoff is $-x^*$ in every equilibrium. This proves (i).

Step 5 (uniqueness and immediate agreement when $\lambda > 0$). By Step 4, payoffs sum to zero in every equilibrium and in every subgame that starts a round. If a proposal were rejected with positive probability, the total payoff after the rejection would be $Q - V < 0$ (exit) or $\mu(Q-V) + \upsilon s$ with $s \leq 0$ (no exit), which is negative when $\mu > 0$. So agreement is immediate. Since continuation payoffs from every round start are $(x^*, -x^*)$, Europe accepts every offer above $\mathcal{R}(x^*)$ and rejects every offer below it, and Washington accepts every demand below P_E and rejects every demand above it. Washington must offer exactly $\mathcal{R}(x^*)$ and Europe must accept it: if Europe rejected $\mathcal{R}(x^*)$, Washington would have no best offer, and any lower offer is rejected and gives Washington strictly

less (Step 3). The same argument pins down Europe's demand P_E and Washington's acceptance of it. Europe's exit choice after rejecting an offer is determined unless $Q = \delta x^*$, and it is never used on the path. This proves the first sentence of (ii). Now let $\lambda = 0$, so $\mu = 0$, $\upsilon = \delta$, $x_A = 0$ and $x_B = (1-\beta)Q/(1-\beta\delta)$. If $Q > 0$, then $x^* = x_B$. Consider the stationary profile in which Europe always makes a demand that Washington rejects (any demand above δx^*) and Washington offers Q, which Europe accepts. Europe's value x satisfies $x = \beta\delta x + (1-\beta)Q$, so $x = x_B = x^*$. Washington's offer is optimal because $\mathcal{R}(x^*) = \max\{Q, \delta x_B\} = Q$ (as $\delta(1-\beta) \leq 1-\beta\delta$), Europe is indifferent between its rejected demand and the acceptable demand δx^*, and Washington is indifferent between accepting and rejecting a demand of δx^*. So the profile is an equilibrium with delay and the payoffs of (i). If $Q \leq 0$, then $x^* = 0$, and perpetual disagreement without exit is an equilibrium with payoffs $(0,0) = (x^*, -x^*)$.

Step 6 (the limit). As $\Delta \to 0$, $\mu = \lambda\Delta + O(\Delta^2)$ and $1 - \upsilon = (r+\lambda)\Delta + O(\Delta^2)$, so $\psi_\Delta \to \psi$ and $x_A \to \psi(Q + \beta(V-Q))$. In x_B the numerator tends to $(1-\beta)Q$ and the denominator to $1-\beta > 0$, so $x_B \to Q$. The maximum is continuous, so $x^* \to P^*$. Finally, $P_U = \mathcal{R}(x^*) \to \max\{Q, P^*\} = P^*$ and $P_E = \mu V + \upsilon x^* \to P^*$. For the threshold, with $r > 0$, $\psi(Q + \beta(V-Q)) > Q$ if and only if $\lambda(Q + \beta(V-Q)) > (r+\lambda)Q$, that is, $Q(r + \lambda\beta) < \lambda\beta V$. □

Exit after a rejected European demand. The footnote to Proposition 1 concerns a game in which Europe may also exit right after Washington rejects Europe's demand. Let $x'_B = \beta V + (1-\beta)Q$. We claim: (1) for each $\Delta > 0$ the stationary equilibrium value is unique and equals $x' = x_A$ if $\delta x_A \geq Q$ and $x' = \min\{x'_B, Q/\delta\}$ otherwise, and, with $r > 0$, $x' \to P^*$ as $\Delta \to 0$; (2) if $\delta x^* > Q$, which for small Δ is the case $Q < \hat{Q}$, the equilibrium payoffs are those of the base game; (3) if $\beta > 0$ and $\delta x_A \leq Q$, which for small Δ is the case $Q > \hat{Q}$, there are nonstationary equilibria in which Europe's payoff is

$$v_H = \max\Big\{x'_B,\ \frac{\beta V + (1-\beta)\mu Q}{1-(1-\beta)\upsilon}\Big\},$$

the largest equilibrium payoff for Europe, and $v_H \to V$ as $\Delta \to 0$.

Proof of the claim. (1) In a stationary equilibrium with values $(v, -v)$, Europe's continuation after a rejection without exit is $\mu Q + \upsilon v$, so Europe exits after a rejection if $Q > \delta v$ and not if $Q < \delta v$. Washington offers $\mathcal{R}(v)$. Washington's rejection of Europe's demand gives Washington $-V$ if Europe then exits and $-\mu V - \upsilon v$ otherwise, so Europe demands V if $Q > \delta v$, $\mu V + \upsilon v$ if $Q < \delta v$, and $pV + (1-p)(\mu V + \upsilon v)$ if $Q = \delta v$ and it exits with probability p. If $v > Q/\delta$, the value equation is $v = \mathcal{T}_A(v)$, so $v = x_A$. If $v < Q/\delta$, it is $v = \beta V + (1-\beta)Q$, so $v = x'_B$. If $v = Q/\delta$, it requires $\mathcal{T}_A(Q/\delta) \leq Q/\delta \leq x'_B$, that is, $x_A \leq Q/\delta \leq x'_B$. The last case excludes the other two. The first two cannot both hold: $x_A > Q/\delta > x'_B$ would require $x'_B < x_A = \psi_\Delta x'_B$ with $\psi_\Delta \leq 1$, so $x'_B < 0$ and hence $Q < 0$, but then $Q/\delta \leq Q \leq x'_B$. Since one of the three conditions always holds, exactly one value exists, and it is the stated one. (If $\delta x_A < Q$ then $Q > 0$: for $Q \leq 0$, $x_A = \psi_\Delta x'_B \geq \min\{x'_B, 0\} \geq Q \geq Q/\delta$.) As $\Delta \to 0$: if $Q < \hat{Q}$, then $\delta x_A > Q$ eventually and $x' = x_A \to P^*$; if $Q > \hat{Q}$, then eventually $x' = \min\{x'_B, Q/\delta\} \to \min\{Q + \beta(V-Q), Q\} = Q = P^*$;

if $Q = \hat{Q}$, both expressions tend to $Q = P^*$.

(2) Steps 4(a) to 4(c) of the proof of Proposition 1 remain valid, because Washington's rejection of a demand now gives it at most $\max\{-V, -\mu V + \upsilon\bar{\pi}\} = -\mu V + \upsilon\bar{\pi}$. So every equilibrium continuation value of Europe is at least x^*. If $\delta x^* > Q$, then $\mu Q + \upsilon v' > Q$ for every continuation value $v' \geq x^*$, exit after a rejection is never optimal, and the game is strategically the base game.

(3) Let L be the stationary equilibrium of (1), with value $v_L = x' \leq Q/\delta$. In state H, Europe demands V and Washington accepts any demand up to V; after Washington rejects a demand, Europe exits, and if it did not, play would continue in L; Washington offers $\mathcal{R}(v_H)$, and after Europe rejects an offer without exiting, play continues in H. Then $v_H = \beta V + (1-\beta)\mathcal{R}(v_H)$, whose solution is the stated maximum (by the argument of Step 2). One-shot deviations: exit after Washington's rejection is optimal for Europe because $\mu Q + \upsilon v_L \leq Q$. Washington is indifferent between accepting V and rejecting it, since exit then gives it $-V$. A demand above V is rejected and gives Europe $Q < V$. A US offer below $\mathcal{R}(v_H)$ is rejected and gives Washington $-V$ or $-\mu V - \upsilon v_H$, both at most $-\mathcal{R}(v_H)$, as in Step 3. Europe's acceptance rule in H is optimal by construction. For the upper bound, Washington can always secure $-V$, so Europe gets at most V when it proposes, and, as in Step 4(d), at most $\mathcal{R}(\bar{x}')$ when Washington proposes, where $\bar{x}'$ is the supremum of Europe's equilibrium payoffs. So $\bar{x}' \leq \beta V + (1-\beta)\mathcal{R}(\bar{x}')$. The right side minus $\bar{x}'$ is strictly decreasing and vanishes at v_H, so $\bar{x}' \leq v_H$. As $\Delta \to 0$, $(\beta V + (1-\beta)\mu Q)/(1-(1-\beta)\upsilon) \to \beta V/\beta = V$. □

Convergence and the order of limits. Since $|\max\{a,b\} - \max\{a',b'\}| \leq \max\{|a-a'|, |b-b'|\}$ and $x_B - Q = \beta(1-\upsilon)(\psi_\Delta V - Q)/(1-\beta\upsilon)$,

$$|x^*(\Delta) - P^*| \leq \max\left\{|\psi_\Delta - \psi|\,|Q + \beta(V-Q)|,\ \frac{\beta(1-\upsilon)}{1-\beta\upsilon}\,|\psi_\Delta V - Q|\right\}, \qquad \frac{\beta(1-\upsilon)}{1-\beta\upsilon} \leq \frac{\beta(r+\lambda)\Delta}{1-\beta}.$$

The error is $O(\Delta)$ where the second term of P^* binds and $O(\Delta/(1-\beta))$ where Q binds, so it is not uniform in β. For fixed Δ, $x^*(\Delta) \to \psi_\Delta V$ as $\beta \to 1$ (both x_A and x_B tend to $\psi_\Delta V$), whereas $P^* \to \max\{Q, \psi V\}$. The limits $\Delta \to 0$ and $\beta \to 1$ therefore do not commute when $Q > \psi V$. The formula for P^* describes bargaining in which US proposals arrive much faster than rupture and discounting, $(1-\beta)/\Delta$ large relative to $r + \lambda$, which is why the statement keeps $\beta < 1$.

A.3 Proof of Corollary 1

Proof of Corollary 1. (a) P^* is a maximum one of whose terms is Q, so $P^* \geq Q$. If $Q \geq 0$ then $P^* \geq 0$, and if $Q < 0$ then $-P^* \leq -Q$; in both cases $-P^* \leq [-Q]^+ = \Lambda$. For $\Delta > 0$, $x^* \geq x_B$ and $x_B - Q = \beta(1-\upsilon)(\psi_\Delta V - Q)/(1-\beta\upsilon)$, so $Q - x^* \leq \beta(1-\upsilon)\,[Q - \psi_\Delta V]^+/(1-\beta\upsilon)$.

(b) $\lambda = 0$ gives $\psi = 0$ and $P^* = \max\{Q, 0\}$. $\beta = 0$ gives $P^* = \max\{Q, \psi Q\}$. If $Q \leq 0$, both terms are nonpositive in each case.

(c) $P^* > 0$ if and only if $Q > 0$ or $\psi(Q + \beta(V-Q)) > 0$, and $\psi > 0$ if and only if $\lambda > 0$.

(d) Where $\psi(Q + \beta(V-Q)) > Q$, $P^* = \psi(1-\beta)Q + \psi\beta V$; differentiate in V, holding Q fixed or letting it vary. Where Q is the larger term, $P^* = Q$. The regime condition is the threshold $\hat{Q}$ of

Proposition 1(iii).

(e) In the formula for P^*, V appears only in $\beta\psi V$, since β, λ and r do not depend on V. This is agenda power under rupture risk. The remaining dependence runs through $Q = \omega - c$ (eq. (3)). In $\omega = \lambda_C \min\{W_C, \bar{m}_C\}$ with $W_C = (1-\theta)W + \rho V$, V enters through the rival's valuation, and $\partial\omega/\partial V = \lambda_C \rho$ where $W_C < \bar{m}_C$ (and 0 where the cap binds). In $c = \kappa S_k + d_p + d_{ai} + d_f + \eta$, the components that depend on V are $d_f = \chi_f f$, where $\chi_f = \mathbf{1}\{\varphi V \leq \gamma + \bar{m}_U\}$ falls from 1 to 0 as V rises past $(\gamma + \bar{m}_U)/\varphi$, which lowers c and raises Q by f (self-harming sanctions); and κS_k, where $\kappa = G(s_X + s_L) - G(s_X - m_S)$ depends on V through $s_X = s_0 + s_B(V)$ (the silicon shield). There $\partial\kappa/\partial s_X = g(s_X + s_L) - g(s_X - m_S)$, whose sign depends on the density g between the two points (Proposition 4(d) gives the case $m_S = \bar{m}_U$, where $\kappa = \bar{\kappa}$).

Nesting. $\lambda = 0$ gives $[Q]^+$ by (b). As $r \to 0$ with $\lambda > 0$, $\psi \to 1$ and $P^* \to \max\{Q, Q + \beta(V - Q)\} = Q + \beta(V - Q)$ because $V > Q$. $\beta = 0$ gives $\max\{Q, \psi Q\}$. Finally, P^* is continuous in $(\beta, \psi) \in [0,1]^2$ and equals $\max\{Q, V\} = V$ at $(\beta, \psi) = (1, 1)$, which is the limit $\beta \to 1$, $\lambda \to \infty$. □

A.4 Proof of Proposition 2

Assumptions. The setup is that of section 5.1. We fix six points. (1) *Payoffs.* Europe's aligned baseline payoff is 0 and its exit payoff is Q. With conceded set S and withdrawn set J_w at the end of the period, Europe's aligned payoff before rent is $\pi^A = -\Delta^A(J_w) - T_{\mathrm{r}}(S) - T_{\mathrm{s}}(S)$ and its exit payoff is $\pi^X = Q - \Delta^X(J_w) - T_{\mathrm{s}}(S)$. The replayed contest gives Europe $\max\{\pi^A, \pi^X\}$ and costs Washington the rent $\left[\pi^X - \pi^A\right]^+ = [Q + d(J_w) + T_{\mathrm{r}}(S)]^+$. Europe's payoff without demands is $[Q]^+$, and its *net loss* is $[Q]^+ - \max\{\pi^A, \pi^X\}$. (2) *Washington retains Europe.* V and $\bar{m}_U$ exceed every rent that can arise. Without this, Europe exits after some histories, its payoff is then at least π^X, and every upper bound on its loss below survives. (3) *Credible set.* J^c is fixed at the start of the period, every subset of J^c is a credible threat, and no instrument outside J^c can be withdrawn during the period. (4) *Execution costs.* Executing a withdrawal of J costs Washington $e(J) \geq 0$, because the baseline is its ex post optimal provision to a loyal ally. (5) *Protocol.* Each ultimatum names a nonempty set of unconceded demands and a nonempty threat set of unwithdrawn instruments. A threat that deters is not executed and stays available, and withdrawn instruments stay withdrawn. Washington may stop at any time. Every ultimatum either adds a concession or spends an instrument, so the game is finite. (6) *Ties.* Europe accepts when indifferent unless stated otherwise. The bounds in (c) hold for every tie-breaking rule.

Proof of Proposition 2. Step 0 (final-state payoffs). Write T_{r} and T_{s} for the totals of the conceded set S. Europe's payoff is $\max\{\pi^A, \pi^X\} = -u(J_w) - T_{\mathrm{s}} + \max\{Q, -d(J_w) - T_{\mathrm{r}}\}$. If $Q \geq 0$, the maximum is Q, because $d \geq 0$ and $T_{\mathrm{r}} \geq 0$. If $Q < 0$, then $-\max\{Q, -d - T_{\mathrm{r}}\} = \min\{\Lambda, d + T_{\mathrm{r}}\}$. In both cases Europe's net loss is

$$L(S, J_w) = T_{\mathrm{s}} + u(J_w) + \min\{d(J_w) + T_{\mathrm{r}}, \Lambda\} = T_{\mathrm{s}} + \min\{\Delta^A(J_w) + T_{\mathrm{r}},\ \Delta^X(J_w) + \Lambda\}.$$

The second form shows that L is nondecreasing in J_w and in S, because Δ^A and Δ^X are nondecreasing

(the conditional part d need not be). Washington's payoff is $W(S) - [Q + d(J_w) + T_{\mathrm{r}}]^+ - e(J_w)$.

(a) A refusal executes J when nothing has been conceded, so the loss is $L(\emptyset, J) = u(J) + \min\{d(J), \Lambda\} = B(J)$. Monotonicity of B follows from Step 0. If $Q \geq 0$, then $\Lambda = 0$ and $B = u$. (After a history with conceded set S_0 and withdrawn set J_0, the bite of J is $L(S_0, J_0 \cup J) - L(S_0, J_0)$, which in general differs from $B(J)$.)

(b) $L(S, \emptyset) = T_{\mathrm{s}} + \min\{T_{\mathrm{r}}, \Lambda\}$, which equals $\min\{T, \Lambda\}$ when all concessions are reversible. Accepting a single ultimatum gives the loss $L(S, \emptyset)$ and refusing it gives $B(J)$, so Europe accepts if and only if $\min\{T_{\mathrm{r}}, \Lambda\} + T_{\mathrm{s}} \leq B(J)$.

(c)(i) Fix a subgame-perfect equilibrium and let Europe deviate to refusing every ultimatum. On the resulting path nothing is conceded and every withdrawn instrument belongs to J^c, so the loss is $L(\emptyset, J_w) = B(J_w) \leq B(J^c)$. Europe's equilibrium loss is at most its loss from any deviation. The argument uses only that withdrawals come from J^c, so it holds for adaptive strategies, for every Q and for every tie-breaking rule. Next, $B(J) - u(J) = \min\{d(J), \Lambda\} \leq \Lambda$. Finally, for a fixed J_w, reversible concessions raise the loss by $\min\{d + T_{\mathrm{r}}, \Lambda\} - \min\{d, \Lambda\} \leq \min\{T_{\mathrm{r}}, \Lambda\} \leq \Lambda$.

(c)(ii) Let (S, J_w) be an equilibrium outcome. By (i) and Step 0, $L(S, \emptyset) \leq L(S, J_w) \leq B(J^c)$, so S is acceptable as a single ultimatum backed by J^c. Since $d \geq 0$ and $e \geq 0$, Washington's equilibrium payoff is $W(S) - [Q + d(J_w) + T_{\mathrm{r}}(S)]^+ - e(J_w) \leq W(S) - [Q + T_{\mathrm{r}}(S)]^+ \leq V_1$.

(c)(iii) By (b), Europe accepts a single ultimatum backed by J^c that asks for a maximizer S^* in the definition of V_1, so the ultimatum yields V_1. Now let Washington also have a divisible sunk demand: any amount $\phi \in [0, \bar{\phi}]$ of it costs Europe ϕ and is worth $\alpha\phi$ to Washington, with $\alpha > 0$. If $L(S, \emptyset) < B(J^c)$, adding $\phi = B(J^c) - L(S, \emptyset)$ keeps the bundle acceptable, leaves the rent unchanged (the demand is sunk) and raises Washington's payoff by $\alpha\phi > 0$. With $\bar{\phi} \geq B(J^c)$, every optimal bundle therefore has $L = B(J^c)$.

(d) For this part we read execution costs as positive for every nonempty set whenever they are positive for each instrument, as they are when e is additive across instruments. *All demands reversible and* $B(J^c) \geq \Lambda$. Every set S satisfies $L(S, \emptyset) = \min\{T_{\mathrm{r}}(S), \Lambda\} \leq \Lambda \leq B(J^c)$, so $V_1 = \max_S\{W(S) - [Q + T_{\mathrm{r}}(S)]^+\}$. Let S^* be an inclusion-maximal maximizer. Consider the subgame after Europe has conceded S^* with nothing withdrawn. Every terminal outcome (S', J_w) of that subgame has $S' \supseteq S^*$ and gives Washington at most $W(S') - [Q + T_{\mathrm{r}}(S')]^+ - e(J_w)$. This is strictly below V_1 if $S' \neq S^*$, by the maximality of S^*, and if $J_w \neq \emptyset$ when all $e(\{j\}) > 0$. So Washington stops, and Europe's loss is $L(S^*, \emptyset) \leq B(J^c)$. Refusing the root ultimatum (S^*, J^c) executes all of J^c, after which no ultimatum can be made, so the refusal loss is exactly $B(J^c)$, and Europe accepts. Washington thus obtains V_1 in every equilibrium, and by (c)(ii) no more. If some $e(\{j\}) = 0$, stopping is still optimal in the subgame, which gives an equilibrium with payoff V_1. *Some best single ultimatum asks for every demand.* Let S^* be such a bundle. After S^* is conceded no ultimatum can be made. Europe accepts (S^*, J^c) because $L(S^*, \emptyset) \leq B(J^c)$, which is the refusal loss. Washington can therefore secure V_1, and by (c)(ii) it obtains no more. *The ratchet.* Let the only instrument have $\Delta^A = \Delta^X = \bar{u}$, so $u = \bar{u} = B(J^c)$ and $d = 0$. Give Washington n sunk demands of cost τ and value $w > 0$ each, and let the execution cost be positive. The rent is always $[Q]^+$. Every

ultimatum is backed by the single instrument, and after a refusal no further ultimatum can be made. Let $b = \lfloor \bar{u}/\tau \rfloor$, and let $\mathcal{N}(m)$ be the number of further concessions from a state with m demands left and the instrument intact. We show by induction on m that $\mathcal{N}(m) = m \bmod (b+1)$. Clearly $\mathcal{N}(0) = 0$. An ultimatum asking for $x \in \{1, \ldots, m\}$ demands leads, if accepted, to $x + \mathcal{N}(m-x)$ concessions in total and an additional loss of $(x + \mathcal{N}(m-x))\tau$, while refusal costs $\bar{u}$. So Europe accepts if and only if $x + \mathcal{N}(m-x) \leq b$. By the induction hypothesis $x + \mathcal{N}(m-x) \equiv m \pmod{b+1}$, and this number is at least $x \geq 1$. Its only possible value in $\{1, \ldots, b\}$ is $\bar{n} = m \bmod (b+1)$, and $x = \bar{n}$ attains it. If $\bar{n} \geq 1$, every accepted ultimatum yields $\bar{n}$ concessions in total, and Washington, which values each concession, obtains $\bar{n}$. If $\bar{n} = 0$, every ultimatum is refused, and Washington stops because a refused ultimatum costs it the execution cost and gains nothing. So $\mathcal{N}(m) = \bar{n}$. With $n = b + 1$ Washington obtains nothing, whereas a single ultimatum would yield $\min\{n, b\} = b$ demands. □

A.5 Proof of Corollary 2

Proof of Corollary 2. For every Q and every $T \geq 0$, $[Q+T]^+ = [Q]^+ + [T-\Lambda]^+$: if $Q \geq 0$ then $\Lambda = 0$, and if $Q < 0$ then $Q + T = T - \Lambda$. So Washington's payoff from a conceded reversible set S with nothing withdrawn is $-[Q]^+ + W(S) - [T(S) - \Lambda]^+$. By Proposition 2(b), (c)(iii) and (d), Washington obtains the best single ultimatum backed by J^c, so it chooses S to maximize this payoff subject to $\min\{T(S), \Lambda\} \leq B(J^c)$, which is the stated program. If $B(J^c) < \Lambda$, the constraint reads $T(S) \leq B(J^c) < \Lambda$ and the rent term vanishes, which gives the knapsack with capacity $B(J^c) = \min\{\Lambda, B(J^c)\}$. Let demands be divisible with $w_k \leq t_k$ for every k, and let $B(J^c) \geq \Lambda$, so that the constraint is slack. At any y with $T > \Lambda$, lowering y_k by a small ε for some k with $y_k > 0$ changes the objective by $\varepsilon(t_k - w_k) \geq 0$, so some optimum has $T \leq \Lambda$. It solves the knapsack with capacity $\Lambda = \min\{\Lambda, B(J^c)\}$. Without divisibility the reduction fails. Take $\Lambda = B(J^c) = 1$ and one demand with $t = 1.5$ and $w = 1.4$. The knapsack value is 0, but Washington obtains $1.4 - 0.5 = 0.9$ by making the demand and paying rent 0.5. The free capacity $\min\{\Lambda, B(J^c)\}$ is nonincreasing in Q because $\Lambda = [-Q]^+$ is and $B(J^c) = u(J^c) + \min\{d(J^c), \Lambda\}$ is. □

A.6 Proof of Proposition 3

Assumptions. Those of Proposition 2. In part (a), China's payoff is $-\sum_{k \in S} \left(z_k + e_k^C \mathbf{1}\{r_k > 0\}\right)$ when Washington's demand set S is conceded, Washington makes a single ultimatum backed by J^c, and $S(r)$ denotes its optimal bundle under the profile r, unique at the profiles compared. In part (b), Washington demands the hedge when it gains weakly.

Proof of Proposition 3. (a)(i) A counter-threat on k is a sunk loss to Europe when k is conceded, which lowers π^A and π^X equally. Step 0 and part (b) of the proof of Proposition 2 then apply with $T_{\mathrm{s}}(S) + \sum_{k \in S} r_k$ in place of $T_{\mathrm{s}}(S)$.

(iv) Counter-threats leave Washington's payoff from every bundle unchanged, since the rent does not depend on r, and by (i) they shrink the set of acceptable bundles.

(ii) If $S(r) = S(0)$, China's payoff under r is $-\sum_{k \in S(0)}(z_k + e_k^C \mathbf{1}\{r_k > 0\})$, which is at most its payoff with no threats.

(iii) Let r^* be optimal for China, $S^* = S(r^*)$, $k \in S^*$ with $r_k^* > 0$, and let r' equal r^* except that $r_k' = 0$. If $S(r') = S^*$, China gains $e_k^C > 0$ by switching to r', which contradicts the optimality of r^*. So $S' = S(r') \neq S^*$. S^* is still acceptable under r' and gives Washington the same payoff, so by uniqueness Washington strictly prefers S' to S^*. Then S' was not acceptable under r^*, or Washington would have chosen it. Acceptable sets are closed under subsets, because the cost in (i) is monotone in S. Hence $S' \not\subseteq S^*$: some demand $j \in S' \setminus S^*$ has been crowded out. For an example, let $B(J^c) = 2$. Demand 1 is sunk with $t_1 = 0.3$, $w_1 = 1$ and $z_1 = 5$, and China cannot threaten it. Demand 2 is sunk with $t_2 = 1.5$, $w_2 = 20$ and $z_2 = 0$, and China can threaten it with $\bar{r}_2 = 0.4$ at $e_2^C = 0.1$. Without threats Washington demands both (cost $1.8 \leq 2$), and China's payoff is -5. Any $r_2 \in (0.2, 0.4]$ makes the pair cost more than 2, so Washington demands only demand 2, China executes, and its payoff rises to -0.1; threats with $r_2 \leq 0.2$ change nothing and cost China 0.1. So China's optimal profiles have $r_2 \in (0.2, 0.4]$, and the threat is executed on the path. All inequalities are strict, so the conclusion persists under small perturbations of the parameters.

(v) Europe accepts only bundles whose cost under (i) is at most $B(J^c)$, so its loss is at most $B(J^c)$ under every profile. If Washington also has the divisible sunk demand of the proof of Proposition 2(c)(iii) and China cannot threaten it, the argument there shows that every optimal bundle has loss exactly $B(J^c)$ under every profile, and the counter-threat changes only the mix of demands. Otherwise the loss can move either way. In the example of (iii), Europe's loss rises from 1.8 to 1.9. For a fall, let $B(J^c) = 2$, let demand 1 be sunk with $t_1 = 1.9$, $w_1 = 10$, $z_1 = 5$, threatenable with $\bar{r}_1 = 0.5$ at $e_1^C = 0.1$, and let demand 2 be sunk with $t_2 = 1$, $w_2 = 5$, $z_2 = 0$ and not threatenable. Without threats Washington demands only demand 1 (the pair costs 2.9), and Europe loses 1.9. With $r_1 = 0.5$ demand 1 costs 2.4, so Washington demands only demand 2, the threat is never executed, China's payoff rises from -5 to 0, and Europe's loss falls to 1.

(b) The hedge is a reversible demand with $t = y_H$. By Proposition 2(b), Europe accepts it if and only if $\min\{y_H, \Lambda\} \leq B(J^c)$. By the identity in the proof of Corollary 2, Washington's payoff changes by $w_H - [y_H - \Lambda]^+$ when the hedge is stopped, so it makes the demand if and only if $w_H \geq [y_H - \Lambda]^+$, and it pays the rent $[y_H - \Lambda]^+$. The demand is free if and only if $y_H \leq \Lambda$ and it is accepted, that is, $y_H \leq \min\{\Lambda, B(J^c)\}$, and its free part $\min\{y_H, \Lambda\}$ vanishes as $Q \to 0$ because Λ does. Blocking the hedge by flow jurisdiction is a threat with net ex post execution cost e_F, so under the commitment technology it is credible if and only if $e_F \leq \bar{m}_U$. Under complete information it deters and is not executed, and it involves no instrument whose withdrawal changes Europe's exit payoff relative to its aligned payoff, so Q is unchanged. Finally, executing a withdrawal of J lowers Europe's aligned payoff by $\Delta^A(J)$ and its exit payoff by $\Delta^X(J)$, so its premium rises to $Q + d(J)$ (Step 0 of the proof of Proposition 2). □

A.7 Proof of Proposition 4

Assumptions. H is convex and increasing on $[\kappa_0, \bar{\kappa}]$ with $H(\kappa_0) = 0$. H' is its right derivative, which is nondecreasing and right-continuous. $\nu \in [0, 1]$ and $S_k > 0$. The thresholds κ_x, κ_1, κ_ν and κ_z are as defined in section 5.3, so that $\kappa_0 \leq \kappa_\nu \leq \kappa_1$.

Proof of Proposition 4. Write $\Pi(\kappa) = \Pi_0(\kappa) - H(\kappa)$ with $\Pi_0(\kappa) = -[Q(\kappa)]^+ + \nu[-Q(\kappa)]^+$, and note that $Q(\kappa) = Q_0 - (\kappa - \kappa_0)S_k = (\kappa_z - \kappa)S_k$. Π_0 is continuous and piecewise linear, with slope S_k for $\kappa < \kappa_z$ and slope νS_k for $\kappa > \kappa_z$. Since $\nu \leq 1$, Π_0 is concave, so Π is concave, and strictly concave if H is strictly convex. Its right derivative,

$$\Pi'(\kappa) = S_k \mathbf{1}\{\kappa < \kappa_z\} + \nu S_k \mathbf{1}\{\kappa \geq \kappa_z\} - H'(\kappa),$$

is nonincreasing and right-continuous. So the set $\mathcal{P} = \{\kappa \in [\kappa_0, \bar{\kappa}] : \Pi'(\kappa) > 0\}$ is an interval $[\kappa_0, p)$ (empty when $p = \kappa_0$), Π is strictly increasing on $[\kappa_0, \min\{p, \bar{\kappa}\}]$ and nonincreasing to the right of p. Hence $\min\{\bar{\kappa}, \sup \mathcal{P}\}$ maximizes Π, with $\sup \emptyset = \kappa_0$, and it is the unique maximizer when Π is strictly concave. In the formulas below a threshold may be $+\infty$, which the outer minimum with $\bar{\kappa}$ handles. By right-continuity of H', $H'(\kappa) < x$ if and only if $\kappa < \kappa_x$.

Case $\kappa_z \leq \kappa_0$. On the whole interval $\Pi' = \nu S_k - H'$, so $\mathcal{P} = [\kappa_0, \kappa_\nu) \cap [\kappa_0, \bar{\kappa}]$ and the maximizer is $\min\{\bar{\kappa}, \kappa_\nu\}$. Because $\min\{\kappa_z, \kappa_1\} \leq \kappa_0 \leq \kappa_\nu$, this is the formula.

Case $\kappa_z > \kappa_0$. Here $\kappa \in \mathcal{P}$ if and only if $\kappa < \min\{\kappa_z, \kappa_1\}$, or $\kappa_z \leq \kappa < \kappa_\nu$. If $\kappa_1 < \kappa_z$, then $\kappa_\nu \leq \kappa_1 < \kappa_z$, the second set is empty, and $\sup \mathcal{P} = \kappa_1 = \max\{\kappa_\nu, \min\{\kappa_z, \kappa_1\}\}$. If $\kappa_1 \geq \kappa_z$, then $\mathcal{P} = [\kappa_0, \max\{\kappa_z, \kappa_\nu\})$ (intersected with $[\kappa_0, \bar{\kappa}]$), and again $\sup \mathcal{P} = \max\{\kappa_\nu, \min\{\kappa_z, \kappa_1\}\}$. In both cases $Q(\kappa^*) = (\kappa_z - \kappa^*)S_k$.

(a) If $\nu = 0$, then $\kappa_\nu = \kappa_0$ because $H' \geq 0$. The formula becomes $\min\{\bar{\kappa}, \max\{\kappa_0, \min\{\kappa_z, \kappa_1\}\}\}$, which equals κ_0 if $Q_0 \leq 0$ (then $\kappa_z \leq \kappa_0$) and $\min\{\kappa_z, \kappa_1, \bar{\kappa}\}$ otherwise. If $Q_0 \geq 0$, then $\kappa^* \leq \kappa_z$, so $Q(\kappa^*) \geq 0$ and $\Lambda^* = 0$. If $Q_0 < 0$, then $\Lambda^* = -Q(\kappa_0) = -Q_0$. In both cases $\Lambda^* = [-Q_0]^+$.

(b) Since $\kappa_z \in [\kappa_0, \bar{\kappa})$, $H'(\kappa_z) < \nu S_k$ means $\kappa_z < \kappa_\nu$, and $\kappa_\nu \leq \kappa_1$. So $\min\{\kappa_z, \kappa_1\} = \kappa_z$ and $\kappa^* = \min\{\bar{\kappa}, \kappa_\nu\} > \kappa_z$. Then $\Lambda^* = -Q(\kappa^*) = (\kappa^* - \kappa_z)S_k > 0$. If $Q_0 < 0$, the first case applies, $\kappa^* = \min\{\kappa_\nu, \bar{\kappa}\} \geq \kappa_0$, and $\Lambda^* = (\kappa^* - \kappa_z)S_k \geq (\kappa_0 - \kappa_z)S_k = -Q_0$.

(c) κ_1, κ_ν and $\bar{\kappa}$ do not depend on Q_0, and $\partial\kappa_z/\partial Q_0 = 1/S_k$. In the interior of each regime the active branch of the formula is locally constant in Q_0, except in regime (i), where it is κ_z. With $Q(\kappa^*) = (\kappa_z - \kappa^*)S_k$, rent $[Q(\kappa^*)]^+$ and slack $[-Q(\kappa^*)]^+$, the derivatives follow. In (i), $Q(\kappa^*) = 0$ and $d\kappa^*/dQ_0 = 1/S_k$. In (ii), and in (iv) with $Q(\bar{\kappa}) > 0$, $\partial Q(\kappa^*)/\partial Q_0 = 1$ and $Q(\kappa^*) > 0$, so the rent rises one for one. In (iii), and in (iv) with $Q(\bar{\kappa}) < 0$, $\partial Q(\kappa^*)/\partial Q_0 = 1$ and $Q(\kappa^*) < 0$, so slack falls one for one.

(d) $\bar{\kappa} = G(s_X + s_L) - G(s_X - \bar{m}_U)$. Differentiating in s_X with s_L and $\bar{m}_U$ fixed gives $g(s_X + s_L) - g(s_X - \bar{m}_U)$. If g is nonincreasing on $[s_X - \bar{m}_U, s_X + s_L]$, the derivative is nonpositive; this holds, for instance, when G is unimodal with its mode at or below $s_X - \bar{m}_U$. If $g(s_X + s_L) > g(s_X - \bar{m}_U)$, the derivative is positive. For example, with G standard normal, $s_L = 1$ and $\bar{m}_U = 2$, $\bar{\kappa} = 0.8186$ at $s_X = 0$ and 0.8664 at $s_X = 0.5$. If s_A is held fixed, $\partial\bar{\kappa}/\partial s_X = -g(s_X - \bar{m}_U) \leq 0$ for every G. The

statements about access value and about V follow from the chain rule with $s_X = s_0 + s_B$ and s_B nondecreasing in the access value and in V. □

A.8 Proof of Proposition 5

Assumptions. The setup is that of section 6.1. (1) *Timing.* The ultimatum is issued at the start of period 1, before the period-1 contest. After a transfer, Europe's period-1 premium is $Q_1 - \Delta_c$ and its period-1 rent is $[Q_1 - \Delta_c]^+$. After a refusal, the contest is played after the threat is carried out. (If the period-1 rent were instead settled before the ultimatum and could not be undone, the terms $[Q_1]^+ - [Q_1 - \Delta_c]^+$ would drop out of both ΔR_U and ΔR_H; the two timings differ only when $Q_1 > 0$.) (2) *Contests.* At every contest Washington retains Europe in the sense of Lemma 1: $Q_1 + d_1 < V$, $Q_1 + d_1 \leq \bar{m}_U$, $Q_2 + \bar{I} < V$ and $Q_2 + \bar{I} \leq \bar{m}_U$. The payment T is a spot payment at signing. (3) *Threat.* The threat withdraws in period 1 a bundle with unconditional part u_1, conditional part d_1 and direct ex post cost $g_1 \geq 0$ to Washington. Its execution cost is $e = g_1 + [Q_1 + d_1]^+ - [Q_1]^+$, and no other period-1 demand draws on the same bite. (4) *Ties.* An indifferent Europe accepts. Washington issues only if it gains strictly (the limit $\iota \to 0^+$ when $\iota = 0$). Europe invests only if that is strictly profitable. After a transfer Washington keeps its baseline provision unless withdrawal is strictly profitable, which it never is. (5) σ, r^c, v_c and ι do not depend on Δ_c; r^c is credible and is executed after any transfer (Corollary 3(d) endogenizes it); $\delta \in (0, 1]$; and for $\Delta_c = \infty$ we set $[-\infty]^+ = 0$ and $I(-\infty) = 0$.

Proof of Proposition 5. Period 2. After the period-1 outcome Europe holds a premium $q \in \{Q_2, Q_2 - \Delta_c\}$. If it invests, the period-2 contest (Lemma 1) gives it $\delta \left[q + \bar{I}\right]^+$, and otherwise $\delta [q]^+$. It invests if and only if $\delta(\left[q + \bar{I}\right]^+ - [q]^+) > K$, that is, if and only if $I(q) = 1$. So Europe's continuation value, in period-1 units and net of K, is $\mathrm{Val}_H(q)$, and Washington's continuation cost is $\mathrm{Rent}_U(q)$. The map $q \mapsto \left[q + \bar{I}\right]^+ - [q]^+$ is nondecreasing, so I is nondecreasing; hence $q + \bar{I}\, I(q)$ and Rent_U are nondecreasing, and $\mathrm{Val}_H(q) = \max\{\delta [q]^+, \delta \left[q + \bar{I}\right]^+ - K\}$ is nondecreasing.

No ultimatum. Europe gets $[Q_1]^+ + \mathrm{Val}_H(Q_2)$ and Washington gets $-[Q_1]^+ - \mathrm{Rent}_U(Q_2)$.

Acceptance. Control passes before the period-1 contest, so Europe's premia are $Q_1 - \Delta_c$ and $Q_2 - \Delta_c$. No stake is triggered, and withdrawing the bundle would cost Washington $g_1 + [Q_1 - \Delta_c + d_1]^+ - [Q_1 - \Delta_c]^+ \geq 0$, so Washington keeps its baseline provision. Europe gets $T - \sigma - r^c + [Q_1 - \Delta_c]^+ + \mathrm{Val}_H(Q_2 - \Delta_c)$, and Washington gets $v_c - \iota - T - [Q_1 - \Delta_c]^+ - \mathrm{Rent}_U(Q_2 - \Delta_c)$.

Refusal. Europe keeps control. If Washington carries out the threat, Europe's aligned payoff falls by $u_1 + d_1$ and its premium rises to $Q_1 + d_1$. By Lemma 1 Europe gets $-(u_1 + d_1) + [Q_1 + d_1]^+$ in period 1, and Washington gets $-g_1 - [Q_1 + d_1]^+$. If Washington does not carry it out, Europe gets $[Q_1]^+$ and Washington gets $-m - [Q_1]^+$, where $m \leq \bar{m}_U$ is the stake. Period 2 is as above with $q = Q_2$ in both cases. Washington carries out the threat if and only if $e \leq m$. Creating a stake costs nothing, so Washington can set $m = \bar{m}_U$, and the threat is carried out if and only if $e \leq \bar{m}_U$. None of these payoffs involves Δ_c, because the threat is carried out only when Europe keeps control. When the threat is carried out, Europe's period-1 loss relative to no ultimatum is

$u_1+d_1-[Q_1+d_1]^+ + [Q_1]^+$. It equals u_1 if $Q_1 \geq 0$, $u_1 - Q_1$ if $-d_1 < Q_1 < 0$, and $u_1 + d_1$ if $Q_1 \leq -d_1$, that is, $u_1 + \min\{d_1, \Lambda_1\} = B_1$ in all cases. So Europe's refusal payoff is $[Q_1]^+ - B_1 + \mathrm{Val}_H(Q_2)$, with $B_1 = 0$ if the threat is not credible.

Europe's decision. Europe accepts if and only if $T - \sigma - r^c + [Q_1 - \Delta_c]^+ + \mathrm{Val}_H(Q_2 - \Delta_c) \geq [Q_1]^+ - B_1 + \mathrm{Val}_H(Q_2)$, that is, if and only if $T \geq \Delta R_H - B_1 + \sigma + r^c$.

Washington's decision. With $T \geq 0$ and an indifferent Europe accepting, the cheapest accepted offer is $T^* = [\Delta R_H - B_1 + \sigma + r^c]^+$. Any lower offer is refused, and no optimal offer would exist if Europe refused at indifference. Relative to no ultimatum, the accepted offer gives Washington

$$v_c - \iota + \Delta R_U - T^* = v_c - \iota + \min\{\Delta R_U,\ \Delta R_U - \Delta R_H + B_1 - \sigma - r^c\} = \Gamma,$$

using $x - [y]^+ = \min\{x, x-y\}$. A refused ultimatum gives Washington $-\iota - e$ relative to no ultimatum when the threat is credible, and $-\iota - m \leq -\iota$ otherwise. Since $e \geq 0$, this is never strictly positive. Hence Washington issues if and only if $\Gamma > 0$, and every ultimatum issued is accepted. Europe's payoff is $T^* - \sigma - r^c + [Q_1 - \Delta_c]^+ + \mathrm{Val}_H(Q_2 - \Delta_c)$, and $T^* - \sigma - r^c = \max\{\Delta R_H - B_1,\ -\sigma - r^c\}$. □

The special case. If $v_c = \iota = 0$, $\Delta_c = \infty$ and $\bar{I} = 0$, then $\Delta R_U = \Delta R_H = [Q_1]^+ + \delta\,[Q_2]^+$, and substituting into Γ gives $\Gamma = \min\{[Q_1]^+ + \delta\,[Q_2]^+,\ B_1 - \sigma - r^c\}$. If also $Q_1 < 0$, then $\Gamma = \min\{\delta\,[Q_2]^+,\ B_1 - \sigma - r^c\}$. These are the expressions in the text after Proposition 5. Since $B_1 \leq u_1 + \Lambda_1$, a bundle with $u_1 = 0$ has $B_1 > \sigma + r^c$ only if $\Lambda_1 > \sigma + r^c$, that is, $Q_1 < -(\sigma + r^c)$.

A.9 Proof of Corollary 3

Assumptions. Those of Proposition 5. In part (d), r^c is replaced by China's announced punishment as described in the statement.

Proof of Corollary 3. (a) The first two statements restate $\Gamma > 0$, using $\min\{a, b\} > x$ if and only if $a > x$ and $b > x$. For the third, $[Q_1]^+ - [Q_1 - \Delta_c]^+ = \min\{[Q_1]^+, \Delta_c\}$, which is positive if and only if $Q_1 > 0$. Let $q = Q_2$ and $q' = Q_2 - \Delta_c < q$. Since I is nondecreasing, $q' + \bar{I}\,I(q') < q + \bar{I}\,I(q)$, so $\mathrm{Rent}_U(q) - \mathrm{Rent}_U(q') > 0$ if and only if $q + \bar{I}\,I(q) > 0$ (with $\delta > 0$). Both terms of ΔR_U are nonnegative, which gives the claim.

(b) The period-1 terms of ΔR_U and ΔR_H coincide, and $\mathrm{Rent}_U(q) - \mathrm{Val}_H(q) = K\,I(q)$, which gives the identity. It is nonnegative because I is nondecreasing. Without the investment option ($\bar{I} = 0$), both ΔR_U and ΔR_H equal $\Delta R^0 = [Q_1]^+ - [Q_1 - \Delta_c]^+ + \delta([Q_2]^+ - [Q_2 - \Delta_c]^+)$, so the second argument of the minimum in Γ is $B_1 - \sigma - r^c$; with the option it is larger by $K[I(Q_2) - I(Q_2 - \Delta_c)] \geq 0$. For the first argument, let $q = Q_2$, $q' = q - \Delta_c$ and $\mathcal{G}(x) = [x]^+ - [x - \Delta_c]^+$, which is nondecreasing. If $I(q) = I(q') = 0$, the period-2 terms with and without the option are equal. If $I(q) = I(q') = 1$, $\mathrm{Rent}_U(q) - \mathrm{Rent}_U(q') = \delta\,\mathcal{G}(q + \bar{I}) \geq \delta\,\mathcal{G}(q)$. If $I(q) = 1 > I(q')$, $\mathrm{Rent}_U(q) - \mathrm{Rent}_U(q') = \delta([q + \bar{I}]^+ - [q']^+) \geq \delta([q]^+ - [q']^+)$. The case $I(q) = 0 < I(q')$ cannot occur. So $\Delta R_U \geq \Delta R^0$. The statement on purchases without bite is (a) with $B_1 = 0$ and the identity just proved. In the example of the text ($\delta = 1$, $Q_1 = 1$, $Q_2 = -\frac{1}{2}$, $\bar{I} = 3$, $K = 2$, $\Delta_c = \infty$, $\sigma = 1$, $r^c = v_c = \iota = 0$,

no threat), $I(Q_2) = 1$ because $[2.5]^+ - 0 > 2$, and $I(-\infty) = 0$. So $\Delta R_U = 1 + 2.5 = 3.5$ and $\Delta R_H = 1 + 0.5 = 1.5$, which give $\Gamma = \min\{3.5,\, 2 + 0 - 1\} = 1$ and $T^* = [1.5 + 1]^+ = 2.5$.

(c) $[Q_1 - \Delta_c]^+$, $\text{Rent}_U(Q_2 - \Delta_c)$ and $\text{Val}_H(Q_2 - \Delta_c)$ are nonincreasing in Δ_c, because $[\cdot]^+$, Rent_U and Val_H are nondecreasing. So ΔR_U and ΔR_H are nondecreasing in Δ_c, and so is $\Delta R_U - \Delta R_H = K[I(Q_2) - I(Q_2 - \Delta_c)]$, because I is nondecreasing. Both arguments of the minimum in Γ are therefore nondecreasing in Δ_c, and T^* is nondecreasing because ΔR_H is. B_1 and e are computed in the refusal branch, where Europe keeps control, so they do not depend on Δ_c.

(d) Given an announcement, Europe and Washington play the game of Proposition 5 with $r^c = r$ if the threat is credible and $r^c = 0$ otherwise. $\Gamma(r)$ is nonincreasing in r. After a transfer China punishes if and only if $c_C(r) \leq m_C$. (i) Suppose control passes on the path. Then $\Gamma(r^c) > 0$ for the effective r^c, so $\Gamma(0) \geq \Gamma(r^c) > 0$. If $r^c > 0$, China punishes and gets $-c_C(r^c) < 0$. If the threat is not credible but $m_C > 0$, China forfeits m_C. In either case announcing $r = 0$ with no stake leaves the transfer in place, because $\Gamma(0) > 0$, and gives China 0, a profitable deviation. So on the path China gets 0, the effective r^c is 0, and the payment is $[\Delta R_H - B_1 + \sigma]^+$. (ii) By (i), a credible $r > 0$ on the path is not followed by a transfer, so Washington does not issue, which by Proposition 5 means $\Gamma(r) \leq 0$. (iii) If $\Gamma(0) \leq 0$, no announcement produces a transfer. If $\Gamma(r_{\max}) > 0$, every announcement produces a transfer and, by (i), China announces no credible punishment. If $\Gamma(r_{\max}) \leq 0 < \Gamma(0)$, announcing $r_{\max}$ with $m_C = \bar{m}_C$ deters and gives China $H_C > 0$, while any announcement that lets the transfer happen gives it at most 0, so China deters. The deterring credible announcements are those with $\Gamma(r) \leq 0$, that is, $r \geq v_c - \iota + \Delta R_U - \Delta R_H + B_1 - \sigma$; here we use $v_c - \iota + \Delta R_U > 0$, which follows from $\Gamma(0) > 0$. (iv) With $H_C = 0$, China gets 0 both from deterring and from announcing nothing and letting the transfer happen. Finally, if $c_C(r) < 0$ for some r, a transfer followed by punishment gives China $-c_C(r) > 0$, which can exceed H_C, so the argument of (i) fails. □

The argument in (d)(i) relies on complete information. Under the private information of Proposition 6, types that accept are punished on the path, so a punishment with negative ex post value is carried out with positive probability whenever Washington issues and $F(\hat{s}) > 0$.

A.10 Proof of Lemma 2

Proof of Lemma 2. Fix $s \in S$. Without the cut, Lemma 1 gives Europe a date-s payoff of $[Q_s]^+$ above its aligned baseline and costs Washington $[Q_s]^+$. With the cut, Europe's aligned payoff falls by $u_s + d_s$ and its exit payoff by u_s, so its premium becomes $Q_s + d_s$. Since Washington still retains Europe, Lemma 1 gives Europe $-(u_s + d_s) + [Q_s + d_s]^+$ and costs Washington $g_s + [Q_s + d_s]^+$. Europe's date-s loss is $u_s + d_s - [Q_s + d_s]^+ + [Q_s]^+$, which equals u_s if $Q_s \geq 0$, $u_s - Q_s$ if $-d_s < Q_s < 0$, and $u_s + d_s$ if $Q_s \leq -d_s$, that is, $u_s + \min\{d_s, \Lambda_s\}$. Washington's extra date-s cost is e_s. Because the cut changes no other date's premium, the date-s effects add up with weights δ^{s-1}, which gives (i) and the first part of (ii). At date s, once the condition of the threat is met, Washington's continuation does not depend on whether it withdraws, so it withdraws if and only if $-e_s \geq -m_s$.

For (iii), $Q_s \geq 0$ gives $\Lambda_s = 0$ and $[Q_s + d_s]^+ - [Q_s]^+ = d_s$. □

Scope. Both conditions in the lemma are needed, and their failures push in opposite directions. (1) If $Q_s + d_s \geq \min\{V_s, \bar{m}_U\}$, the cut drives Europe out at s. By Lemma 1 Europe's date-s loss is then $u_s + d_s - [\min\{Q_s + d_s, V_s, \bar{m}_U\}]^+ + [\min\{Q_s, V_s, \bar{m}_U\}]^+$, which exceeds the formula. (2) If Europe responds to the cut, for instance with the autonomy investment of section 6.1 when it learns of the cut before deciding, the bite is smaller. Take $\delta = 1$, $Q_2 = -1$, $d_2 = 1$, $u_2 = 0$, $\bar{I} = 3$ and $K = 2.5$. Without the cut Europe does not invest and gets 0. With it the premium is 0, Europe invests and gets $3 - 2.5 - 1 = -0.5$, so the bite is 0.5, while the formula gives 1.

A.11 Proof of Corollary 4

Assumptions. Those of Proposition 5 at every date, with $v_c = 0$, $\Delta_c = \infty$ and no investment option. A refusal leaves Europe in control at the next date. Washington may issue ultimatums at several dates.

Proof of Corollary 4. Let $R_s = \sum_{\tau \geq s} \delta^{\tau - s} [Q_\tau]^+$ be the value at date s of the rents from s on.

Step 1. We show by backward induction that if no date in $\{s, \ldots, N\}$ is eligible, no ultimatum is issued in the subgame that starts at s with Europe in control. Suppose this holds from $s+1$. Then at s the continuation after a refusal and after no ultimatum is the same: rents worth δR_{s+1} to Europe and $-\delta R_{s+1}$ to Washington. As in the proof of Proposition 5, with the period-2 terms replaced by δR_{s+1}, $\Delta_c = \infty$ and no investment, the cut is carried out after a refusal if and only if $e_s \leq \bar{m}_U$; Europe's refusal loss is b_s if so and 0 otherwise; and an accepted ultimatum gives Washington $-\iota + \min\{R_s, b_s^{\text{eff}} - \sigma - r^c\}$ relative to no ultimatum, where b_s^{eff} is b_s or 0. A refused ultimatum gives at most $-\iota$. Since s is not eligible, $R_s = 0$ or $b_s^{\text{eff}} \leq \sigma + r^c$, so Washington's gain from issuing is at most $-\iota \leq 0$, and it does not issue.

Step 2. Suppose some date is eligible and let t^* be the last one. By Step 1, no ultimatum is issued after t^*, so in the subgame at t^* with Europe in control Washington's gain from issuing is $-\iota + \min\{R_{t^*}, b_{t^*} - \sigma - r^c\}$. Eligibility gives $b_{t^*} > \sigma + r^c$ and $R_{t^*} > 0$ (some $Q_{t'} > 0$ with $t' \geq t^*$, and $\delta > 0$). Let $\bar{\iota} = \min\{R_{t^*}, b_{t^*} - \sigma - r^c\} > 0$. For $\iota < \bar{\iota}$, Washington issues at t^* if Europe still holds control there. Hence on the path an ultimatum is issued at t^* or earlier. If no date is eligible, Step 1 at $s = 1$ shows that none is issued.

Step 3. At any date, a refused ultimatum leaves Europe in control at the next date and costs Washington ι plus the execution cost $e_t \geq 0$ or the forfeited stake, so it is never strictly better than issuing nothing. Hence every ultimatum issued is accepted. With $\iota = 0$ and issuing only on a strict gain, Steps 1 to 3 apply with $\bar{\iota}$ replaced by any positive number.

Step 4. If $u_t = 0$, then $b_t = \min\{d_t, \Lambda_t\} > \sigma + r^c \geq 0$ requires $\Lambda_t > 0$, that is, $Q_t < 0$. So a date $t' \geq t$ with $Q_{t'} > 0$ satisfies $t' > t$. □

The condition $t' \geq t$ cannot be replaced by $t' > t$ when some $u_t > 0$. With $N = 2$, $\delta = 1$, $Q = (1, -5)$, $u_1 = 3$, $d_1 = g_1 = 0$ and $\sigma + r^c = 1$, no date has a later date with positive premium,

yet at date 1 Washington gains $\min\{[Q_1]^+ + \delta\,[Q_2]^+,\, u_1 - \sigma - r^c\} = 1 > 0$ and issues, because the transfer precedes the date-1 contest and removes the date-1 rent.

A.12 Proof of Remark 1

Assumptions. Those of Proposition 5. In the example of part (c), $\Delta_c = \infty$, there is no investment option, and $\bar{m}_U \geq 1$, so that the cut is credible after binding.

Proof of remark 1. (a) With $\Delta_b = \infty$, $[Q_t - \Delta_b]^+ = 0$, and Rent_U and Val_H vanish at both premia, before and after a transfer, so $\Delta R_U = \Delta R_H = 0$. Europe's period-1 slack is unbounded, so the bite of the threat is $u_1 + \min\{d_1, \infty\} = u_1 + d_1$, and its execution cost is $g_1 + [-\infty + d_1]^+ - 0 = g_1$. Proposition 5 then gives Γ^b and the payment $\left[0 - B_1^b + \sigma + r^c\right]^+$. Binding pre-empts the ultimatum if and only if $\Gamma^b \leq 0$, that is, $v_c \leq \iota$ or $v_c - \iota + B_1^b \leq \sigma + r^c$.

(b) After binding, Europe pays $\sigma^b + r^b$ and earns no rents. If $\Gamma^b > 0$ it also accepts the ultimatum and gets $\left[\sigma + r^c - B_1^b\right]^+ - \sigma - r^c = -\min\{\sigma + r^c, B_1^b\}$. Without binding, Proposition 5 with $Q_1 < 0$, $\Delta_c = \infty$ and no investment gives $\Delta R_H = \delta\,[Q_2]^+$, and Europe's payoff is $\max\{\delta\,[Q_2]^+ - B_1,\, -\sigma - r^c\}$ if $\Gamma > 0$ and $[Q_1]^+ + \delta\,[Q_2]^+ = \delta\,[Q_2]^+$ if $\Gamma \leq 0$. Comparing gives the statements. In the last case, $-\sigma^b - r^b - \mathbf{1}\{\Gamma^b > 0\}\min\{\sigma + r^c, B_1^b\} \leq 0 \leq \delta\,[Q_2]^+$.

(c) Without binding, $Q_1 = -1$ gives $B_1 = \min\{5, 1\} = 1$ (if the cut is credible; otherwise $B_1 = 0$) and $\Delta R_U = \Delta R_H = [Q_1]^+ + \delta\,[Q_2]^+ = 10$, so $\Gamma \leq \min\{10,\, 1 - 1.5\} < 0$. With $\Delta_b = 3$, the premia are -4 and 7, the cut's execution cost is $0 + [-4 + 5]^+ - 0 = 1 \leq \bar{m}_U$, $B_1 = \min\{5, 4\} = 4$ and $\Delta R_U = \Delta R_H = 7$. So $\Gamma = \min\{7,\, 4 - 1.5\} = 2.5$ and $T^* = [7 - 4 + 1.5]^+ = 4.5$.

(d) Washington's gain from demanding an arrangement is its Γ, by Proposition 5 applied with that arrangement's parameters. It demands the arrangement with the larger gain and issues only if that gain is positive. Since $\sigma_d < \sigma$, the second argument of the minimum in Γ_{dual} can be positive when that of Γ_{full} is not. □

By the argument of Corollary 3(d), if China's ex post value of punishing a binding rule is negative, a binding rule that Europe adopts on the path is not punished, and r^b matters only through deterrence. We have not checked this step numerically.

A.13 Proof of Proposition 6

Assumptions. Those of Proposition 5, with the conditions in the statement. Washington may also offer to buy control without staking the threat (a stake of zero), which is the purchase offer of part (b). Ties between accepting and refusing have probability zero.

Proof of Proposition 6. Acceptance. A type that accepts gets $T - \sigma - r^c$; no rents remain because $\Delta_c = \infty$. Refusing a threat-backed ultimatum gives $[Q_1]^+ - B_1 + R = R - B_1$ by the refusal step of the proof of Proposition 5, and refusing a purchase offer gives R. So Europe accepts if and only if $\sigma \leq T + \tilde{b} - R$, respectively $\sigma \leq T - r^c - R$.

Washington's gain. Relative to no offer, an accepted offer gives Washington $v_c + R - T - \iota$. A refused threat-backed ultimatum gives $-e - \iota$, since $e \leq \bar{m}_U$ and Washington carries out the threat; a refused purchase offer gives $-\iota$. A stake in $(0, e)$ is never executed and is forfeited after a refusal, so it is dominated by a stake of zero. Write the cutoff as $s = T + \tilde{b} - R$ for the threat and $s = T - r^c - R$ for the purchase offer. Washington's gain is then $\Pi(s; \tilde{b} + v_c, e) - \iota$ subject to $s \geq \tilde{b} - R$, respectively $\Pi(s; v_c - r^c, 0) - \iota$ subject to $s \geq -r^c - R$, where the constraints are $T \geq 0$.

Shape of Π. For $s \leq 0$, $\Pi(s; A, x) = -x$. For $s \geq \bar{\sigma}$, $\Pi = A - s$, which is decreasing. On $(0, \bar{\sigma})$, $\partial\Pi/\partial s = f(s)(A - s) - F(s) + f(s)x = f(s)\left[A + x - s - F(s)/f(s)\right]$. Log-concavity of F means that $(\log F)' = f/F$ is nonincreasing, so F/f is nondecreasing and the bracket is strictly decreasing in s. As $s \downarrow 0$, $\log F(s) \to -\infty$; a concave function with a bounded right derivative near 0 would stay bounded, so $f/F \to \infty$ and $F/f \to 0$. Hence the bracket is positive near 0 if and only if $A + x > 0$, it changes sign at most once, and Π is nondecreasing on $(-\infty, s^*(A, x)]$ and nonincreasing on $[s^*(A, x), \infty)$, with s^* as defined in the statement.

(a) A function that rises up to s^* and falls after it attains its maximum over $s \geq L$ at $\max\{L, s^*\}$. With $L = \tilde{b} - R$ and $s^* \leq \bar{\sigma}$, this is $\hat{s}$, and the payment is $T = \hat{s} - \tilde{b} + R \geq 0$. Refusal, and with it the execution of the threat, has probability $1 - F(\hat{s})$. For uniform F, $F(s)/f(s) = s$, so $s^* = (\tilde{b} + v_c + e)/2$ when this is below $\bar{\sigma}$.

(b) Here $L = -r^c - R \leq 0$ and Π is constant on $(-\infty, 0]$, so the constraint never binds and the cutoff is s_0. If $v_c \leq r^c$, then $\Pi(s; v_c - r^c, 0) = F(s)(v_c - r^c - s) \leq 0$ for $s \geq 0$, and $\Pi = 0$ for $s \leq 0$, so $\Pi_0 \leq -\iota$.

(c) Washington compares no offer (0), the best threat-backed ultimatum (Π_B) and the best purchase offer (Π_0). If $v_c \leq r^c$, then $\Pi_0 \leq -\iota \leq 0$, so Washington makes an offer if and only if $\Pi_B > 0$, and every offer is threat-backed; $\Pi_B > 0$ is the stated inequality.

(d) The bracket is increasing in $A + x$, so $s^*(A, x)$ is nondecreasing in $A + x = B_1 - r^c + v_c + e$. Hence $\hat{s} = \max\{B_1 - r^c - R,\, s^*\}$ is nondecreasing in B_1, v_c and e, nonincreasing in r^c and R, and independent of ι, and $P_b = 1 - F(\hat{s})$ moves the opposite way. Without log-concavity: $\partial\Pi/\partial A = F(s)$ and $\partial\Pi/\partial x = F(s) - 1$ are nondecreasing in s, so Π has increasing differences in (s, A) and in (s, x). The feasible set $[B_1 - r^c - R, \infty)$ rises with B_1 (as does A), falls with r^c (as does A) and falls with R (with A unchanged), and it does not depend on v_c or e. Π is continuous and decreasing beyond $\bar{\sigma}$, so the set of maximizers is nonempty and compact. By Topkis's monotonicity theorem it moves in the same direction as the parameter in the strong set order, so its largest and smallest elements are monotone.

(e) Two facts. First, $\hat{s} \geq s_0$, because $\hat{s} \geq s^*(\tilde{b} + v_c, e) \geq s^*(v_c - r^c, 0) = s_0$, as $(\tilde{b} + v_c + e) - (v_c - r^c) = B_1 + e \geq 0$. Second, the direction in which Washington can switch. In terms of the payment, the threat-backed gain is $\pi_B(T) = F(T + \tilde{b} - R)(v_c + R - T + e) - e - \iota$. Only $T \leq v_c + R + e$ can be optimal, since a larger T makes acceptance worse for Washington than refusal. On that range $\pi_B(T)$ is nondecreasing in B_1 and nonincreasing in e (its derivative in e is $F - 1 \leq 0$), so Π_B is nondecreasing in B_1 and nonincreasing in e. In the cutoff form, R does not enter $\Pi(s; \tilde{b} + v_c, e)$ and only relaxes the constraint $s \geq \tilde{b} - R$, so Π_B is nondecreasing in R. Π_0 depends on none of B_1, e and

R. For v_c: in the cutoff forms the feasible sets do not depend on v_c, and each objective is affine in v_c with slope $F(s)$, so by the envelope theorem $\partial\Pi_B/\partial v_c = F(\hat{s}) \geq F(s_0) = \partial\Pi_0/\partial v_c$ wherever the derivatives exist, and $\Pi_B - \Pi_0$ is nondecreasing in v_c. So as B_1, R or v_c rises, Washington can switch only from the purchase offer to the threat, and as e rises only from the threat to the purchase offer. Within each option the cutoff moves as in (d); for the purchase offer, s_0 is nondecreasing in v_c and independent of B_1, e and R. Combining: as B_1 or v_c rises, the cutoff never falls (a switch moves it from s_0 up to $\hat{s}$), so the refusal probability is nonincreasing. As e rises, the breakdown probability is nonincreasing within the threat and drops to zero at a switch. As R rises, it is nondecreasing within the threat and rises from zero at a switch. For r^c: where the threat-backed payment is positive, the constraint does not bind, and the envelope theorem gives $\partial\Pi_B/\partial r^c = -F(\hat{s}) \leq -F(s_0) = \partial\Pi_0/\partial r^c$; the purchase payment $s_0 + r^c + R$ is positive when $v_c > r^c$, because then $s_0 > 0$. So switches go from the threat to the purchase offer, the cutoff falls from $\hat{s}$ to s_0, and the refusal probability rises; within each option it is nondecreasing in r^c by (d) and because s_0 is nonincreasing in r^c. For the remaining directions, take uniform F on $[0, 10]$, $R = 0$ and $\iota = 0$. Raising e from 0.2 to 20 (with $B_1 = 1$, $r^c = 0$, $v_c = 8$) moves Washington from the threat (refusal probability 0.54) to the purchase offer (refusal probability 0.6). Raising B_1 from 1 to 2 (with $e = 1$, $v_c = 8$, $r^c = 0$) moves it from the purchase offer (breakdown probability 0) to the threat (0.45). Raising v_c from 8 to 10 (with $B_1 = e = 1$, $r^c = 0$) moves it from the purchase offer to the threat (breakdown probability from 0 to 0.4). Raising r^c from 0 to 2 (with $B_1 = e = 1$, $v_c = 10$) moves it from the threat (breakdown probability 0.4) to the purchase offer (0). □

A.14 Proof of Remark 2

Assumptions. The benefit x_h is part of the baseline: it is provision that Washington finds ex post optimal for a loyal Netherlands, such as a preferential tier of the AI instrument, and its withdrawal on exit is credible (its execution cost is at most $\bar{m}_U$). The conditions of Lemma 1 hold before and after the benefit, except where the statement treats the case $Q_{NL} > \min\{V, \bar{m}_U\}$.

Proof of remark 2. Credible withdrawal on exit adds x_h to the conditional part of Washington's instruments for the Netherlands, so c_{NL} rises by x_h and Q_{NL} falls by x_h. Lemma 1 gives the payment $[Q]^+$ at premium Q. For $x \geq 0$: if $Q \leq 0$, then $[Q]^+ - [Q - x]^+ = 0 = \min\{x, [Q]^+\}$ and $[x - Q]^+ - [-Q]^+ = x = \left[x - [Q]^+\right]^+$; if $Q > 0$, then $Q - [Q - x]^+ = \min\{x, Q\}$ and $[x - Q]^+ - 0 = \left[x - [Q]^+\right]^+$. With $Q = Q_{NL}$ and $x = x_h$, these are the stated changes in the payment and in the slack. If $Q_{NL} > \min\{V, \bar{m}_U\}$, Lemma 1(ii) applied to $Q'_{NL} = Q_{NL} - x_h$ gives alignment if and only if $Q_{NL} - x_h \leq \bar{m}_U$ and $Q_{NL} - x_h < V$. If instead x_h is a promised increment above the baseline, it is a transfer paid on the path: the Netherlands aligns if and only if $x_h + \tau \geq Q_{NL}$, so Washington's cheapest total outlay is $x_h + [Q_{NL} - x_h]^+ = \max\{x_h, Q_{NL}\} \geq [Q_{NL}]^+$. □

A.15 Proof of Remark 3

Assumptions. Under a licence, the firm compares shipping, worth $y - f^F$, with complying with US measures, worth $-p_C - p_B$, and ships when indifferent. The penalty $f^F = \chi_f f$ is the credible one, so it is zero when the threat to penalize the firm is not credible. p_B is positive only if the US measures are listed in the annex of the Blocking Statute. The compulsion cost ζ excludes the firm's own net loss, which already enters ω and c through y and d_f.

Proof of remark 3. The firm ships if and only if $y - f^F \geq -p_C - p_B$, that is, $y + p_C + p_B \geq f^F$. If it ships, exit needs nothing further and Europe's premium is Q. If it does not, exit requires the state to compel it at cost ζ, which lowers Europe's exit payoff, and hence its premium, by ζ. So Europe's effective premium is $Q - \zeta\,\mathbf{1}\{y + p_C + p_B < f^F\}$. By Lemma 1, Europe exits only if its effective premium is positive, so it compels the firm only if $Q - \zeta > 0$. □

A.16 Proof of Remark 4

Assumptions. The determination is a Council implementing act adopted by qualified majority on a Commission proposal (European Parliament and Council of the European Union, 2023). Under Article 16(4) TEU and Article 238(3)(a) TFEU, a qualified majority needs at least 55 percent of the members (15 of 27) representing at least 65 percent of the population, and a blocking minority must include at least four members, failing which the qualified majority is deemed attained (European Union, 2016). Abstention and absence count as not voting in favour. Response measures are Commission implementing acts under the examination procedure; in the appeal committee the Commission may adopt unless a qualified majority votes against (European Parliament and Council of the European Union, 2011). A Council objection to a delegated act needs at least 72 percent of the members (20 of 27) representing at least 65 percent of the population (Article 238(2) TFEU). Member i withholds its vote from the countermeasure if it is paid $\tau_i \geq q_i$, payments are enforceable and contingent on the vote, and no one bids for adoption. Population shares are those of Council Decision 2025/2427 (Council of the European Union, 2025).

Proof of remark 4. The blocking family. Let N be the set of members that do not vote in favour and Y its complement. The determination fails if and only if $|N| \geq 4$ and either $|Y| < 15$ or Y holds less than 65 percent of the population. Since $|Y| < 15$ is $|N| \geq 13$, and Y holds less than 65 percent if and only if N holds more than 35 percent, N blocks if and only if $|N| \geq 13$, or $|N| \geq 4$ and N holds more than 35 percent of the population. This is the family $\mathcal{B}$. It is upward closed: adding members raises both the count and the population.

The cost of blocking. A member with $q_i > 0$ must be paid q_i to withhold its vote, and a member with $q_i \leq 0$ withholds it for nothing, so blocking with a coalition J costs $\sum_{i \in J} [q_i]^+$, and the cheapest block costs $\Phi(0) = \min_{J \in \mathcal{B}} \sum_{i \in J} [q_i]^+$. Adding the members of $F = \{i : q_i \leq 0\}$ to any blocking coalition costs nothing and preserves blocking, because $\mathcal{B}$ is upward closed. So some cheapest blocking coalition contains F, and $\Phi(0)$ is the cost of the cheapest completion of F into a member

of $\mathcal{B}$. If $F \in \mathcal{B}$, then $\Phi(0) = 0$. Conversely, if $\Phi(0) = 0$, a cheapest blocking coalition consists of members with $[q_i]^+ = 0$, that is, it is a subset of F, and $F \in \mathcal{B}$ by upward closure.

The coalitions. The six frontline states have fewer than 13 members and 15.09 percent of the population, so they do not block. The thirteen eastern and Nordic members of table 2 hold 26.77 percent and block by number.

Later stages. Defeating a response measure in the appeal committee requires a qualified majority against, hence at least 15 members, and a Council objection to a delegated act requires at least 20. Either coalition has at least 13 members, so it belongs to $\mathcal{B}$, and paying its members to vote against costs at least $\sum_{i \in J} [q_i]^+ \geq \Phi(0)$. □

A.17 Proof of Proposition 7

Assumptions. Rounds start at $t_k = k\Delta$. After a rejection, if Washington proposed, Europe may exit, which gives Europe Q and Washington $-V$ in normalized flow units. Otherwise rupture occurs with probability $\mu = 1 - e^{-\lambda\Delta}$, with the same values. Otherwise the status quo runs for the period with normalized flows $-c_E(t_k)$ and $-c_U(t_k)$ (weight $1 - \delta$, with $\delta = e^{-r\Delta}$), and play moves to round $k + 1$. The costs are measured in the same normalized flow units as Q and V. They are bounded, each interval of unit length contains at most N_d of their discontinuity points, and between discontinuities they are uniformly continuous. "Markov" means that strategies depend only on the round index (calendar time) and the current offer. We write $\upsilon = \delta(1 - \mu) = e^{-(r+\lambda)\Delta}$, $c_{i,k} = c_i(t_k)$ and $n_k = n(t_k)$, and $\bar{a}$ denotes a constant that does not depend on Δ and may change from line to line.

Proof of Proposition 7. Recursion. Fix Δ and a Markov perfect equilibrium, and let x_k and y_k be Europe's and Washington's values at the start of round k. In a Markov equilibrium the continuation after a rejection does not depend on the rejected offer. If Washington rejects Europe's demand in round k, its value is $\varrho_k = -\mu V + (1 - \mu)[-(1 - \delta)c_{U,k} + \delta y_{k+1}]$. If Europe rejects Washington's offer and does not exit, its value is $d_k = \mu Q + (1 - \mu)[-(1 - \delta)c_{E,k} + \delta x_{k+1}]$, so its value of rejecting is $\max\{Q, d_k\}$. A responder accepts an offer exactly equal to its rejection value, since otherwise the proposer would have no best offer. So the proposer's best acceptable offer gives Europe $-\varrho_k$ when Europe proposes and Washington $-\max\{Q, d_k\}$ when Washington proposes.

Every outcome is an agreement (whose payoffs sum to zero), exit (summing to $Q - V < 0$) or disagreement flows (summing to at most zero), so $x_k + y_k \leq 0$ for every k. The proposer's gain from its best acceptable offer over an unacceptable one is, when Europe proposes,

$$\begin{aligned} -\varrho_k - d_k &= \mu(V - Q) + (1 - \mu)(1 - \delta)(c_{E,k} + c_{U,k}) - (1 - \mu)\delta(x_{k+1} + y_{k+1}) \\ &\geq \mu(V - Q) + (1 - \mu)(1 - \delta)(c_{E,k} + c_{U,k}), \end{aligned}$$

and when Washington proposes it is the same expression if Europe would not exit after rejecting, and $V - Q > 0$ if it would. The gain is strictly positive if $\lambda > 0$ (because $Q < V$) or $c_{E,k} + c_{U,k} > 0$, and then agreement in round k is immediate. If the gain is zero, then $\mu = 0$, $c_{E,k} + c_{U,k} = 0$ and

$x_{k+1} + y_{k+1} = 0$, and an unacceptable offer gives both players the same payoffs as an acceptable one, so delay is payoff equivalent. In either case the payoffs in round k sum to zero, so $y_k = -x_k$ for every k. Substituting $y_{k+1} = -x_{k+1}$,

$$\begin{aligned} x_k = \mathcal{T}_k(x_{k+1}) &\equiv \beta\big[\mu V + (1-\mu)\big((1-\delta)c_{U,k} + \delta x_{k+1}\big)\big] + (1-\beta)\max\{Q, d_k\} \\ &= \max\{\mathcal{T}_k^A(x_{k+1}), \mathcal{T}_k^B(x_{k+1})\}, \end{aligned}$$

$$\mathcal{T}_k^A(x) = \mu M + (1-\mu)(1-\delta)n_k + \upsilon x, \qquad \mathcal{T}_k^B(x) = (1-\beta)Q + \beta\big[\mu V + (1-\mu)(1-\delta)c_{U,k} + \upsilon x\big],$$

where $M = Q + \beta(V - Q) = \beta V + (1-\beta)Q$ and $n_k = \beta c_{U,k} - (1-\beta)c_{E,k}$.

Part (i). Each $\mathcal{T}_k$ is nondecreasing with modulus $\upsilon < 1$, so the map $(x_k)_k \mapsto (\mathcal{T}_k(x_{k+1}))_k$ is a contraction on bounded sequences in the supremum norm. Equilibrium values are bounded, so every Markov perfect equilibrium has the same values, the unique fixed point. The profile built from the fixed point (each proposer makes its best acceptable offer, each responder accepts exactly the offers at or above its rejection value, and Europe exits after rejecting if and only if $Q > d_k$) is an equilibrium by the one-shot deviation principle. The statements on immediate agreement and payoff-equivalent delay were shown above.

Part (ii), policies. For a sequence $\sigma = (\sigma_k)$ with $\sigma_k \in \{\mathcal{T}_k^A, \mathcal{T}_k^B\}$, let v^σ be the bounded solution of $v_k^\sigma = \sigma_k(v_{k+1}^\sigma)$. Since $\sigma_k \le \mathcal{T}_k$ and $\mathcal{T}_k$ is a monotone contraction, $v^\sigma \le x$, and the sequence that picks the larger map at each round attains x. So $x_0 = \max_\sigma v_0^\sigma$.

The limit problem. Let $\mathcal{Z}(s) = \lambda M + r\, n(s)$, let $\mathcal{J}_t(\tau) = \int_t^\tau e^{-(r+\lambda)(s-t)}\mathcal{Z}(s)\,ds + e^{-(r+\lambda)(\tau-t)}Q$ for $\tau \in [t, \infty]$, and let $\mathcal{V}(t) = \sup_{\tau \ge t}\mathcal{J}_t(\tau)$. $\mathcal{V}$ is bounded, $\mathcal{V} \ge Q$ (take $\tau = t$), and, by taking $\tau \ge t + \Delta$, $\mathcal{V}(t) \ge \int_t^{t+\Delta} e^{-(r+\lambda)(s-t)}\mathcal{Z}(s)\,ds + \upsilon\,\mathcal{V}(t+\Delta)$. The last inequality and boundedness give $\mathcal{V}(t+\Delta) \le \mathcal{V}(t) + \bar{a}\Delta$.

One-step errors. Since $\mu = \lambda\Delta + O(\Delta^2)$, $(1-\mu)(1-\delta) = r\Delta + O(\Delta^2)$ and $\int_0^\Delta e^{-(r+\lambda)s}ds = \Delta + O(\Delta^2)$,

$$\mu M + (1-\mu)(1-\delta)n_k = \int_{t_k}^{t_{k+1}} e^{-(r+\lambda)(s-t_k)}\mathcal{Z}(s)\,ds + \varepsilon_k, \qquad |\varepsilon_k| \le \bar{a}\Delta^2 + r\Delta\,\mathrm{osc}_{[t_k, t_{k+1}]}n.$$

Let ϖ be the modulus of continuity of n between discontinuities. In a round that contains no discontinuity, $\mathrm{osc}\, n \le \varpi(\Delta)$. A unit interval of time $[m, m+1)$, with m a nonnegative integer, contains at most N_d discontinuities, so at most $2N_d$ of its rounds contain one; for those rounds $\mathrm{osc}\, n \le 2\sup|n|$, and their index is at least $\lfloor m/\Delta \rfloor$. Hence, for weights $w_k \le \bar{\upsilon}^k$ with $\bar{\upsilon} = \max\{\upsilon, \beta\} < 1$,

$$\sum_k w_k|\varepsilon_k| \le \frac{\bar{a}\Delta^2 + r\Delta\varpi(\Delta)}{1-\bar{\upsilon}} + 4N_d\, r \sup|n|\,\Delta \sum_{m=0}^{\infty} \bar{\upsilon}^{\lfloor m/\Delta \rfloor} \;\longrightarrow\; 0 \quad (\Delta \to 0).$$

For the first term, $1 - \bar{\upsilon} \ge 1 - \upsilon$ and $\Delta/(1-\upsilon) \to 1/(r+\lambda)$, so it is $O(\Delta) + \varpi(\Delta)\, r/(r+\lambda)$. For the second, $\bar{\upsilon}^{\lfloor m/\Delta \rfloor} \le \bar{\upsilon}^{-1}(\bar{\upsilon}^{1/\Delta})^m$ and, for $\Delta \le 1$, $\bar{\upsilon}^{1/\Delta} \le \max\{e^{-(r+\lambda)}, \beta\} < 1$, so the sum over m is bounded and the term is $O(N_d\Delta)$.

Lower bound. Fix $\tau < \infty$ and let $K = \lceil \tau/\Delta \rceil$. Consider the policy that uses $\mathcal{T}^A$ in rounds

$k < K$ and $\mathcal{T}^B$ afterwards. Using $\mathcal{T}^B$ in every round from K on gives values with $v_k - Q = \beta[\mu V + (1-\mu)(1-\delta)c_{U,k} - (1-\upsilon)Q] + \beta\upsilon(v_{k+1} - Q)$ for $k \geq K$, where the bracket is $O(\Delta)$, so $|v_K - Q| \leq \bar{a}\Delta/(1-\beta)$. The policy's value is therefore $\sum_{k<K} \upsilon^k[\mu M + (1-\mu)(1-\delta)n_k] + \upsilon^K v_K = \mathcal{J}_0(K\Delta) + \sum_{k<K} \upsilon^k \varepsilon_k + \upsilon^K(v_K - Q)$, which tends to $\mathcal{J}_0(\tau)$ as $\Delta \to 0$. Hence $\liminf x_0 \geq \mathcal{J}_0(\tau)$ for every finite τ, and the case $\tau = \infty$ is a limit of finite τ. So $\liminf x_0 \geq \mathcal{V}(0)$.

Upper bound. Fix any policy σ and let $e_k = [v_k^\sigma - \mathcal{V}(t_k)]^+$. By the one-step error bound and the inequality for $\mathcal{V}$, $\mathcal{V}(t_k) \geq \mathcal{T}_k^A(\mathcal{V}(t_{k+1})) - \varepsilon_k$, so after a $\mathcal{T}^A$ step $e_k \leq \upsilon e_{k+1} + |\varepsilon_k|$. Since $Q \leq \mathcal{V}(t_k)$ and $\mathcal{V}(t_{k+1}) \leq \mathcal{V}(t_k) + \bar{a}\Delta$, $\mathcal{T}_k^B(\mathcal{V}(t_{k+1})) \leq (1-\beta)\mathcal{V}(t_k) + \beta\mathcal{V}(t_k) + \bar{a}\Delta$, so after a $\mathcal{T}^B$ step $e_k \leq \beta e_{k+1} + \bar{a}\Delta$. Iterating, e_0 is at most the sum of the step errors, each weighted by the product of the earlier factors, plus a remainder that vanishes because every factor is at most $\bar{\upsilon} < 1$ and e is bounded. The $\mathcal{T}^A$ errors carry weights at most $\bar{\upsilon}^k$ and contribute a term that tends to zero, as shown above. The m-th $\mathcal{T}^B$ step carries a weight at most β^{m-1}, so all of them together contribute at most $\bar{a}\Delta/(1-\beta)$. Hence $e_0 \to 0$ uniformly in σ, $\limsup x_0 \leq \mathcal{V}(0)$, and $P^* = \mathcal{V}(0)$, which is the first expression in (ii).

Part (iii) and the second expression. By definition of $\hat{C}_i$ and N, $\int_\tau^\infty e^{-(r+\lambda)s}\mathcal{Z}(s)\,ds = e^{-(r+\lambda)\tau}[\psi M + (1-\psi)N(\tau)] = e^{-(r+\lambda)\tau}x^\circ(\tau)$, because $\lambda/(r+\lambda) = \psi$ and $r/(r+\lambda) = 1-\psi$. So $\mathcal{J}_0(\tau) = x^\circ(0) - e^{-(r+\lambda)\tau}(x^\circ(\tau) - Q)$ and $\mathcal{J}_0(\infty) = x^\circ(0)$, which gives the second expression for P^*, and shows that $P^* = x^\circ(0)$ if and only if $x^\circ(\tau) \geq Q$ for every τ. Next, $(r+\lambda)(x^\circ(t) - Q) = \lambda\beta(V - Q) + r(N(t) - Q)$, so $x^\circ(t) \geq Q$ if and only if $Q(r + \lambda\beta) \leq \lambda\beta V + rN(t)$, which is the condition $Q \leq \bar{Q}$ once the infimum over t is taken. Applying the same argument from any date t shows that the limit value at t is $\mathcal{V}(t) = x^\circ(t)$ for every t exactly under this condition, so the exit option is never used. With constant costs x° is constant and $P^* = x^\circ + [Q - x^\circ]^+ = \max\{Q, x^\circ\}$. The condition at $t = 0$ alone is not sufficient. Take $\beta = \lambda = 0$, $r = \frac{1}{2}$, $c_U \equiv 0$, $c_E = 0$ on $[0, 2)$ and $c_E = 2$ afterwards, and $Q = -1$. Then $x^\circ(t) = -\hat{C}_E(t)$, so $x^\circ(0) = -2e^{-1} \approx -0.736 \geq Q$, but $x^\circ(t) = -2 < Q$ for $t \geq 2$. The supremum in the second expression is attained at $t = 2$, so $P^* = x^\circ(0) + e^{-1}(Q + 2) = -e^{-1} \approx -0.368$, which exceeds $x^\circ(0)$. □

A.18 Proof of Corollary 5

Assumptions. Baseline disagreement flows are c_E^0 and c_U^0, as in Proposition 7; they may include Washington's cost of cutting a loyal Europe's AI access from date zero. In a round with interruption, Europe's disagreement flow rises by ℓ, and Washington's rises by $(1-h)c_S$ only if the buffer stock is exhausted at the start of the round. The stock falls by Δ in each round with interruption and is unchanged otherwise, starting at t_U. We characterize Markov perfect equilibria, in which strategies depend on calendar time, the stock and the current offer. Let $D_0 = \beta(1-h)c_S - (1-\beta)\ell$, the value of D at $t_U = 0$, and let $x_0^\circ(t)$ be $x^\circ(t)$ of Proposition 7 under the baseline costs, with $x_0^\circ = x_0^\circ(0)$. Variants (a) and (c) are proved, like part (i), for the case in which exit never binds without interruption.

Proof of Corollary 5. Limit problem. The state is the round and the stock b. Interruption is chosen

before the proposer is drawn, and both proposers' offers depend on the choice only through the round's disagreement flows and the next state, so Europe chooses it to maximize its own value, and its value satisfies a recursion of the form in the proof of Proposition 7 with an additional maximization over $i_k \in \{0,1\}$. The convergence argument of that proof, applied state by state, gives as the limit value the supremum over measurable $i : [0,\infty) \to \{0,1\}$ and stopping times τ of

$$\mathcal{J}(i,\tau) = \int_0^\tau e^{-(r+\lambda)s}\big(\mathcal{Z}_0(s) + r\,\tilde{n}(s)\big)\,ds + e^{-(r+\lambda)\tau}Q,$$
$$\tilde{n}(s) = i(s)\big[-(1-\beta)\ell + \beta(1-h)c_S\mathbf{1}\{b(s)=0\}\big],$$

where $\mathcal{Z}_0 = \lambda M + r(\beta c_U^0 - (1-\beta)c_E^0)$ and $b(s) = [t_U - \int_0^s i]^+$. Let T_b be the first date with $b = 0$; $T_b \geq t_U$, and $T_b = \infty$ if the stock is never exhausted.

Key bound. For every i and τ, $r\int_0^\tau e^{-(r+\lambda)s}\tilde{n}(s)\,ds \leq (1-\psi)\,[D]^+$, and if $T_b \leq \tau$ and $D_0 \geq 0$, the left side is at most $(1-\psi)e^{-(r+\lambda)(T_b - t_U)}D$. If $T_b > \tau$, the integrand is $-(1-\beta)\ell\, i \leq 0$. Otherwise, before T_b the set of interruption dates has measure t_U and the integrand is $-(1-\beta)\ell\, i$. Among sets of measure t_U in $[0,T_b]$, the discounted measure $\int_0^{T_b} e^{-(r+\lambda)s}i(s)\,ds$ is smallest for $[T_b - t_U, T_b]$, where it equals $e^{-(r+\lambda)T_b}(e^{(r+\lambda)t_U} - 1)/(r+\lambda)$. After T_b the integrand is at most $[D_0]^+$. So the left side is at most

$$\frac{r}{r+\lambda}\,e^{-(r+\lambda)T_b}\Big[-(1-\beta)\ell\big(e^{(r+\lambda)t_U} - 1\big) + [D_0]^+\Big].$$

If $D_0 \geq 0$, the bracket equals $\beta(1-h)c_S - (1-\beta)\ell e^{(r+\lambda)t_U} = e^{(r+\lambda)t_U}D$, so the bound is $(1-\psi)e^{-(r+\lambda)(T_b-t_U)}D \leq (1-\psi)\,[D]^+$, since $T_b \geq t_U$. If $D_0 < 0$, the bracket is nonpositive, and $D \leq D_0 < 0$.

Part (i). Since exit never binds without interruption, $\sup_\tau \mathcal{J}(0,\tau) = \mathcal{J}(0,\infty) = x_0^\circ$. By the key bound, $\mathcal{J}(i,\tau) \leq \mathcal{J}(0,\tau) + (1-\psi)\,[D]^+ \leq x_0^\circ + (1-\psi)\,[D]^+$. If $D > 0$, the policy $i \equiv 1$, $\tau = \infty$ attains this: it exhausts the stock at $T_b = t_U$ and adds $\frac{r}{r+\lambda}\big[-(1-\beta)\ell(1-e^{-(r+\lambda)t_U}) + D_0 e^{-(r+\lambda)t_U}\big] = (1-\psi)D$. If $D \leq 0$, $i \equiv 0$ is optimal and interruption does not raise P^*. Since $r > 0$, $1-\psi > 0$, and D is strictly decreasing in t_U when $\beta(1-h)c_S > 0$, so $D > 0$ if and only if $t_U < \bar{t}$.

Part (ii). With constant baseline costs, $\alpha = \mathcal{Z}_0 - (r+\lambda)Q$ is constant and $x_0^\circ - Q = \alpha/(r+\lambda)$. Then $\mathcal{J}(i,\tau) - Q = \int_0^\tau e^{-(r+\lambda)s}(\alpha + r\tilde{n})\,ds$. If $T_b > \tau$, this is at most $\int_0^\tau e^{-(r+\lambda)s}\alpha\,ds \leq [\alpha]^+/(r+\lambda)$. If $T_b \leq \tau$, apply the bathtub step on $[0,T_b]$ and bound the integrand on $[T_b,\tau]$ by $[\alpha + rD_0]^+$, which gives

$$\mathcal{J} - Q \leq \frac{1}{r+\lambda}\Big[\alpha\big(1 - e^{-(r+\lambda)T_b}\big) + e^{-(r+\lambda)T_b}\Big([\alpha + rD_0]^+ - r(1-\beta)\ell\big(e^{(r+\lambda)t_U} - 1\big)\Big)\Big].$$

If $\alpha + rD_0 \geq 0$, then, since $D_0 - (1-\beta)\ell(e^{(r+\lambda)t_U} - 1) = e^{(r+\lambda)t_U}D$, the bound equals $\alpha/(r+\lambda) + (1-\psi)e^{-(r+\lambda)(T_b-t_U)}D$. This is affine in $e^{-(r+\lambda)(T_b-t_U)} \in (0,1]$, so it is at most $\max\{\alpha/(r+\lambda),\ \alpha/(r+\lambda) + (1-\psi)D\}$. If $\alpha + rD_0 < 0$, the bound is at most $[\alpha]^+/(r+\lambda)$. Hence $\mathcal{J} \leq \max\{Q,\ x_0^\circ,\ x_0^\circ + (1-\psi)D\}$, and the three terms are attained by $\tau = 0$, by $(i \equiv 0, \tau = \infty)$ and by $(i \equiv 1, \tau = \infty)$. Without interruption the value is $\max\{Q, x_0^\circ\}$ by Proposition 7, so interruption raises P^* if and only if $(1-\psi)D > [Q - x_0^\circ]^+$. This requires $D > 0$, the condition of part (i), and when exit binds $(Q > x_0^\circ)$

it requires more.

Part (iii). In the setting of part (i), exit never binds from any date, and the problem from state (t, b) is the problem of part (i) with t_U replaced by b. Attaining the bound $x_0^\circ(t) + (1-\psi)\,[D(b)]^+$, where $D(b) = \beta(1-h)c_S e^{-(r+\lambda)b} - (1-\beta)\ell$, requires interrupting continuously from t when $D(b) > 0$, because any delay multiplies the gain by $e^{-(r+\lambda)(T_b - t - b)} < 1$; and when $D(b) < 0$ every interruption lowers the value. So at stock b Europe interrupts if and only if $\beta(1-h)c_S e^{-(r+\lambda)b} > (1-\beta)\ell$. Along an interruption path the stock falls one for one with time, and the left side rises as b falls, so the condition keeps holding. Without interruption b stays at t_U, so a condition that fails at $t = 0$ fails in every round. (With constant costs and exit binding, the condition at stock b is $(1-\psi)D(b) > [Q - x_0^\circ]^+$ by part (ii), which is also monotone in b, so the decision is time consistent there as well.)

(a) Calendar buffer. Now $\tilde{n}(s) = i(s)\big[-(1-\beta)\ell + \beta(1-h)c_S \mathbf{1}\{s \geq t_U\}\big]$, which does not depend on past choices and can be maximized date by date: $i(s) = 1$ if and only if $s \geq t_U$ and $D_0 > 0$. With $\tau = \infty$ optimal, this adds $r\int_{t_U}^{\infty} e^{-(r+\lambda)s}\,[D_0]^+\,ds = (1-\psi)e^{-(r+\lambda)t_U}\,[D_0]^+$, and interrupting before t_U strictly lowers the value when $\ell > 0$.

(b) Choice after a rejection. Europe then chooses i ex post to maximize its own disagreement value $\mu Q + (1-\mu)[-(1-\delta)(c_E^0 + \ell i) + \delta x(b_i')]$, where $b_1' = [b - \Delta]^+$ and $b_0' = b$; the expression is the same after Europe rejects an offer and after Washington rejects a demand. At $b = 0$ interruption leaves the state unchanged and costs Europe $(1-\mu)(1-\delta)\ell > 0$, so $i = 0$, and $x(0) = y_0$, the value without interruption. Suppose $x(b-\Delta) = y_0$. If Europe interrupted at b, its value there would be x_1, the value of the recursion of Proposition 7 with Europe's disagreement flow raised by ℓ for one round and continuation y_0; since the two recursions differ only in Europe's disagreement value, which enters through the exit-floor term, $y_0 - x_1 \in [0, (1-\beta)(1-\mu)(1-\delta)\ell]$. Interrupting would then change Europe's disagreement value by $(1-\mu)[-(1-\delta)\ell + \delta(y_0 - x_1)] \leq (1-\mu)(1-\delta)\ell\,[-1 + \delta(1-\beta)(1-\mu)] < 0$, so interruption is not a best response. Not interrupting at b gives the value y_0 and is a best response. So $i = 0$ is the unique consistent choice and $x(b) = y_0$, and induction over b completes the argument.

(c) Choice after the proposer is drawn. When Europe proposes, its accepted demand equals Washington's disagreement value. Interruption raises Washington's disagreement cost by $(1-\mu)(1-\delta)(1-h)c_S\mathbf{1}\{b = 0\}$ plus $(1-\mu)\delta[x(b-\Delta) - x(b)] \geq 0$, where the second term is nonnegative because a lower stock only brings forward the date from which interruption costs Washington something and Europe nothing. Europe's own cost ℓ does not enter, because the demand is accepted. So Europe interrupts in all of its proposer rounds. Under the policy that interrupts in those rounds only, the stock falls by Δ in each of them; by the law of large numbers, in the frequent-offer limit it falls at rate β per unit of time and is exhausted at t_U/β, after which the flow gain is $r\beta(1-h)c_S$ with no cost. This policy therefore adds $r\int_{t_U/\beta}^{\infty} e^{-(r+\lambda)s}\beta(1-h)c_S\,ds = (1-\psi)\beta(1-h)c_S e^{-(r+\lambda)t_U/\beta}$, which bounds the gain from below and is positive whenever $\beta(1-h)c_S > 0$ and $t_U < \infty$. □

B Numerical Verification

We checked every result in appendix A against solutions computed by brute force from the primitives of the model. The scripts are in the folder `code/checks` and run with `python3 -I` (some need numpy or scipy); each finishes in under thirty seconds. No solver uses the formula it checks. Equilibria are found by enumeration, backward induction or direct optimization, and the closed forms are computed separately and compared with the result. Where a statement depends on a tie-breaking rule, the script either solves the game under that rule or computes every equilibrium outcome with no tie-breaking at all. Comments inside the scripts number the results as in our verification notes; the file `code/checks/README.md` maps them to the numbering used here.

Lemma 1. `v2_m1_lemma1.py` works in exact rational arithmetic and solves three versions of the auction. In the clock auction, 300 random draws on an integer lattice ($\lambda_C \in \{1, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{1}{3}, \frac{5}{6}\}$, caps binding or not, Q negative, zero or positive) enumerate all pairs of drop levels on a grid of mesh 1/12. Dropping at one's maximal credible bid is weakly dominant in every draw, and Europe's payoff matches $[\min\{Q, V, \bar{m}_U\}]^+$ exactly (draws with $Q = V$ excluded). In the alternating-raise auction, 800 games (400 draws, each with either bidder moving first) are solved for the full set of pure subgame-perfect outcomes by set-valued backward induction, with no tie-breaking. The winner matches the lemma in all 580 games with Q more than one grid step below $\bar{z}_U$ and in all 44 with Q more than one step above it; the largest payoff to Europe across equilibria is within one grid step of $[\min\{Q, V, \bar{m}_U\}]^+$ in all 800; and in 108 games the smallest is lower by more than a grid step (for example $Q = \frac{7}{2}$, with equilibrium payoffs $0, 1, 2, 3, 4$), which is the price indeterminacy of the claim in appendix A. Single-valued backward induction on larger grids under four tie-breaking rules (70 draws each) finds no failure of the winner, and when the losing bidder raises whenever indifferent the price is within one grid step of $[Q]^+$. Edge cases: $Q < 0$, $Q = 0$, $Q = \bar{m}_U < V$, $Q = V$, $\bar{m}_U < Q < V$ (exit in every equilibrium, and every $\tau \in [Q, V)$ makes both Europe and Washington better off), $Q > V$, a binding cap $\bar{m}_C < W_C$, and $\lambda_C = \frac{1}{2}$.

Proposition 1 and Corollary 1. `v2_m1_prop1.py` runs finite-horizon backward induction in which each proposer compares every offer on a 601-point grid, together with the responder's point of indifference, with the payoff of an unacceptable offer; Europe compares exit with continuation; and Washington's value is tracked separately rather than set to minus Europe's. Over 500 runs (250 draws with two random terminal values each, $\Delta \in \{1, 0.5, 0.2, 0.1\}$, $\beta = 0$ and $\lambda = 0$ included), the largest gap between Europe's value and $x^*(\Delta)$ is 1.7×10^{-11}, the largest sum of the two values is 2.9×10^{-11}, and with $\lambda > 0$ no proposer ever weakly prefers delay. For the limit, backward induction at $\Delta = 0.004, 0.002, 0.001$ with Richardson extrapolation on 120 draws ($\beta \leq 0.9$) gives a largest error against P^* of 7.5×10^{-6}. Uniqueness is checked in two ways: limits of backward induction from six terminal conditions, including infeasible ones, differ by at most 4.5×10^{-12} (60 draws), and iterating the four supremum and infimum maps of Step 4 from the box $[-10V, 10V]^2$ converges to $(x^*, x^*, -x^*, -x^*)$ within 4.4×10^{-12} (300 draws). At $\lambda = 0$ the stationary profiles with delay

pass a one-shot deviation check. Using the numerical limits, parts (a) to (c) of Corollary 1 hold in every draw (120 draws; 43 with $\lambda = 0$; 59 with $\beta = 0$); central finite differences in V match $\beta\psi$ to 6.6×10^{-10} where the second term binds (25 draws); the threshold $\hat{Q}$ classifies every draw; and the nesting limits are approached as stated. For the alternative timing of the footnote, the stationary value matches its piecewise formula to 1.6×10^{-14} on 300 draws (42 with mixing), the nonstationary profile that gives Europe close to V passes a one-shot deviation check in all 67 draws in which it was built, and at $\Delta = 10^{-5}$ the stationary value is within 5×10^{-5} of P^*. Near $\beta = 1$ (with $Q = 1.5$, $V = 2$, $\lambda = r = 1$, so $\psi V < Q$), the gap $P^* - x^*(\Delta)$ follows the approximation in appendix A and is still 0.33 at $\beta = 0.999$ and $\Delta = 10^{-3}$. Finally, $Q = -1$, $V = 1$, $\beta = \frac{1}{4}$, $\lambda = r = 1$ give $P^* = -\frac{1}{4}$ both by backward induction and by the formula.

Proposition 2, Corollary 2 and Proposition 3(b). `v2_m3_prop2.py` draws instances with one to four instruments and one to four demands. Set functions are additive (40 percent of draws) or general monotone, built so that the conditional part is often not monotone; Q is negative, zero or positive; demands are reversible or sunk; execution costs are zero or positive; and integer draws create ties. Payoffs come only from the final-state primitives. The full set of subgame-perfect outcomes of the adaptive protocol is computed by set-valued backward induction over states (instruments not yet withdrawn, demands conceded), with every ultimatum allowed and Europe either accepting when indifferent or, in a separate run, free to respond either way. A separate solver finds the best single ultimatum. The bite $B(J)$ and the loss $\min\{T, \Lambda\}$ match to 8.9×10^{-16} over 3,000 instances, and the acceptance rule of part (b) has no mismatch in 123,300 pairs of threat set and bundle. The bound of part (c) is never violated in 4,000 adaptive instances (1,013 with $Q > 0$; largest excess 1.1×10^{-16}). Washington's equilibrium payoff never exceeds V_1 and falls strictly below it in 325 of 2,000 continuous instances. Where a sufficient condition of part (d) holds (1,593 instances), the adaptive and single-ultimatum values coincide in every case, and the ratchet count $n \bmod (b+1)$ matches in all 300 draws; with $n = 1, \ldots, 6$ identical demands ($\tau = 1$, $B(J^c) = 1.5$) the adaptive protocol yields $2, 0, 2, 0, 2, 0$ against 2 for a single ultimatum. When a refusal can make a new instrument credible, Europe's loss exceeds the bite of the initial credible set by up to 4.9 but never that of the enlarged set, which is why the proposition fixes J^c. For Corollary 2, a plain knapsack with capacity $\min\{\Lambda, B(J^c)\}$ misses Washington's optimum in 521 of 1,046 instances with reversible demands only, and the stated program in none. For Proposition 3(b), the linkage condition has no mismatch in 3,000 draws with the hedge as the only demand. The hand-picked cases include $B(J^c) = b\tau$ exactly, $Q > 0$, the example of a deliberately burned instrument and the knapsack counterexample.

Proposition 3(a). `v2_m3_prop2e.py` lets China choose counter-threats from a grid (four or five levels on each demand it can threaten) before Washington's single bundled ultimatum. Over 4,000 instances the acceptance rule of (a)(i) has no mismatch in 13,286 checks. China gains in 615 instances, in every one of them by changing Washington's demand set. A counter-threat is executed on the path in 140 instances (151 threats); dropping any one of them would change Washington's

demand set, and each crowds out another demand. Washington never gains. Europe's loss rises in 89 instances, falls in 405 and is unchanged in 3,506, and with a divisible sunk demand that China cannot threaten it equals $B(J^c)$ to 8.9×10^{-16} with and without threats. The numerical example of the text is reproduced, and in 300 instances of the adaptive protocol every executed counter-threat also changes the demand set.

Proposition 4. `v2_m3_prop3.py` draws 1,500 problems with H from six convex families (quadratic with and without a linear term, exponential, power with exponent 1.5 to 4, kinked piecewise linear, and linear, which produces ties), $\kappa_0 \in [0, 0.3]$, $\bar{\kappa} - \kappa_0 \in [0.05, 0.75]$, $S_k \in [0.5, 10]$, Q_0 negative, zero or positive, and ν equal to 0, 1 or uniform on $[0, 1]$. Each problem is maximized on a grid of 200,001 points with golden-section polishing, and the thresholds come from bisection on H'. The formula for κ^* is never beaten (largest payoff shortfall 8.9×10^{-16}; largest gap in the maximizer 3.3×10^{-8} when it is unique), and all five regimes occur (the κ_0 corner 381 times, $\kappa_1 < \kappa_z$ 257, $\kappa_\nu > \kappa_z$ 173, κ_z 187, the cap $\bar{\kappa}$ 502). Finite differences in Q_0 over strictly convex draws give exactly the derivatives of part (c) in each regime. For part (d), 3,000 draws of $(s_X, s_L, \bar{m}_U)$ per distribution show $\bar{\kappa}$ rising with s_X in 1,218 normal, 1,198 logistic, 974 exponential and 973 uniform cases, and the derivative formula matches finite differences to 3×10^{-11}.

Proposition 5 and Corollary 3(a) to (c). `v2_m4_prop4.py` solves the two-period game by backward induction from primitives: Europe's investment by comparing its two payoffs, the period rents from the bidding benchmark, Washington's execution decision after a refusal by comparing ex post payoffs that include the forfeited stake, its withdrawal decision after acceptance, its stake in $\{0, \bar{m}_U\}$ and payment on a grid, Europe's response, and the issue decision by strict comparison with no ultimatum. The draws are nine hand-picked edge cases (both premia zero; $\sigma + r^c = B_1$; $\Gamma = 0$; $e = \bar{m}_U$; the investment tie; $g_1 = 0$; finite $\Delta_c < Q_2$; an unconditional bite with $Q_1 > 0$; a purchase without bite), 20,000 draws of small integers and dyadic δ on a grid of mesh 1/64, so that all ties are exact, and 10,000 continuous draws on a grid of mesh 10^{-3} (319 draws within 3×10^{-3} of $\Gamma = 0$ set aside). The issue decision never differs from the proposition. The errors in Γ, T^* and Europe's payoff are exactly zero on the integer draws and below the grid mesh on the continuous ones. Of 7,715 draws with an ultimatum, none has a refusal on the path. Under the opposite tie rule for investment, the answer changes in 139 knife-edge draws, and the formulas with a weak inequality in I match all of 5,000 further draws. In 2,000 draws the execution decision after a refusal is the same for eight values of Δ_c. On the same draws: no issued ultimatum violates the conditions of Corollary 3(a); the identity of (b) holds to 2.7×10^{-15}, and Γ with the investment option is never below Γ without it; the purchase without bite gives $\Gamma = 1$ and $T^* = 2.5$; in 3,000 draws solved for $\Delta_c \in \{0, 0.5, 1, 2, 3, 4, 6, 9, \infty\}$, Γ never falls as Δ_c rises (part (c)); and the special case after Proposition 5 matches in 1,504 draws, exactly on integer draws.

Corollary 3(d). `v2_m4_prop4_china.py` adds a first stage in which China announces a punishment r on a grid (step $\frac{1}{2}$, up to $\bar{r} \leq 6$) and a stake in $\{0, \bar{m}_C\}$, with $c_C(r) = \gamma r$, $\gamma \in \{\frac{1}{2}, 1, 2\}$

and $H_C \in \{0, \frac{1}{4}, 1\}$, and solves the full game for each announcement. In 5,000 integer draws, every China-optimal announcement satisfies (i) to (iii), and the payment on the path matches $[\Delta R_H - B_1 + \sigma]^+$ (529 draws with a transfer, 394 with deterrence). With $H_C = 0$ the outcome is not unique in 183 draws, as (iv) allows. With a self-rewarding punishment ($\gamma = -\frac{1}{2}$), China punishes a transfer on the path in 968 draws.

Lemma 2 and Corollary 4. `v2_m4_cuts_windows.py`, part A, draws 20,000 games with one to five dates, random cut sets, unconditional parts zero or positive, and δ equal to one or random. Europe's loss and Washington's cost are computed by comparing, date by date, the outcomes of the bidding contest with and without the cut. In the 13,927 draws inside the lemma's scope the largest errors are 3.6×10^{-15} for the bite and 1.8×10^{-15} for the cost, and the identities of part (iii) hold to 1.3×10^{-15}. In all 6,073 draws outside the scope the true loss exceeds the formula, and the investment example gives a bite of 0.5 against the formula's 1. Part B solves the multi-date game by backward induction in exact rational arithmetic, with execution decided by comparing Washington's ex post payoffs and a stake in $\{0, \bar{m}_U\}$. It draws 60,000 games with two to six dates, integer premia in $[-6, 4]$, $\delta \in \{1, \frac{1}{2}, \frac{9}{10}\}$ and unconditional parts possible in 24,111, each solved with $\iota = 0$ and strict issuing and with $\iota = 10^{-9}$. The condition with $t' > t$ misclassifies 1,030 draws, all with some $u_t > 0$; the condition of Corollary 4 matches every draw for both values of ι. Of the 20,199 games with an ultimatum, 13,923 issue it at the last eligible date and 1,202 at a date that is not itself eligible. No refused ultimatum is ever optimal.

Remark 1. `v2_m4_selfbind.py` solves 6,000 integer draws with $Q_1 < 0$, $\Delta_c = \infty$ and no investment, with and without binding, using the solver of `v2_m4_prop4.py` with shifted premia and the bite recomputed from primitives. Part (a) has no mismatch. The payoff comparison of part (b) matches all 6,000 draws, whereas a comparison of $-\sigma^b - r^b$ with Europe's payoff under the ultimatum alone fails in 999, none of them a case in which Washington would issue and binding pre-empts it. Partial binding ($\Delta_b \in \{1, \ldots, 5\}$) invites an ultimatum that would not otherwise be issued in 210 draws, and the example of part (c) gives no ultimatum and a payoff of 10 to Europe without binding, and an ultimatum with $T^* = 4.5$ and a payoff of 3 before σ^b with $\Delta_b = 3$. Washington's choice between full control and a dual key matches part (d) in every draw (dual key chosen in 370).

Proposition 6. `v2_m4_prop5.py` maximizes Washington's gain directly over the payment $T \geq 0$ (a grid of 4,001 points, local refinement and an explicit corner at $T = 0$), for both the threat-backed ultimatum and the purchase offer, without using the first-order condition. It covers six families of distributions for σ: uniform, truncated normal, Beta(2,2), triangular, truncated exponential with a rate of either sign, and a bimodal normal mixture that is not log-concave, with 1,500 draws each ($\bar{\sigma} \in [0.5, 8]$, $B_1 \in (0, 6)$, r^c zero or in $[0, 3]$ so that $\tilde{b} < 0$ occurs, $e \in (0, 6)$, v_c zero or in $[0, 4]$, ι zero or in $[0, 1]$, R zero or in $[0, 6]$). For the five log-concave families, after setting aside 40 to 53 draws per family within 10^{-6} of the issuance threshold, the issuance condition of part (c)

matches in every draw, the largest error in the cutoff is 3.4×10^{-7}, and the largest error in P_b is 2.2×10^{-7}. The corner $T = 0$ binds in 293 to 388 draws per family. The purchase offer strictly beats the threat in 104 to 133 draws per family and never when $v_c \leq r^c$. For the bimodal family the first-order characterization fails in 124 draws, as expected. For the comparative statics, each parameter is raised by a random step in $[0.05, 1.5]$ in 500 draws per family, keeping comparisons in which Washington issues before and after. For threat-backed ultimatums there is no violation of part (d) in any family, the bimodal one included. With the purchase option, violations occur only in the directions that part (e) says can fail, and the four examples in the proof are reproduced.

Remarks 2 to 4. `v2_m2_remarks_GFE.py` checks the three remarks. For remark 2 it finds by brute force, on a grid of mesh 10^{-3}, the smallest staked promise that keeps the gatekeeper aligned and the largest reversible concession it accepts, before and after the benefit, in 20,000 draws that include $Q_{NL} = 0$, $x_h = 0$ and $x_h = |Q_{NL}|$; the largest errors are 4.4×10^{-16} for the payment and 8.9×10^{-16} for the slack, and a benefit paid as a transfer never saves Washington anything. For remark 3, 200,000 integer draws, many of them ties, reproduce the firm's choice and Europe's effective premium in every case. For remark 4 the script reads the population shares of Council Decision 2025/2427 (Council of the European Union, 2025) from `pop_check.py`, which must sit in the same folder. The first-stage blocking rule matches the treaty rule, including the four-member clause, on all 403,168 sets of members tested, and the rule for keeping a delegated act in force matches on 241,708 sets. The script computes the coalitions of table 2 and the cost $\Phi(0)$ exactly, taking the cheapest thirteen members for the route by number and solving a dynamic program over member count and population for the route by population; the program matches brute force on 30 random fourteen-member councils to 1.8×10^{-15}.

Council arithmetic for the scenario. `council_revealed_blocking.py` uses the exact populations of Council Decision 2025/2427 and the EUROPOP2023 baseline projections for 2029 to 2031, and applies the qualified-majority rule in integer arithmetic. For the coalition named in the *Europe 2031* scenario it reports the population share (28.06 percent in 2026, 27.76 percent in 2031) and the members whose addition would complete a block; for two orderings of members, by revealed positions in 2025 and 2026 and by structural exposure, the first blocking prefix; blocking under informal supermajority norms; and exact Shapley–Shubik and normalized Banzhaf indices, computed by counting swing coalitions and checked against a Monte Carlo estimate from 200,000 random orderings. Section 7.3 and appendix C use these numbers.

Proposition 7. `v2_m2_prop6_holdup.py` solves the discrete game by backward induction on the grid of offer dates, without using the formula. Costs are random piecewise-constant paths with up to three breaks, constant after a random date of at most 4; the exact stationary fixed point of the discrete recursion after that date, exit floor included, is the terminal value. It draws 6,000 parameter sets with $r \in (0.05, 1.5)$, $\lambda = 0$ in 20 percent and $\beta = 0$ in 15 percent of draws, and Q below, above, at and around $\bar{Q}$, and it extrapolates from $\Delta = 0.002$ and 0.001. Where $Q \leq \bar{Q}$,

the largest error against $x^\circ(0)$ is 3.1×10^{-7} (2.1×10^{-4} at $Q = \bar{Q}$ exactly), and the test $Q \leq \bar{Q}$ classifies all 6,000 draws as the test $Q \leq \inf_t x^\circ(t)$ does. Where exit binds, the backward-induction value exceeds $x^\circ(0)$ in every draw and converges to the stopping value at first order (largest errors from 0.14 at $\Delta = 0.004$ to 0.0097 at $\Delta = 0.00025$, mean errors from 3.6×10^{-3} to 2.3×10^{-4}), with the largest errors at β near 0.95, where the exit-floor branch carries a factor $1/(1-\beta)$. The counterexample in the proof is reproduced (-0.3679 by backward induction against -0.7358 for $x^\circ(0)$), and the edge cases with zero or constant costs return Proposition 1.

Corollary 5. `v2_m2_interruption.py` solves four versions of the game by backward induction over the buffer stock, with the exit floor and constant baseline costs: the stock buffer of the corollary and the three variants (a) to (c). It draws 3,000 parameter sets ($\lambda = 0$ in 25 percent, $\beta = 0$ in 10, $h = 0$ in 20, $\ell = 0$ in 5, $t_U \in [0, 3]$; exit binds without interruption in 977) and extrapolates from $\Delta = 0.002$ and 0.001. For the stock buffer, the largest error against part (ii) is 8.1×10^{-4}; the decision agrees with $(1-\psi)D > [Q - x_0^\circ]^+$ in all 2,834 draws that are not near a tie, and with the condition of part (i) in all 1,862 such draws in which exit does not bind; in 188 draws exit binds and the condition of part (i) holds, yet interruption does not raise P^*; and every one of the 1,048 draws that interrupt at the full stock also interrupts at every lower stock. For variant (a), the largest error is 8.1×10^{-4}, no draw interrupts before t_U, and in 435 draws interruption raises P^* although the condition of part (i) fails. For variant (b), the gain is exactly zero in all 2,852 draws with $\ell > 0$, and the Markov equilibrium is unique at every stock. For variant (c), in 1,070 draws in which exit does not bind, the value is never below that of the stock-buffer timing, and the gain is positive in 551 draws in which the condition of part (i) fails. In a hand-picked case ($r = 0.2$, $\lambda = 0.3$, $\beta = 0.5$, $\ell = 0.5$, $c_S = 2$, $h = 0$, so $\bar{t} = 2.773$), the gain is 0.2115 at $t_U = 0.5$ and 0.0472 at $t_U = 2$, as the closed form gives.

C Europe 2031 Read Through the Model

Europe 2031 stages four rounds of US–European bargaining over ASML (Juijn et al., 2026). Its authors say the storyline is not meant to be a forecast, and we do not treat it as evidence. With as many free parameters as our model has, the statement that an episode is an equilibrium carries little information on its own. Table 5 therefore states, for each move, the result that applies and the conditions under which the move is an equilibrium, so that a reader can see what the story assumes about the parameters.

2028. The scenario's Washington demands that The Hague end ASML's remaining sales and servicing of immersion tools in China and threatens the foreign direct product rule; "The Dutch cave", and nothing is negotiated in return (Juijn et al., 2026). The demand concerns the hedge, and the threat uses flow jurisdiction, which by Proposition 3(b) is credible whenever its net ex post cost is within Washington's stake and does not depend on Europe's slack. The text concedes that "such penalties would also harm the US". In our model that self-harm attaches to penalties on the whole

firm (χ_f), and jurisdiction over China-bound flows stays credible for every V. What the episode needs is a nonpositive premium and no rupture risk. The crisis comes during a compute crunch, when V is high, so if the dispute carried a hazard of rupture, Corollary 1(c) would give Europe a positive transfer whenever $Q + \beta(V - Q) > 0$. The scenario attributes the outcome to a European position that "fragments before it has properly formed". Licensing is a Dutch decision (section 7.1), and solidarity among member states reaches the gatekeeper only through transfers, so a united position would not by itself have changed Q.

2029. The Frontier Inference Services Rule caps Tier 2 countries at 25 percent of any provider's frontier inference, which the text says means "roughly halving European allocations", while the Netherlands joins the Five Eyes, Japan, South Korea and Taiwan in Tier 1 (Juijn et al., 2026). Cutting a loyal Europe lowers the conditional part of the AI instrument, because conditional parts are measured from what a loyal ally receives (section 4.3). Washington spends slack to keep compute, and Europe's premium drifts upward. The real AI Diffusion Rule, rescinded before it took effect, is our evidence that such tiering is costly for Washington (section 4.3). Tier 1 for the Netherlands is the targeted benefit of remark 2. The failed Council vote is the least secure episode. The arithmetic in section 7.3 shows that the states the scenario names cannot block, so the story needs further members with $q_i \leq 0$ or an informal norm that keeps a measure large states dislike from reaching a vote. The measure itself, a restriction on exports and servicing to US fabs in Arizona, is a hold-up instrument aimed at the servicing Washington can most easily compel (section 7.4).

The stockpile. Washington's claim that it has "amassed enough EUV machines and maintenance know-how" through Taiwanese partners to outlast Europe (Juijn et al., 2026) is a claim about t_U and h in Corollary 5. It lowers c_S, what Washington's own fabs would lose in a dispute, and leaves V, its loss if China gained the tools, where it was. The claim and the ultimatum are therefore compatible: a Washington that can do without ASML's servicing still has a motive to keep China from gaining the chokepoint. Under complete information the claim would be either true or irrelevant. As a bluff it belongs to the incomplete-information extension that section 9.1 leaves aside.

2031. The vice president's offer is a preventive demand of the kind Proposition 5 describes, issued "now, while the option still exists" (Juijn et al., 2026). On our reading, Chinese credit to a Portuguese bank, to Greece and to a Spanish region raises ω, and rationing has lowered d_{ai}, so Washington expects the premium to turn positive and acts while its threat still bites (Corollary 4). Three details of the offer bear on the conditions. A controlling vote in a Dutch-American holding stays under Dutch law and has low enforceability Δ_c, which by Corollary 3(c) lowers what control is worth to Washington. Capital for production on US soil raises Δ_c. Cash transfers indexed to US AI windfalls are promises paid after control has passed, credible only up to $\bar{m}_U$ (section 4.2). The Tier 3 threat is a cut Washington would make permanent, and by Lemma 2(iii) a longer cut adds no bite once Europe's future premium is positive.

In the complete-information game an issued ultimatum is accepted, so Europe's refusal requires the setting of Proposition 6, in which Washington does not know what a transfer would cost the governments concerned. Refusal is then more likely when China's punishment of a transfer is credible and large and when Europe's future premium is high. Beijing's threat is conditioned on signing, so by Corollary 3(d) it is never carried out if control passes, and it protects European ownership only by deterring the demand. The scenario's further claim that signing with neither power means absorbing "the full displeasure of both great powers at once" needs a Chinese punishment of non-alignment that the text does not state.

The epilogue's counterfactual coalition of the Netherlands, Germany, France, Norway, the United Kingdom, Canada, Japan and South Korea (Juijn et al., 2026) lies outside the model, which has one rival and lets holders bargain one at a time (section 9.1).

Table 3: Observable implications of the model and of rival accounts

Implication (result)	This model	Shared threat (large η)	Hierarchy	Autonomy–security trade-off	Conflict expectations	Cost minimization
Threshold (Corollary 1)	Holder is paid only if $Q > 0$, or if talks risk rupture and it has agenda power	No payment; holder aligns for its own reasons	No payment; compliance is owed	Protection is the payment	Allies concede without payment	No prediction about payment
Linkage only with slack (Proposition 3(b))	Linkage against secondary demands appears when the holder has slack and fades as $Q \to 0$	Little coercion of any kind	Commands need no threats	Protection is linked to autonomy throughout	Coercion of allies rare but effective, whatever the rival offers	Cheapest credible instrument, unchanged as Q moves
One budget across issue areas (Proposition 2(c), Corollary 2)	Demands on trade, defence and controls are substitutes; the budget shrinks as the rival's offer rises	Issue areas independent	Authority bounded issue by issue by legitimacy	One autonomy price for security, set within the alliance	No cross-issue constraint	Instruments priced separately
Counter-threats (Proposition 3(a))	Change the patron's demand mix and tax the patron; executed only to crowd out other demands	Holder absorbs them and keeps aligning	Patron shields its subordinate from third parties	No prediction	Rival's coercion of an adversary yields little	No prediction
Rival bids (Proposition 4(c)(i))	A stronger bid elicits a more conditional guarantee, not a payment, while assurance is cheap	No response needed	Patron must provide more order to keep its authority	Better terms for the protégé	No prediction	No prediction
Preventive control (Proposition 5, Corollary 4)	Control is demanded while the holder is weak but expected to strengthen	No demand	More control where dependence is deeper, with no window	No prediction	No prediction	Control demanded whenever cheapest, with no window
Frontline states (remark 4)	Members with large conditional exposure join blocking coalitions at no cost, though they cannot block alone	Members most hostile to China and Russia block	Most dependent members block	Members receiving most security block	No prediction	No prediction

Note. Entries for the rival accounts are our derivations from Lake (2009), Morrow (1991), Drezner (1999) and Stigler (1970), not claims their authors make about these cases. Q is the exit premium (definition 1).

Table 4: US demands on holders, coded through the model

Holder, period	US demand	Instrument threatened	Compensation	Rival bid or counter-threat	Outcome	Reading in the model
Netherlands (ASML), 2019–2026	End EUV, then advanced and older immersion DUV and metrology sales to China; from 2026, servicing	Flow jurisdiction: zero de minimis (2023), SME FDP rule (2024), MATCH Act (proposed)	None; Supplement No. 4 status, a threat left unexercised	China's market, 26–36% of sales (2023–2025); Nexperia ban executed (2025)	Aligned on sales with lags; kept the licensing pen; servicing in China continued into 2026	$Q \leq 0$ through jurisdiction (remark 3); hedge tolerated (Corollary 2); security channel untested
Japan (Tokyo Electron), 2023–2024	Control advanced equipment; in 2024, restrict sales and servicing in China	Flow jurisdiction: FDP rule not used but not ruled out	None found; Supplement No. 4 status	China threatened "severe economic retaliation"; Toyota warned of lost access to critical minerals	23 equipment types licensed (2023), "not aligning"; no new measures (2024); servicing continued into 2026	Counter-threat raised the cost of the servicing concession; threat not executed (Proposition 3(a))
South Korea (Samsung, SK hynix fabs in China), 2025	No expansion or upgrade of Chinese fabs	Flow jurisdiction: VEU status revoked	None; annual licences to operate (2026); trade deal silent on controls	None documented	Accepted; fabs operate under licence	Hedge policed by flow jurisdiction; threat rebuilt as licences, not executed (Proposition 3(b))
Germany (Zeiss, Trumpf), 2023–2024	Join equipment controls; "primary target is Zeiss"	None documented	None	None documented	Reportedly declined; slower licensing of laser exports to China	Demand of low marginal value while Dutch licences govern the tools that carry Zeiss optics (Corollary 2)
Taiwan, 2025	Half of US-used chips made in the US ("50-50")	Floated publicly while tariff talks were under way; no explicit threat found	None	China a threat, not a bidder	Rejected publicly	Relocation demand with high enforceability (Corollary 3(c)); outside scope
UAE, 2025–2026	Align security regulations, prevent diversion, invest in US data centres	Conditional linkage: chip access	Client pays in investment; receives chip approvals and Country Group A:5	Not documented in our sources	Licence-free access (July 2026), citing defence partnership and Operation Epic Fury	Roles reversed: Washington holds the chokepoint; explicit security-for-access reward; outside scope
West Germany, 1961–1967 (offsets)	Offset the foreign-exchange cost of US troops	Conditional linkage: troop levels, explicit in 1966	Holder paid: arms purchases, $500m in US securities, gold pledge; US placed orders in the UK	German option to convert dollars into gold, given up	Redeployment of up to 35,000 US personnel agreed (1967); Erhard fell (1966)	Linkage extracting sunk payments; repeated renegotiation (Proposition 2(d)); domestic cost outside the model
United Kingdom, 1957 (CoCom China differential)	Keep stricter controls on China than on the Soviet bloc	None; Washington chose no sanctions	None; US-backed compromise failed	China trade; 12 of 16 members for abolition	Differential abolished; US kept its own embargo	Secondary demand left without linkage (Corollary 2)
West Germany, 1975–1977 (Brazil nuclear deal)	Defer or stop enrichment and reprocessing transfers	High-level pressure ("major crisis"); no explicit linkage, though Bonn feared for the US commitment	Offered to Brazil (fuel guarantee), not to Bonn	Brazil as buyer; Bonn cited its contract and the NPT	Contract kept; informal restraint on future transfers by 1978	Concession on the forward margin; sunk deal kept
Germany and Spain, 2026 (posture, access)	Germany: none stated; cut followed Merz's criticism of US	Posture cut executed (Germany); trade threats (Spain)	None	None	About 5,000 troops ordered out of Germany; Spain kept its	Executed instrument is spent (section 5.3);

Table 5: Europe 2031 moves and the conditions under which they are equilibria

Scenario move	Results	Equilibrium when
May 2028: Washington demands an end to ASML's immersion sales and servicing in China and threatens the foreign direct product rule; the Dutch comply and get nothing in return	Proposition 3(b), Corollary 1(c)	The jurisdictional threat is credible ($e_F \leq \bar{m}_U$), so it deters without being carried out and leaves Q unchanged. A zero transfer needs $Q \leq 0$ and either no rupture risk in the dispute ($\lambda = 0$) or $Q + \beta(V - Q) \leq 0$.
April 2029: the Frontier Inference Services Rule puts the Netherlands in Tier 1 and most of Europe in Tier 2	section 4.3, remark 2, Corollary 2	Rationing a loyal Europe lowers $x_{ai}(A)$, hence d_{ai} and Washington's slack. On our reading Washington rations only if the compute it keeps is worth more to it than the slack it loses. Tier 1 lowers the gatekeeper's price only if it is baseline provision credibly withdrawn on exit.
May 2029: an Anti-Coercion Instrument measure restricting ASML's exports and servicing to US fabs in Arizona falls short of a qualified majority	remark 4, Corollary 5	The named opponents and abstainer do not block (section 7.3). Failure needs more members with $q_i \leq 0$ (six more, or a large state) or consensus norms that keep the measure from a vote. Even adopted, the measure adds little where Washington can compel servicing (h near one).
March 2031: Washington says that through Taiwanese partners it can forgo ASML longer than Europe can forgo frontier AI	Proposition 7, Corollary 5	The claim is that the US buffer t_U exceeds $\bar{t}$. It holds for a measure confined to US soil if h is near one; for a broad interruption it depends on h in Taiwan and Korea. It concerns c_S and leaves V unchanged.
March 2031: a joint Dutch-American holding with a US controlling vote, capital for US production and windfall-indexed transfers, made public with a Tier 3 threat	Proposition 5, Corollary 3, Corollary 4, Lemma 2	Issued iff $\Gamma > 0$: a premium expected to turn positive and a bite above $\sigma + r^c$ while slack lasts, or $v_c > \iota$. A vote in a Dutch company has low Δ_c, which lowers Γ. Transfers due after the transfer of control are credible only up to $\bar{m}_U$. A permanent cut adds no bite at dates with $Q \geq 0$.
March 2031: Europe refuses; Beijing says rare-earth terms will be reviewed if the Dutch sign	Proposition 6, Corollary 3(d)	Under complete information an issued ultimatum is accepted, so refusal needs private sovereignty costs. China's threat is never carried out if control passes; if credible and large enough ($\Gamma(r_{\max}) \leq 0$) it deters the demand.